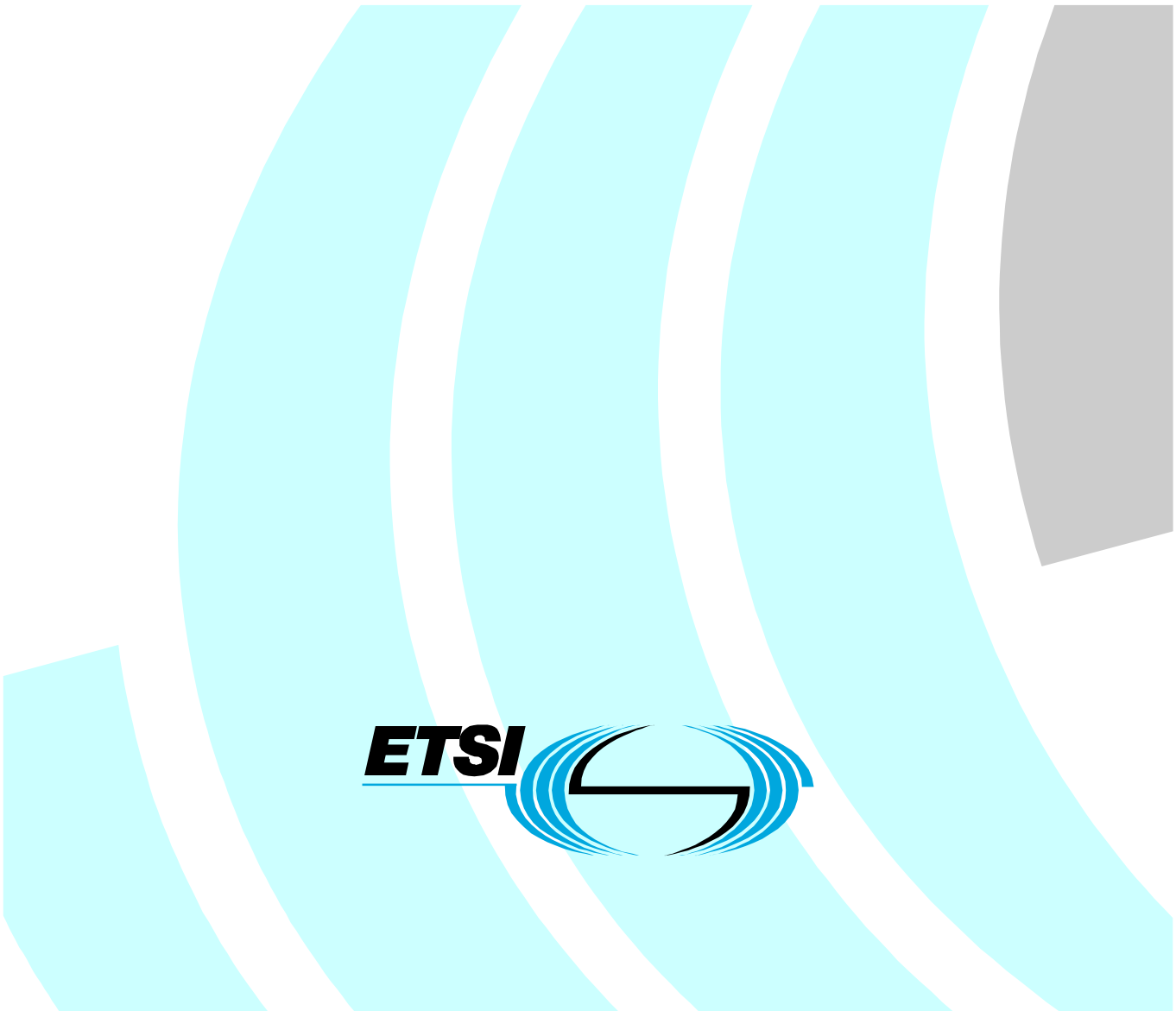
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Technical Specification

Methods for Testing and Specification (MTS); Internet Protocol Testing (IPT): IPv6 Mobility; Interoperability Test Suite



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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Methods for Testing and Specification (MTS).

Introduction

IPv6 is the next generation Internet. It gives vastly increased address space and true end-to-end communication. It has improved security and mobility features and allows "plug-and-play" connection to the network. The complexity of implementing IPv6 technology and the relative openness of IETF standards means that wide-ranging and effective testing of IPv6 products will be one of the key factors in ensuring the deployment, interoperability, security and reliability of the IPv6 infrastructure.

The present document specifies interoperability tests for IPv6 Mobility. The test suite results from and analysis of RFC 3775 and RFC 3776, the extraction of the requirements contained in these documents, and a selection of the requirements which could be tested by interoperability means.

The methodology and framework used to analyse the RFCs, to extract the requirements, write the Test Purposes, and the test descriptions is described in TS 102 351 [1]. The reader is strongly encouraged to read TS 102 351 [1] in order to make the best usage of the present document.

1 Scope

The present document specifies the interoperability Test Descriptions (TDs) with integrated Test Purposes (TPs) for the IPv6 Mobility standards. The TDs are presented in the tabular form specified in TS 102 424 (see Bibliography) and the TPs are defined using the TPLan notation also described in ES 202 553 (see Bibliography).

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication and/or edition number or version number) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.

Referenced documents which are not found to be publicly available in the expected location might be found at http://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication ETSI cannot guarantee their long term validity.

[1] ETSI TS 102 351: "Methods for Testing and Specification (MTS); Internet Protocol Testing (IPT)".

[2] IETF RFC 3775: "Mobility Support in IPv6".

[3] IETF RFC 3776: "Using IPsec to Protect Mobile IPv6 Signalling Between Mobile Nodes and Home Agents".

[4] ETSI TS 102 559: "Methods for Testing and Specification (MTS); Internet Protocol Testing (IPT): IPv6 Mobility; Requirements Catalogue".

3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

CoA	Care of address
CN	Correspondent Node
EUT	Equipment Under Test
HA	Home Agent
MN	Mobile Node
MTU	Maximum Transmission Unit
QE	Qualified Equipment
TD	Test Description
TP	Test Purpose
TPLan	Test Purpose Language
TSS	Test Suite Structure

4 IPv6 Mobility Interoperability Test Specification

4.1 Test Descriptions

Test Description presentation and concepts are explained in TS 102 351 [1].

Requirements referred to within the Test Description (example: RQ_001_1016) are all contained in TS 102 559 [4], the IPv6 Mobility "Requirements catalogue".

4.1.1 Index of test grouping

In the present document, tests have been grouped according to the original RFC clause numbers from which they were extracted.

In the table below, each close number (example: "5.1") refers to the clause number of the RFC from which the requirements covered by the Tests Descriptions were extracted.

NOTE: Test Descriptions covering requirements coming from more than one clause are repeated in the relevant clause.

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5.1 Binding Updates to Home Agents

Test Description			
Identifier:	TD_MOB_1016_01	**Test Purpose:**	TP_MOB_1016_01
Summary:	'Home Agent supports transport_mode_ESP protection of Binding Update messages'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1016		

```
with {     QE1 disconnected
        and QE1 configured to protect any Binding_Update packet
                                   using transport_mode_ESP
      }
  ensure that
      { when {     QE1 is connected to a foreign_network
           and QE1 receives a packet from QE4
                              indicating that a response is required }
        then { QE1 sends response directly to QE4 }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is disconnected QE1 is configured to protect Binding Update packets using transport mode ESP

Step	Test Sequence	Verdict	
		Pass	Fail
1	Connect QE1 to Foreign Network 1		
2	Wait a few seconds		
3	Cause QE4 to send an Echo Request to the Home Address of QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1017_01	**Test Purpose:**	TP_MOB_1017_01
Summary:	'Mobile Node uses transport_mode_ESP to protect Binding Update messages'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1017		

```
with { QE1 configured to accept only a Binding_Update
           secured using transport_mode_ESP
      }
  ensure that
      { when {     EUT moves to a foreign_network
           and QE4 is requested to send a packet to EUT
                              indicating that a response is required }
        then { QE4 indicates receipt of the response }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected to Home Network 1 QE1 configured to serve as Home Agent for EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Configure QE1 to accept only ESP protected binding update messages		
2	Move EUT to foreign network 1		
3	Wait a few seconds to allow EUT to attempt to register with QE1		
4	Cause QE4 to send an Echo Request to the Home Address of EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

5.2.5 Return Routability Procedure

Test Description			
Identifier:	TD_MOB_1047_01	**Test Purpose:**	TP_MOB_1047_01
Summary:	'Correspondent Node completes Return Routability Procedure before accepting a Binding Update from a mobile node'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1047		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
         and QE1 unreachable from QE4 -- QE4 cannot complete RRP
       }
  ensure that
       { when { QE4 is requested to send a packet to the EUT
                    indicating that a response is required }
         then { EUT sends no response directly to QE4 }
       }
```

Pre-test conditions:	Not testable by interoperability means Not possible for a QE to send a BU without completing the RRP !		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

5.2.6 Authorizing Binding Management Messages

Test Description			
Identifier:	TD_MOB_1068_01	**Test Purpose:**	TP_MOB_1068_01
Summary:	'Correspondent Node does not establish any binding with a lifetime greater than MAX_RR_BINDING_LIFETIME'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1068		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
    }
  ensure that
       { when {     EUT establishes a binding to QE4}
         then {     EUT and QE4 are able to communicate directly
                                within the binding_lifetime
                and EUT and QE4 are unable to communicate directly
                                after MAX_RR_BINDING_LIFETIME }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to perform Route Optimization QE4 is configured to perform Route Optimization QE4 is connected to Foreign Network 1 and registered to its Home Agent QE1 QE3 is configured with the same (home) address as QE4 QE3 is disconnected from Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Wait a few seconds (to let QE4 and EUT perform the Return Routability Procedure and establish Bindings)		
4	Disconnect QE1 from Home Network 1		
5	Cause EUT to send an Echo Request to the home address of QE4		
6	Check: does EUT receive an Echo Reply from QE4? (this ensures that EUT and QE4 have established a Binding)	Yes	No
7	Connect QE3 to Home Network 1		
8	Disconnect QE4 from Foreign Network 1		
9	Cause EUT to send an Echo Request to the home address of QE4		
10	Check: does EUT receive an Echo Reply from QE4?	No	Yes
11	Wait 420 seconds==MAX_RR_BINDING_LIFETIME (Binding between EUT and QE4 Bindings has expired)		
12	Cause EUT to send an Echo Request to the Home Address of QE4		
13	Check: does EUT receive an Echo Reply from QE3 (which is a clone of QE4)?	Yes	No
Observations:	Once the Bindings have expired the Correspondent Node can no longer communicate directly with the Mobile Node and will send the Echo Request to the Home Address of the Mobile Node via the Home Agent. In our case, QE3 respond because it is configured with the same address as QE4.		

Test Description			
Identifier:	TD_MOB_1068_02	**Test Purpose:**	TP_MOB_1068_02
Summary:	'Mobile Node does not establish any binding with a lifetime greater than MAX_RR_BINDING_LIFETIME'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1068		

```
--NOT interop TESTABLE
-- ==> there is no Interoperability method to guarantee that a
-- Mobile Node (on a foreign network) will use its Home Address
-- to send a new packet (instead of using its Care-of-Address).

  with {     EUT configured to perform route_optimization
         and EUT away_from_home
         and EUT registered to QE1
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when {     EUT establishes a binding to QE4 }
         then {     EUT and QE4 are able to communicate directly
                                within the binding_lifetime
                and EUT and QE4 are unable to communicate directly
                                after MAX_RR_BINDING_LIFETIME }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEANS There is no Interoperability method to guarantee that a Mobile Node (on a foreign network) will use its Home address to send a new packet (instead of using its Care-of-Address)		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

8.4 IPv6 Home Agents

Test Description			
Identifier:	TD_MOB_1803_01	**Test Purpose:**	TP_MOB_1803_01
Summary:	'Home Agent discards packets received from a non registered mobile node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_05
References:	RQ_001_1803		

```
with
    {
          EUT disconnected from home_network 1
      and QE2 configured to use the same IP_address as EUT
      and QE1 away_from_home
      and QE1 registered to QE2
      and QE1 configured to perform route_optimization
      and QE4 configured to perform route_optimization
    }
  ensure that
       { when {     QE2 is disconnected from home_network
                and EUT is connected to home_network
                and QE1 is requested to send a packet to QE4
                                indicating that a response is required }
         then {     QE3 indicates no receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is disconnected QE2 serves as Home Agent on Home Network 1 with the same address as EUT QE1 is connected to Foreign Network 1 and registered to its Home Agent QE2 QE1 is configured to perform Route Optimization QE4 is configured to perform Route Optimization

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE2		
2	Connect EUT to Home Network 1		
3	Cause QE1 to send an Echo Request to QE4		
4	Check: does QE1 receive an Echo Reply?	No	Yes
Observations:	Ensure that QE1 use its Home Address when pinging QE4 (Home Address Option present in IPv6 packet).		

8.5 IPv6 Mobile Nodes

Test Description			
Identifier:	TD_MOB_1810_01	**Test Purpose:**	TP_MOB_1810_01
Summary:	'MN is able to receive tunnelled multicast packets from HA'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1810		

```
with {      EUT subscribed to a global_multicast_group
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { QE3 is requested to send a packet
                                to the global_multicast_group }
         then { QE3 indicates no receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1 EUT has registered to multicast group FF1E:1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign Network 1		
2	Cause QE3 to send an Echo Reply to multicast address FF1E::1		
3	Check: does QE3 receive an Echo Reply?	Yes	No
Observations:			

9 Correspondent Node Operation

Test Description			
Identifier:	TD_MOB_1013_01	**Test Purpose:**	TP_MOB_1013_01
Summary:	'Home Agent supports the protection of Binding Update message'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1013		

```
with
    {        QE1 at_home
         and QE1 configured to protect any Binding_Update
    }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE4 is requested to send a packet to QE1
                                  indicating that a response is required }
         then { QE4 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 is configured to protect Binding Update messages

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (for QE1 to establish a binding with EUT)		
3	Cause QE4 to send an Echo Request to the home address of QE1		
4	Check: does QE4 receive an Echo Reply?	Yes	No

Observations:	If EUT does not support protection of Binding Update, then it is unable to accept the binding.

Test Description			
Identifier:	TD_MOB_1013_02	**Test Purpose:**	TP_MOB_1013_02
Summary:	'Mobile Node supports the protection of Binding Update message'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1013		

```
with {     EUT at_home
         and QE1 configured to accept only a secured Binding_Update
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE4 is requested to send a packet to EUT
                                  indicating that a response is required }
         then { QE4 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to home network 1 QE1 is configured to accept only protected Binding Update messages

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds (for EUT to establish a binding with QE1)		
3	Cause QE4 to send an Echo Request to the home address of EUT		
4	Check: does QE4 receive an Echo Reply?	Yes	No

Observations:	

Test Description			
Identifier:	TD_MOB_1082_01	**Test Purpose:**	TP_MOB_1082_01
Summary:	'Home Agent uses a Security Association to protect integrity and authenticity of Mobile_Prefix_Solicitations and Advertisements'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1014, RQ_001_1082, RQ_001_1607		

```
with {     QE1 at_home
         and QE1 configured to protect
                          Mobile_Prefix_Solicitations
         and QE1 configured to accept only secured
                          Mobile_Prefix_Advertisements
         and QE1 configured to have a short prefix_lifetime --(suggest 30s)
       }
  ensure that
       { when {        QE1 moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to QE1
                             indicating that a response is required }
         then {        QE4 indicates receipt of the response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT QE1 is configured to protect Mobile Prefix Solicitation messages QE1 is configured to only accept protected Mobile Prefix Advertisement messages EUT is configured to advertise a preferred prefix lifetime of 20 sec EUT is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that initial lifetime of the home address of QE1 has expired		
2	Cause QE4 to send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1082_02	**Test Purpose:**	TP_MOB_1082_02
Summary:	'Mobile Node uses a Security Association to protect integrity and authenticity of Mobile Prefix Solicitations and Advertisements'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1014, RQ_001_1082, RQ_001_1663, RQ_001_1664		

```
with {     EUT at_home
         and QE1 configured to accept only secured
                           Mobile_Prefix_Solicitations
         and QE1 configured to protect Mobile_Prefix_Advertisements
         and EUT configured to have a short prefix_lifetime (--suggest 30s)'
       }
  ensure that
       { when {        EUT moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to the EUT
                             indicating that response is required }
         then {        QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE1 is configured to only accept protected Mobile Prefix Solicitation messages QE1 is configured to protect Mobile Prefix Advertisement messages QE1 is configured to advertise a preferred prefix lifetime of 20 sec QE1 is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that EUT has send a Mobile Prefix Solicitation message and received a Mobile Prefix Advertisement message		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

9.3.1 Receiving Packets with Home Address Option

Test Description			
Identifier:	TD_MOB_1412_01	**Test Purpose:**	TP_MOB_1412_01
Summary:	'Correspondent Node receives packets directly from the Mobile Node when a binding exists or Mobile Node believes in its existence'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1412		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                    indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 EUT is configured to support route optimization QE4 is configured to support route optimization QE4 has established a binding with EUT

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnect QE1 from Home Network 1		
2	Cause QE4 to send an Echo Request to EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No

Observations:	Actually it would be sufficient to check whether EUT actually receives the ECHO request from QE4. However I don't know yet how we should test that.

Test Description			
Identifier:	TD_MOB_1413_01	**Test Purpose:**	TP_MOB_1413_01
Summary:	'Correspondent Node drops packets containing Home Address option (but no Binding Update) when there is no corresponding Binding Cache entry'		
Roles:	Correspondent_Node, Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1413, RQ_001_1413, RQ_001_1427		

```
with {      EUT configured to perform route_optimization
          and QE4 configured to perform route_optimization
          and QE4 away_from_home
          and QE4 registered to QE1
          and EUT having no Binding_Cache_entry for QE4
          and QE4 having a Binding_Cache_entry for EUT
       }
  ensure that
       { when { EUT receives a packet directly from QE4
                    indicating that a response is required }
         then { EUT sends no response to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT configured to perform route optimization QE3 configured to perform route optimization QE4 configured to perform route optimization EUT disconnected QE3 is connected to Home Network 2 QE3 is configured to have the same address as EUT QE4 connected to Foreign Network 1 and registered to its Home Agent QE1 QE4 has established a binding with QE3

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE3 from Home Network 2		
2	Connect EUT to Home Network 2		
3	Cause QE4 to send an Echo Request to EUT		
4	Check: does QE4 receive an Echo Reply from EUT	No	Yes
Observations:	Step 3 : QE4 MUST use its Home Address when pinging EUT (Home Address Option present in IPv6 packet).		

Test Description			
Identifier:	TD_MOB_1413_02	**Test Purpose:**	TP_MOB_1413_02
Summary:	'Correspondent Node drops packets containing Home Address option (but no Binding Update) and send a Binding Error message to the Mobile Node, when there is no corresponding Binding Cache entry'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1413		

```
with {      EUT configured to perform route_optimization
          and QE4 configured to perform route_optimization
          and QE4 away_from_home
          and QE4 registered to QE1
          and EUT having no Binding_Cache_entry for QE4
          and QE4 having a Binding_Cache_entry for EUT
                }
  ensure that
       { when { EUT receives a packet directly from QE4
                      indicating that a response is required }
      then {        EUT sends no response to QE4
             before EUT and QE4 are able to communicate }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN Mobile Node may ignore the Binding Error from Correspondent Node		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1414_01	**Test Purpose:**	TP_MOB_1414_01
Summary:	'When using Route Optimization, a Correspondent Node drops any packet in which the Home Address option (but not Binding Update) does not match the Home Address recorded in its Binding Cache'		
Roles:	Correspondent_Node, Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1414, RQ_001_1414, RQ_001_1427		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT receives a packet from QE4
                      containing a Home_Address_Option
                      not set to the Home_Address of QE4 }
         then { EUT discards the packet }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to perform route optimization QE4 is configured to perform route optimization QE3 is configured to perform route optimization QE5 is configured to perform route optimization QE4 is connected to Foreign Network 1 and registered with QE1 QE4 has established a binding with EUT QE5 is disconnected QE5 is configured with the same link-layer Address as QE4 QE5 is configured with a Home Address which is not the same as QE4 QE5 is configured to be a Mobile Node to Home Network 1 QE5 is configured to register with its Home Agent QE1 QE3 is disconnected QE3 is configured with the same addresses as EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnect EUT		
2	Disconnect QE4		
3	Reboot QE1 (mobility support only)		
4	Connect QE3 on Home Network 2		
5	Connect QE5 on Foreign Network 1		
6	Wait a few seconds (for QE5 to register with its Home Agent QE1)		
7	Cause QE3 to send an Echo Request to the Home Address of QE5		
8	Check: does QE3 receive an Echo Reply from QE5	Yes	No
9	Disconnect QE3 from Home Network 2		
10	Reconnect EUT to Home Network 2		
11	Cause QE5 to send an Echo Request to EUT		
12	Check: does QE5 receive an Echo Reply from EUT?	No	Yes
Observations:	Step 7 causes QE5 to establish a binding with QE3. QE3 and QE5 share the same Care-Of Address, but do not have the same Home Address.		

Test Description			
Identifier:	TD_MOB_1414_02	**Test Purpose:**	TP_MOB_1414_02
Summary:	'When using Route Optimization, a Correspondent Node drops any packet in which the Home Address option (but not Binding Update) does not match the Home Address recorded in its Binding Cache and send a Binding Error mess		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1414		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
       }
  ensure that
       { when {        EUT receives a packet from QE4
                           containing a Home_Address_Option
                           not set to the Home_Address of QE4 }
         then {        EUT discards the packet
                 before EUT and QE4 are able to communicate }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN Mobile Node may ignore the Binding Error from Correspondent Node		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1415_01	**Test Purpose:**	TP_MOB_1415_01
Summary:	'A Correspondent Node drops packets in which the Source Address does not match the care-of address recorded in the corresponding Binding Cache Entry'		
Roles:	Correspondent_Node, Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1415, RQ_001_1415, RQ_001_1427		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT receives a packet from QE4
                    containing a Home_Address_Option
                    set to the Home_Address of QE4
                  and containing a Source_Address
                    not set to the Care_Of_Address of QE4 }
         then { EUT discards the packet }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to perform route optimization QE3 is configured to perform route optimization QE4 is configured to perform route optimization QE5 is configured to perform route optimization QE4 is connected to Foreign Network 1 and registered with QE1 QE4 has established a binding with EUT QE5 is disconnected QE5 is configured to be a Mobile Node to Home Network 1 QE5 is configured to use QE1 as Home Agent QE5 is configured with the same Home Address address as QE4 QE5 is configured with a Link-Layer address which is not the same as QE4 QE3 is disconnected QE3 is configured with the same addresses as EUT

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnect EUT		
2	Disconnect QE4		
3	Reboot QE1 (restart mobility support)		
4	Connect QE3 on Home Network 2		
5	Connect QE5 on Foreign Network 1		
6	Wait a few seconds (for QE5 to register with its Home Agent QE1)		
7	Cause QE3 to send an Echo Request to the Home Address of QE5		
8	Check: does QE3 receive an Echo Reply from QE5?	Yes	No
9	Disconnect QE3 from Home Network 2		
10	Connect EUT to Home Network 2		
11	Cause QE5 to send an Echo Request to EUT		
12	Check: does QE5 receive an Echo Reply from EUT?	No	Yes

Observations:	Step 7 causes QE5 to establish a binding with QE3. QE3 and QE5 share the Home Care-Of Address, but have different Care-Of addresses.

Test Description			
Identifier:	TD_MOB_1415_02	**Test Purpose:**	TP_MOB_1415_02
Summary:	'A Correspondent Node drops packets in which the Source Address does not match the care-of address recorded in the corresponding Binding Cache Entry and sends a Binding Error message to the Mobile Node'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1415		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
       }
  ensure that
       { when {        EUT receives a packet from QE4
                           containing a Home_Address_Option
                           set to the Home_Address of QE4
                       and containing a Source_Address
                           not set to the Care_Of_Address of QE4 }
         then {        EUT discards the packet
                 before QE4 and the EUT are able communicate }
       }
--*********************************************************************--
--* This TP is phrased as a conformance test                          *--
--*********************************************************************--
--TP id   : TP_MOB_1661_01
--summary : 'MN sends Mobil Prefix Solicitation to HA when its home address
--           becomes invalid'
--RQ ref  : RQ_001_1661
--Role    : Mobile_Node
--config  : CF_MOB_03
--TD ref  : TD_MOB_1661_01

--  with {     EUT away_from_home
--           and EUT registered to QE1
--       }
--  ensure that
--       { when { EUT 'has its home address which becomes invalid' }
--         then { EUT 'update the prefix lifetime information for its home address' }
--       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN Mobile Node may ignore the Binding Error from Correspondent Node		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

9.3.2 Sending Packets to a Mobile Node

Test Description			
Identifier:	TD_MOB_1423_01	**Test Purpose:**	TP_MOB_1423_01
Summary:	'Correspondent Node sends packets directly to Mobile Node when a binding already exists'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1423		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
         and EUT having established a binding with QE4
       }
  ensure that
       { when { QE4 sends a packet to the EUT
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 EUT is configured to support route optimization QE4 is configured to support route optimization QE4 has established a binding with EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE1 from Home Network 1		
2	Cause QE4 to send an Echo Request to EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

9.3.4 Receiving ICMP Error Messages

Test Description			
Identifier:	TD_MOB_1429_01	**Test Purpose:**	TP_MOB_1429_01
Summary:	'Correspondent Node deletes a binding cache entry if it receives persistent ICMP Destination Unreachable messages'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1429		

```
-- EUT=CN2
-- QE1=HA1
-- QE4=MN1

  with {     QE4 'away from home and registered to QE1'
         and EUT 'having established a binding with QE4'
       }
  ensure that
    {
      when {     QE4 'returns home'
             and QE4 'is unable to update its binding with EUT'
             and EUT is requested to 'send packets to QE4' }
      then {     EUT 'is temporarily unable to communicate with QE4'
             and EUT 'is able to communicate again with QE4 before the end of the binding lifetime' }
    }
```

Pre-test conditions:	QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 EUT and QE4 have established binding QE4 is configured not to ask Binding Acknowledgements The tester is aware of the lifetime of these bindings and of the remaining binding lifetime when he starts this test

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE4 from Foreign Network 1		
2	Cause router R1 to disable its interface connected to Foreign Network 1		
3	Connect QE4 to Home Network 1		
4	Wait a few seconds to let QE4 perform the "return home" procedure. (a protocol analyzer can be used to ensure that the "return home" procedure is finished) Note that EUT will not receive Binding Update messages from QE4 (because R1 is disconnected from Foreign Network 1) and still thinks that QE4 is in Foreign Network 1		
5	Cause R1 to enable its interface connected to Foreign Network 1		
6	Cause EUT to send an Echo Request message to the Home Address of QE4 (the IPv6 destination address of the Echo Request sent should still be the former care-of-address of QE4 in Foreign Network 1)		
7	Check: does EUT receive an Echo Reply from QE4? (EUT will receive an ICMP "Destination Unreachable" message from router R3)	No	Yes
8	Cause EUT to continuously send Echo Request messages to the Home Address of QE4 (EUT will receive persistent ICMP "Destination Unreachable" messages from router R3)		
9	Check: after several unsuccessful Echo Request messages, does EUT receive Echo Replies from QE4?	Yes	No
10	Check: when EUT receives the first Echo Reply from QE4, have the previous EUT <--> QE4 Bindings expired?	No	Yes

Observations:	NOTE 1: If EUT received no Binding Update message when EUT returned home, and EUT manages to re-establish communication with QE4 before the expiration of the EUT <--> QE4 bindings, then this implies that the ICMP Destination Unreachable messages caused EUT to delete these bindings and to use the Home Address of QE4 instead its previous care-of-address as IPv6 destination address. NOTE 2: RFC 3775 [2] states in clause 9.3.4 that "persistent ICMP messages lead to the deletion of the Binding", yet there is no detail about the amount of ICMP error messages that will cause the deletion to occur. Hence this TD might lead to inconclusive verdicts. NOTE 3: Step 2: beware of interface buffers, which can keep and then retransmit packets emitted before the interface is disabled! Do not unplug, but use "ifconfig" or equivalent.

9.3 Packet Processing

Test Description			
Identifier:	TD_MOB_1344_01	**Test Purpose:**	TP_MOB_1344_01
Summary:	'Correspondent Node discard received packets containing Type 2 Routing Headers'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1344		

```
with {     EUT disconnected
         and QE4 configured to have the same link_local_address as EUT
         and QE4 connected to Home_Network 2
         and QE4 registered to QE1
         and QE4 having established a binding with QE3
       }
  ensure that
       { when {     QE4 is disconnected
                and EUT is connected to Home_Network 1
                and QE3 is requested to send a packet to the EUT
                        indicating that a response is required
                        -- QE3 sends packets containing
                        -- Routing Headers Type 2
                and QE5 is requested to send a packet to the EUT
                        indicating that a response is required }
                        -- QE5 sends packets not containing
                        -- Routing Headers Type 2
         then {     QE3 indicates no receipt of the response from the EUT
                and QE5 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is disconnected QE4 is configured with the same link-layer address as EUT QE4 is connected to Home Network 2 and registered with its Home Agent QE1 QE4 has established a binding with QE3

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnect QE4		
2	Connect EUT to Home Network 2		
3	Cause QE3 to send an Echo Request to EUT		
4	Check: does QE3 receive an Echo Reply from EUT?	No	Yes
5	Cause QE5 to send an Echo Request to EUT		
6	Check: does QE5 receive an Echo Reply from EUT?	Yes	No
Observations:			

9.4.1 Receiving Home Test Init Messages

Test Description			
Identifier:	TD_MOB_1348_01	**Test Purpose:**	TP_MOB_1348_01
Summary:	'Correspondent Node is able to participate in Return Routability Procedure'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1050, RQ_001_1051, RQ_001_1348		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE4 is configured to perform route optimization EUT is configured to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 so sent an Echo Request to EUT		
2	Check: does QE4 receive a reply from EUT?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routability Procedure and establish Bindings.		
4	Disconnect QE1, the Home Agent of QE4		
5	Cause EUT to send an Echo Request to the Home Address of QE4		
6	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:	Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. Cannot send the Echo Request from QE4 because nothing guarantees that QE4 would use its Home Address.		

9.4.2 Receiving Care-of Test Init Messages

Test Description			
Identifier:	TD_MOB_1348_01	**Test Purpose:**	TP_MOB_1348_01
Summary:	'Correspondent Node is able to participate in Return Routability Procedure'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1050, RQ_001_1051, RQ_001_1348		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE4 is configured to perform route optimization EUT is configured to perform route optimization

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 so sent an Echo Request to EUT		
2	Check: does QE4 receive a reply from EUT?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routability Procedure and establish Bindings		
4	Disconnect QE1, the Home Agent of QE4		
5	Cause EUT to send an Echo Request to the Home Address of QE4		
6	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:	Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. Cannot send the Echo Request from QE4 because nothing guarantees that QE4 would use its Home Address.		

9.4 Return Routability Procedure

Test Description			
Identifier:	TD_MOB_1348_01	**Test Purpose:**	TP_MOB_1348_01
Summary:	'Correspondent Node is able to participate in Return Routability Procedure'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1050, RQ_001_1051, RQ_001_1348		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                    indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE4 is configured to perform route optimization EUT is configured to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 so sent an Echo Request to EUT		
2	Check: does QE4 receive a reply from EUT?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routability Procedure and establish Bindings		
4	Disconnect QE1, the Home Agent of QE4		
5	Cause EUT to send an Echo Request to the Home Address of QE4		
6	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:	Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. Cannot send the Echo Request from QE4 because nothing guarantees that QE4 would use its Home Address.		

9.5.1 Receiving Binding Updates

Test Description			
Identifier:	TD_MOB_1817_01	**Test Purpose:**	TP_MOB_1817_01
Summary:	'Correspondent Node registers the binding when it receives a BU from Mobile Node'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1817		

```
with { QE4 away_from_home
     and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { QE4 is requested to send a packet directly to EUT
                                indicating that a response is required
           }
         then { QE4 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and registered to its Home Agent QE1 QE4 is configured to perform Route Optimization EUT is configured to perform Route Optimization

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to the Home address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Disconnect QE1 from Home Network 1		
4	Cause EUT to send an Echo Request to the Home Address of QE4		
5	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1065_01	**Test Purpose:**	TP_MOB_1065_01
Summary:	'Correspondent Node sends a Binding_Acknowledgement message upon receipt of a Binding Update message containing A-Flag set to 1'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1065, RQ_001_1350		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured 'to request Binding_Acknowledgements from
                             correspondent nodes'
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends the response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and has registered to its Home Agent QE1 QE4 configured to perform route optimization EUT is configured to perform route optimization QE4 is configured to ask Correspondent Node for Binding Acknowledgement QE4 and EUT have never communicated yet and don't know each other (no binding established, no cache entries)

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routing Procedure and establish Bindings.		
4	Disconnect QE1 from Home Network 1		
5	Cause EUT to sent an Echo Request to the Home Address of QE4		
6	Check: does EUT receive an Echo Reply from QE4?	Yes	No

Observations:	NOTE 1: Configuring QE4 to ask Binding Acknowledgement from Correspondent Nodes ensures that its Binding Update message will have the A-Flag set to 1. Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. NOTE 2: Cannot send the Echo Request from QE4 (as described in Tesp Purpose) because nothing guarantees that QE4 would use its Home Address.

9.5.2 Requests to Cache a Binding

Test Description			
Identifier:	TD_MOB_1457_01	**Test Purpose:**	TP_MOB_1457_01
Summary:	'Correspondent Node creates a new entry in its binding cache when receiving a valid Binding Update'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1457		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and registered to its Home Agent QE1 QE4 is configured to perform route optimization EUT is configured to perform route optimization

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive a reply from QE4?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routing Procedure and establish Bindings		
4	Disconnect QE1, the Home Agent of QE4		
5	Cause EUT to send an Echo Request to the Home Address of QE4		
6	Check: does EUT receive a reply from QE4?	Yes	No

Observations:	NOTE 1: Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. NOTE 2: Cannot send the Echo Request from QE4 (as described in Tesp Purpose) because nothing guarantees that QE4 would use its Home Address.

Test Description			
Identifier:	TD_MOB_1458_01	**Test Purpose:**	TP_MOB_1458_01
Summary:	'Correspondent Node updates a binding cache entry on receipt of a valid Binding Update'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1458		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     QE4 moves to another foreign_network
                and EUT is requested to send a packet to QE4
                    indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT and QE4 have established bindings

Step	Test Sequence	Verdict	
		Pass	Fail
1	move QE4 to Foreign Network 2		
2	wait a few seconds (to let QE4 update its bindings with QE1 and with EUT)		
3	Disconnect QE1 from Home Network 1		
4	Cause EUT to send an Echo Request to the Home Address of QE4		
5	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1486_01	**Test Purpose:**	TP_MOB_1486_01
Summary:	'Correspondent Node deletes any Binding Cache entry after the expiration of the binding lifetime'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1461, RQ_001_1486		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
       }
  ensure that
       { when {    EUT establishes a binding to QE4}
         then {    EUT and QE4 are able to communicate directly
                         within the binding_lifetime
                and EUT and QE4 are unable to communicate directly
                         after the binding_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to perform Route Optimization QE4 is configured to perform Route Optimization QE4 is connected to Foreign Network 1 and registered to QE1 EUT is configured to have a lifetime at one minute

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	QE1 is disconnected from Home Network 1		
4	Cause QE4 to send an Echo Request to EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
6	Wait a minute (to let the binding between QE4 and EUT expire)		
7	Cause QE4 to send an Echo Request to EUT		
8	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
Observations:	Ensure that QE4 use its Home Address when pinging EUT (Home Address Option present in IPv6 packet).		

Test Description			
Identifier:	TD_MOB_1465_01	**Test Purpose:**	TP_MOB_1465_01
Summary:	'Correspondent Node deletes any existing related bindings on receipt of a Binding Update requesting to delete a binding'		
Roles:	Correspondent_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1465, RQ_001_1465, RQ_001_1466, RQ_001_1470		

```
with {     EUT away_from_home
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT returns home }
         then { QE4 and the EUT are able to communicate with EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT is configured to perform route optimization QE4 is configured to perform route optimization QE4 has established a binding to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Home Network 1		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

9.5.4 Sending Binding Acknowledgements

Test Description			
Identifier:	TD_MOB_1065_01	**Test Purpose:**	TP_MOB_1065_01
Summary:	'Correspondent Node sends a Binding_Acknowledgement message upon receipt of a Binding Update message containing A-Flag set to 1'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1065, RQ_001_1350		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured 'to request Binding_Acknowledgements from
                             correspondent nodes'
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                indicating that a response is required }
         then { EUT sends the response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and has registered to its Home Agent QE1 QE4 configured to perform route optimization EUT is configured to perform route optimization QE4 is configured to ask Correspondent Node for Binding Acknowledgement QE4 and EUT have never communicated yet and don't know each other (no binding established, no cache entries)

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Wait a few seconds to allow QE4 and EUT to perform Return Routing Procedure and establish Bindings		
4	Disconnect QE1 from Home Network 1		
5	Cause EUT to sent an Echo Request to the Home Address of QE4		
6	Check: does EUT receive an Echo Reply from QE4?	Yes	No
Observations:	NOTE 1: Configuring QE4 to ask Binding Acknowledgement from Correspondent Nodes ensures that its Binding Update message will have the A-Flag set to 1. Disconnecting the Home Agent QE1 guarantees that traffic will not go through if Route Optimization didn't succeed. NOTE 2: Cannot send the Echo Request from QE4 (as described in Tesp Purpose) because nothing guarantees that QE4 would use its Home Address.		

9.5.5 Sending Binding Refresh Requests

Test Description			
Identifier:	TD_MOB_1483_01	**Test Purpose:**	TP_MOB_1483_01
Summary:	'Correspondent Node tries to refresh a binding cache entry before it expires'		
Roles:	Correspondent_Node, Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1483, RQ_001_1483		

```
with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
         and EUT established 'in continuous unbounded communication with QE4'
       }
  ensure that
       { when { QE4 sends a packet to the EUT
                    after the binding_lifetime
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is connected to Foreign Network 1 and registered to QE1 EUT having established a binding with QE4 EUT is configured to have a binding lifetime at one minute

Step	Test Sequence	Verdict	
		Pass	Fail
1	wait 45 seconds after the registration between EUT and QE4		
2	Cause QE4 to send an series of Echo Request during 30 seconds to EUT		
3	After one minute and 5 seconds, disconnect QE1 from Home Network 1		
4	Check: are more one Echo Request of the series dropped?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1485_01	**Test Purpose:**	TP_MOB_1485_01
Summary:	'Correspondent Node stops retransmitting Binding Refresh Requests messages when it receives a Binding Update'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1485		

```
--*********************************************************************--
--*  However we wrap this up, it is a conformance test, not interop      *--
--*********************************************************************--

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     EUT 'tries to refresh its binding with QE4'
                and EUT receives 'a message indicating that
                                  the refresh is successful' }
         then { EUT 'stops sending refresh requests to QE4' }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN depending of the implementation to reduce the handover

Step	Test Sequence	Verdict	
		Pass	Fail
Observations:			

Binding Cache Management (was: 9.6 Cache Replacement Policy)

Test Description			
Identifier:	TD_MOB_1390_01	**Test Purpose:**	TP_MOB_1390_01
Summary:	'Correspondent Node maintains a separate binding cache for each of its unicast routable addresses'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1390		

```
with {     EUT configured to have 2 global_addresses
         and QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding to QE4
                  using its first global_address
         and QE1 disconnected from the network
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                    containing a source_address
                    set to its second global_address }
         then { QE4 does not indicate receipt of the packet  }
       }
--TP id   : TP_MOB_1429_01
--summary : 'Correspondent Node deletes a binding cache entry if it receives
--           persistent ICMP Destination Unreachable messages'
--RQ ref  : RQ_001_1429
--Role    : Correspondent_Node
--config  : CF_MOB_04
--TD ref  : TD_MOB_1429_01


--*********************************************************************--
--* The imprecision in this requirement and the lack of clear guidance  *--
--* on how many ICMP DU messages will cause the deletion of the     *--
--* binding cache entry make this almost impossible to test by IOP      *--
--* methods. It should be removed from the IOP test suite.              *--
--*********************************************************************--

--  with {     QE4 away_from_home
--         and QE4 registered to QE1
--         and EUT having established a binding with QE4
--       }
--  ensure that
--       { when {     QE4 returns home
--              and QE4 is unable to update its binding to the EUT
--              and EUT is requested to send packets to QE4 }
--         then {     EUT 'is temporarily unable to communicate with QE4'
--              and EUT is able to communicate with QE4
--                    before the binding_lifetime expires }
--       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEANS That relates to internal operations		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1486_01	**Test Purpose:**	TP_MOB_1486_01
Summary:	'Correspondent Node deletes any Binding Cache entry after the expiration of the binding lifetime'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1461, RQ_001_1486		

```
with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
       }
  ensure that
       { when {    EUT establishes a binding to QE4}
         then {    EUT and QE4 are able to communicate directly
                         within the binding_lifetime
              and EUT and QE4 are unable to communicate directly
                         after the binding_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to perform Route Optimization QE4 is configured to perform Route Optimization QE4 is connected to Foreign Network 1 and registered to QE1 EUT is configured to have a lifetime at one minute		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause EUT to send an Echo Request to the Home Address of QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	QE1 is disconnected from Home Network 1		
4	Cause QE4 to send an Echo Request to EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
6	Wait a minute (to let the binding between QE4 and EUT expire)		
7	Cause QE4 to send an Echo Request to EUT		
8	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
Observations:	Ensure that QE4 use its Home Address when pinging EUT (Home Address Option present in IPv6 packet).		

10 Home Agent Operation

Test Description			
Identifier:	TD_MOB_1807_01	**Test Purpose:**	TP_MOB_1807_01
Summary:	'If HA support the multicast group membership control protocols, HA must be capable to determine which multicast data packets to forward via the tunnel to MN'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1561, RQ_001_1807		

```
-- Config-HA
  with {      QE1 connected to home_network
           and QE1 subscribed to a global_multicast_group
           and QE3 subscribed to another global_multicast_group
       }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 registers to EUT }
         then {     QE1 receives packets from its global_multicast_group }
       }
--*************************************************--
--* RFC3776  -  Security for Mobility
--*************************************************--
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEANS You do not know if the Home Agent do not forward the multicast packets or if the Mobile Node drops this packets		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1557_01	**Test Purpose:**	TP_MOB_1557_01
Summary:	'HA tunnels multicast packets with a global scope to the mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1372, RQ_001_1557, RQ_001_1557		

```
with {      QE1 subscribed to a global_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { QE3 is requested to send a packet to the global_multicast_group
                                indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 has registered to multicast group FF1E::1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Cause QE3 to send an Echo Request to the multicast address FF1E::1		
3	Check: does QE3 receives an Echo Reply	Yes	No
4	Disconnect EUT		
5	Cause QE3 to send an Echo Request to the multicast address FF1E::1		
6	Check: does QE3 receives an Echo Reply	No	Yes
Observations:	FF1E::1 is a global scope, non reserved, multicast group.		

10.5 Dynamic Home Agent Address Discovery

Test Description			
Identifier:	TD_MOB_1804_01	**Test Purpose:**	TP_MOB_1804_01
Summary:	'Home Agent is able to participate in Dynamic Home Agent Address Discovery'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1368, RQ_001_1804		

```
with {     QE1 at_home
         and QE1 not configured to have its Home_Agent_address available
       }
  ensure that
       { when {     QE1 moves to a foreign_network
              and QE4 is requested to send a packet to QE1
                                indicating that a response is required }
         then {    QE4 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is disconnected from Home Network 1 QE1 is unaware of the address of its Home Agent

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds		
3	Cause QE4 to send an Echo Request to the Home Address of QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

10.6 Sending Prefix Information to the Mobile Node

Test Description			
Identifier:	TD_MOB_1592_01	**Test Purpose:**	TP_MOB_1592_01
Summary:	'HA schedules the delivery of the prefix information when a new prefix is added to its home subnet interface'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1370, RQ_001_1592, RQ_001_1805		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {         EUT generates a new prefix to the home_address of QE1
                before (QE4 sends a packet to QE1
                             using the new home_address
                         and indicating that a response is required) }
         then {         QE4 indicates receipt of a response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Foreign Network 1 QE1 registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to generate a new prefix added on its home subnet interface		
2	Wait for at least 3.5 seconds		
3	Check: does QE1 indicate a new home address?	Yes	No
Observations:	MIN_DELAY_BETWEEN_RAS (3,0 s) + MAX_RA_DELAY_TIME (0,5 s) = 3,5 s (cf. RFC 2561).		

10.6.2 Scheduling Prefix Deliveries

Test Description			
Identifier:	TD_MOB_1812_01	**Test Purpose:**	TP_MOB_1812_01
Summary:	'HA deliveries new prefix information when lifetime information (valid or preferred lifetime) of the prefix changes'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1812		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
         and EUT configured to have a short prefix_lifetime on the
                         Home_Network of QE1 --(suggest 30s)
       }
  ensure that
       { when { EUT is requested to configure a new prefix_lifetime
                    for QE1 }
         then { QE1 and QE4 are able to communicate
                    using the new prefix
                    after the original prefix_lifetime expires }
       }
--TP id   : TP_MOB_1602_01
--summary : 'If the bindings of MN expire before BU then HA stops sending MPAs'
--RQ ref  : RQ_001_1602
--Role    : Home_Agent
--config  : CF_MOB_02
--TD ref  : TD_MOB_1602_01

--XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX--
--X Not interoperability                                                      X--
--XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXxxxxx--

--  with {      QE1 away_from_home
--         and QE1 registered to EUT
--       }
--  ensure that
--       { when {    QE1 'is disconnected' from 'the foreign network'
--            and EUT 'indicates the binding of QE1 expired'
--           }
--         then { EUT sends 'no other messages to QE1' before 'new registration' }
--       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Foreign Network 1 QE1 registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Configure EUT with new, higher, valid lifetime		
2	Wait for expiration of the original prefix lifetime		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive an Echo Reply	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1592_01	**Test Purpose:**	TP_MOB_1592_01
Summary:	'HA schedules the delivery of the prefix information when a new prefix is added to its home subnet interface'		
Roles:	Home_Agent	**Configuration:**	7
References:	RQ_001_1370, RQ_001_1592, RQ_001_1805		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {         EUT generates a new prefix to the home_address of QE1
                before (QE4 sends a packet to QE1
                            using the new home_address
                        and indicating that a response is required) }
         then {         QE4 indicates receipt of a response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Foreign Network 1 QE1 registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to generate a new prefix added on its home subnet interface		
2	Wait for at least 3.5 seconds		
3	Check: does QE1 indicate a new home address?	Yes	No

Observations:	MIN_DELAY_BETWEEN_RAS (3,0 s) + MAX_RA_DELAY_TIME (0,5 s) = 3,5 s (cf. RFC 2561).

Test Description			
Identifier:	TD_MOB_1600_01	**Test Purpose:**	TP_MOB_1600_01
Summary:	'HA retransmits periodically unsolicited Advertisement until the receipt of a Mobile_Prefix_Solicitation from MN'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1600, RQ_001_1813		

```
with {     QE1 away_from_home
         and QE1 registered to EUT }
  ensure that
       { when {        QE1 is disconnected from the foreign_network
                before EUT generates a new prefix to the home_address of QE1
                before QE1 is connected to the foreign_network after 1 minute
                before QE4 sends a packet to QE1
                            using the new home_address
                       and indicating that a response is required }
         then {        QE4 indicates receipt of a response from QE1 }
       }
--TP id   : TP_MOB_1591_01
--summary : 'HA delivers new prefix information when the state of the flags
--           changes for the prefix of the MN registered'
--RQ ref  : RQ_001_1591
--Role    : Home_Agent
--config  : CF_MOB_02
--TD ref  : TD_MOB_1591_01

--XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX--
--X Not interoperability                                                      X--
--XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX--

--  with {     QE1 away_from_home
--         and QE1 registered to EUT
--       }
--  ensure that
--       { when { EUT generates 'a new state of the flags on the prefix of its home subnet interface' }
--         then { QE1 indicates 'a new state on its home address' }
--       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected in Foreign Network 1 QE1 is registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect EUT		
2	Cause EUT to generate a new prefix added on its home subnet interface		
3	Wait for one minute		
4	Reconnect EUT to home network		
5	Check: does QE1 indicate a new home address?	Yes	No
Observations:	It is not clear to me why a delay of 'one' minute is suggested here.		

Test Description			
Identifier:	TD_MOB_1603_01	**Test Purpose:**	TP_MOB_1603_01
Summary:	'HA transmits again unsolicited Mobile_Prefix_Advertisement after the prolongation of the binding of the Mobile Node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1603		

```
with {      QE1 away_from_home
          and QE1 is registered to the EUT
          and QE1 and the EUT configured to have a short binding_lifetime
                                                 -- (Suggest 30s)
       }
  ensure that
       { when {        QE1 is disconnected from the foreign_network
                 before the binding_lifetime expires between QE1 and the EUT
                 before QE1 is connected to another foreign_network }
         then { QE1 and QE4 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Foreign Network 1 QE1 registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE1		
2	Wait until QE1's binding to EUT expires		
3	Cause EUT to generate a new prefix added on its home subnet interface		
4	connect QE1 to foreign network 2		
5	Check: does QE1 indicate a new home address?	Yes	No
Observations:			

10.6.3 Sending Advertisements

Test Description			
Identifier:	TD_MOB_1082_01	**Test Purpose:**	TP_MOB_1082_01
Summary:	'Home Agent uses a Security Association to protect integrity and authenticity of Mobile_Prefix_Solicitations and Advertisements'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1014, RQ_001_1082, RQ_001_1607		

```
with {     QE1 at_home
         and QE1 configured to protect
                          Mobile_Prefix_Solicitations
         and QE1 configured to accept only secured
                          Mobile_Prefix_Advertisements
         and QE1 configured to have a short prefix_lifetime --(suggest 30s)
       }
  ensure that
       { when {        QE1 moves to a foreign_network
                 before its prefix_lifetime expires
                 before QE4 sends a packet to QE1
                             indicating that a response is required }
         then {        QE4 indicates receipt of the response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT QE1 is configured to protect Mobile Prefix Solicitation messages QE1 is configured to only accept protected Mobile Prefix Advertisement messages EUT is configured to advertise a preferred prefix lifetime of 20 sec EUT is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that initial lifetime of the home address of QE1 has expired		
2	Cause QE4 to send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

10.6 Sending Prefix Information to the Mobile Node

Test Description			
Identifier:	TD_MOB_1592_01	**Test Purpose:**	TP_MOB_1592_01
Summary:	'HA schedules the delivery of the prefix information when a new prefix is added to its home subnet interface'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1370, RQ_001_1592, RQ_001_1805		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {         EUT generates a new prefix to the home_address of QE1
                before (QE4 sends a packet to QE1
                            using the new home_address
                        and indicating that a response is required) }
         then {         QE4 indicates receipt of a response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Foreign Network 1 QE1 registered to EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to generate a new prefix added on its home subnet interface		
2	Wait for at least 3.5 seconds		
3	Check: does QE1 indicate a new home address?	Yes	No
Observations:	MIN_DELAY_BETWEEN_RAS (3,0 s) + MAX_RA_DELAY_TIME (0,5 s) = 3,5 s (cf. RFC 2561).		

10.3.2 Primary Care-of Address De-Registration

Test Description			
Identifier:	TD_MOB_1526_01	**Test Purpose:**	TP_MOB_1526_01
Summary:	'Home Agent rejects De-Registration Binding Update if no entry exists in its Binding Cache for the Mobile Node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_05
References:	RQ_001_1526, RQ_001_1526, RQ_001_1527, RQ_001_1535		

```
with {     EUT disconnected from Home_Network
         and QE2 configured to have the same IP_address as EUT
         and QE1 connected to a foreign_network
         and QE1 registered to QE2
       }
  ensure that
       { when {     QE2 is disconnected from Home_Network
                  and EUT is connected to Home_Network
                  and QE1 returns home }
         then {     QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE2 is configured to serve the Home Network 1 QE2 and EUT are configured with the same IP address EUT is disconnected from Home Network 1 QE1 is connected to Foreign Network 1 and registered to its Home Agent QE2 QE2 is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	QE2 is disconnected from Home Network 1		
2	EUT is connected to Home Network 1		
3	Move QE1 to Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1528_01	**Test Purpose:**	TP_MOB_1528_01
Summary:	'When accepting Care of Address de-registration Binding Update, Home Agent deletes any existing entry in its Binding Cache for the Mobile Node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1528, RQ_001_1529		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
         and QE4 able to communicate with QE1
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE1 returns home }
         then { QE4 and QE1 are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured not to perform route optimization QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Home Network 1		
2	Wait a few seconds (to let QE1 deregister to EUT)		
3	Disconnected EUT from Home Network 1		
4	Cause QE4 to send an Echo Request directly to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply directly from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1528_01	**Test Purpose:**	TP_MOB_1528_01
Summary:	'When accepting Care of Address de-registration Binding Update, Home Agent deletes any existing entry in its Binding Cache for the Mobile Node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1528, RQ_001_1529		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
         and QE4 able to communicate with QE1
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE1 returns home }
         then { QE4 and QE1 are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured not to perform route optimization QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Home Network 1		
2	Wait a few seconds (to let QE1 deregister to EUT)		
3	Disconnected EUT from Home Network 1		
4	Cause QE4 to send an Echo Request directly to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply directly from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1534_01	**Test Purpose:**	TP_MOB_1534_01
Summary:	'Home Agent stops intercepting packets for the mobile node on acceptance of the Primary Care of Address De-Registration Binding Update'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1534		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {     QE1 returns home
                and EUT is disconnected from the home_network }
         then { QE4 and QE1 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Home Network 1		
2	Wait a few seconds (to let QE1 register to EUT)		
3	EUT is disconnected from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

10.3.1 Primary Care-of Address Registration

Test Description			
Identifier:	TD_MOB_1818_01	**Test Purpose:**	TP_MOB_1818_01
Summary:	'A Binding Update contains a Home Address destination option.'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1818		

```
with
    { QE1 at_home
    }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 sends a Binding_Update
                     containing no Home_Address_Destination_Option }
         then { QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	Not testable by interoperability means		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:	It is not possible to send a Binding Update containing no Home Address!		

Test Description			
Identifier:	TD_MOB_1825_01	**Test Purpose:**	TP_MOB_1825_01
Summary:	'When Home Agent processes a Binding Update secured by IPsec, the contents of Home Address Option is not checked'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_05
References:	RQ_001_1825		

```
with {     QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
         and EUT configured to accept any secured Binding_Update from QE1
         and QE1 configured to send only a secured Binding_Update to EUT
       }
  ensure that
       { when { QE1 sends a Binding_Update secured by IPsec
                        containing a Home_Address_Option
                               not indicating its Home_Address
           }
         then { EUT accepts the packet }
       }
```

Pre-test conditions:	Not testable by IOP means		
Step	Test Sequence	Verdict	
		Pass	Fail
Observations:			

Test Description			
Identifier:	TD_MOB_1826_01	Test Purpose:	TP_MOB_1826_01
Summary:	'Home Agent updates the entry in its Binding Cache upon receipt of a Binding Update'		
Roles:	Home_Agent	Configuration:	CF_MOB_02
References:	RQ_001_1826		

```
with {     QE1 away_from_home
         and QE1 registered to EUT
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE1 moves to another foreign_network }
         then { QE3 and QE1 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT QE1 is configured not to perform Route Optimization		
Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 2		
2	Cause QE3 to send an Echo request to the Home Address of QE1		
3	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1494_01	Test Purpose:	TP_MOB_1494_01
Summary:	'Home Agent returns a type "132" Binding Acknowledgment upon receipt of a Binding Update which Home Address is not on link'		
Roles:	Home_Agent	Configuration:	CF_MOB_02
References:	RQ_001_1494		

```
with {    QE1 disconnected from the home_network
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when {     QE1 is connected to a foreign_network
                and EUT receives a Binding_Update
                        containing a Home_Address not on_link }
         then {     QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN no node can answer to an address which is not on link		
Step	Test Sequence	Verdict	
		Pass	Fail
Observations:			

Test Description			
Identifier:	TD_MOB_1496_01	Test Purpose:	TP_MOB_1496_01
Summary:	'Correspondent Node sends a type "131" Binding Acknowledgment when receiving a Home Registration Binding Update'		
Roles:	Correspondent_Node, Correspondent_Node, Mobile_Node	Configuration:	CF_MOB_04
References:	RQ_001_1496, RQ_001_1496		

```
with {     EUT at_home
         and QE5 at_home
         and QE5 configured 'to use EUT as Home Agent'
       }
  ensure that
       { when { QE5 moves to a foreign_network }
         then { QE3 and QE5 are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 2 QE5 is connected to Home Network 2 QE5 is configured to use EUT as Home Agent		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE5 to Foreign Network 2		
2	Cause QE3 to send an Echo Request to the Home Address of QE5		
3	Check: does QE3 receive an Echo Reply from QE5?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1497_01	**Test Purpose:**	TP_MOB_1497_01
Summary:	'Home Agent creates a new entry in its Binding Cache upon receipt of a Binding Update'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1497		

```
with
    {
          QE1 at_home
      and QE1 configured not to perform Route_Optimization
    }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 is able to communicate with QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 is configured not to perform Route Optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (to let QE1 register to EUT)		
3	Cause QE4 to send an Echo Request to the Home Address of QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1500_01	**Test Purpose:**	TP_MOB_1500_01
Summary:	'Home Agent maintains a Binding Cache entry from receipt of a Binding Update until the Binding Lifetime expires'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1497, RQ_001_1500, RQ_001_1522		

```
with {     QE1 at_home
         and QE1 configured not to perform Route_Optimization
    }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate
                          within the Binding_Lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 is configured not to perform Route Optimization EUT is configured to set the binding lifetime to 2 minutes QE3 is disconnected from Home Network 1 QE3 is configured with the same Ethernet address as QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (to let QE1 register with its Home Agent)		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive en Echo Reply from QE1	Yes	No
5	Disconnect QE1 from Foreign Network 1		
6	Wait a period a little less long than the Binding Lifetime		
7	Connected QE3 to Home Network 1		
8	Wait for the expiration of the binding lifetime		
9	Cause QE4 to send an Echo Request to QE1		
10	Check: does QE4 receive en Echo Reply from QE1?	No	Yes
Observations:	DAD occurs at step 7		

Test Description			
Identifier:	TD_MOB_1501_01	**Test Purpose:**	TP_MOB_1501_01
Summary:	'Home Agent performs Duplicate Address Detection on home link before sending a Binding Acknowledgment'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1501, RQ_001_1502, RQ_001_1503		

```
with {     QE1 at_home
         and QE3 disconnected from home_network
         and QE3 having the same home_address as QE1
       }
  ensure that
       { when {     QE1 is disconnected from the home_network
                and QE3 is connected to the home_network
                and QE1 is connected to a foreign_network }
         then { QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE3 is manually configured with the same Home Address as QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (to let QE1 register to EUT)		
3	Disconnect QE3 from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1503_01	**Test Purpose:**	TP_MOB_1503_01
Summary:	'Home Agent returns a type "134" Binding Acknowledgement when Duplicate Address Detection for the Home Address / Link Local Address fails'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1503		

```
with {     QE1 disconnected from home_network
         and QE3 connected to home_network
         and QE3 configured to have the same link_local_address as QE1
       }
  ensure that
       { when { QE1 is connected to a foreign_network }
         then { QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is disconnected from Home Network 1 QE3 is connected on Home Network QE3 is manually configured with the same link-local address as QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	QE1 is connected to Foreign Network 1		
2	Cause QE4 to send an Echo Request to QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1507_01	**Test Purpose:**	TP_MOB_1507_01
Summary:	'Home Agent removes binding when the associated prefix valid lifetime expires'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1507		

```
with {     QE1 away_from_home
         and QE1 registered to the EUT
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { the prefix_lifetime expires at the EUT }
         then { QE4 and QE1 are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to accept only secured Mobile Prefix Solicitations. QE1 is configured to send unsecured Mobile Prefix Solicitations. QE1 is configured not to perform Route Optimization QE1 is connected to Foreign Network 1 and registered to its Home Agent R1 is configured to advertise a valid prefix lifetime of 1 minute

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait a minute (to let the prefix A become invalid)		
2	Cause QE4 to send an Echo Request to QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1510_01	**Test Purpose:**	TP_MOB_1510_01
Summary:	'Home Agent returns a Binding Acknowledgment in response to a Binding Update'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1510, RQ_001_1501, RQ_001_1510, RQ_001_1511, RQ_001_1516, RQ_001_1517, RQ_001_2002, RQ_001_2013, RQ_001_2029		

```
with { QE1 at_home }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (to let the QE1 register with its Home Agent)		
3	Cause QE4 to send an Echo Request to the Home Address of QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1500_01	**Test Purpose:**	TP_MOB_1500_01
Summary:	'Home Agent maintains a Binding Cache entry from receipt of a Binding Update until the Binding Lifetime expires'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1497, RQ_001_1500, RQ_001_1522		

```
with {     QE1 at_home
         and QE1 configured not to perform Route_Optimization
    }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate
                          within the Binding_Lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 is configured not to perform Route Optimization EUT is configured to set the binding lifetime to 2 minutes QE3 is disconnected from Home Network 1 QE3 is configured with the same Ethernet address as QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (to let QE1 register with its Home Agent)		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive en Echo Reply from QE1?	Yes	No
5	Disconnect QE1 from Foreign Network 1		
6	Wait a period a little less long than the Binding Lifetime		
7	Connected QE3 to Home Network 1		
8	Wait for the expiration of the binding lifetime		
9	Cause QE4 to send an Echo Request to QE1		
10	Check: does QE4 receive en Echo Reply from QE1?	No	Yes
Observations:	DAD occurs at step 7.		

10.3 Processing Bindings

Test Description			
Identifier:	TD_MOB_1366_01	**Test Purpose:**	TP_MOB_1366_01
Summary:	'Home Agent returns a Binding_Acknowledgement message in response to a Binding Update message'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1366		

```
with {     QE1 at_home
       }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 receives a packet from QE4
                                indicating that a response is required }
         then {     QE4 indicates receipt of the response}
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to foreign network 1		
2	Wait a few seconds		
3	Cause QE4 to send an Echo Request to the Home Address of QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

10.4.1 Intercepting Packets for a Mobile Node

Test Description			
Identifier:	TD_MOB_1536_01	**Test Purpose:**	TP_MOB_1536_01
Summary:	'HA intercepts packets for registered mobile node on the home network and uses reverse tunnel encapsulation'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1364, RQ_001_1536, RQ_001_1551, RQ_001_1568		

```
with {    QE1 away_from_home
       and QE1 registered to EUT
       and QE1 configured not to perform Route_Optimization
      }
  ensure that
      { when { QE3 is requested to send a packet to QE1
                              indicating that a response is required }
        then { QE3 indicates receipt of the response }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and has registered to its Home Agent EUT QE1 is configured NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the Home Address of QE1		
2	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:	Very Similar to TD_MOBILITY_1547_01.		

Test Description			
Identifier:	TD_MOB_1537_01	**Test Purpose:**	TP_MOB_1537_01
Summary:	'HA multicasts Neighbour Advertisement in Home network in order to intercept packets for mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1537, RQ_001_1537, RQ_001_1539, RQ_001_1540, RQ_001_1541, RQ_001_1542, RQ_001_1543, RQ_001_1544		

```
with {      QE1 at home
         and QE1 having established communication to QE3
       }
  ensure that
      { when {      QE1 moves to a foreign_network
          }
        then { QE3 updates its Neighbor_Discovery_cache_entry
                              for the link_local_address of QE1 }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE3 has already sent packets to the home address of QE1 QE4 is configured with the same link-layer address as QE1 and with the Home Address of QE1 Configure R2 to advertise Prefix A instead of Prefix C		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (for QE1 to register with its Home Agent EUT)		
3	Disconnect EUT		
4	Connect Home Network 1 and Home Network 2 together		
5	Cause QE3 to send an Echo Request to the home address of QE1		
6	Check: does EUT receive an Echo Reply?	No	Yes
Observations:	Explanation: If EUT behaves correctly (i.e. sends a NA) then QE3 will send the Echo Request to the link-layer address of EUT and will not receive any Echo Reply. If EUT behaves incorrectly (i.e. sends no NA) then QE3 will send the Echo Request to the link-layer address of QE1, which is equal to QE4's one and will receive any Echo Reply. HN1 and HN2 are connected together to avoid QE4 performing DAD and advertising itself NOTE: Go through steps 1 to 5 quickly enough to avoid neighbour cache expiry on QE3.		

Test Description			
Identifier:	TD_MOB_1538_01	**Test Purpose:**	TP_MOB_1538_01
Summary:	'HA multicast Neighbour Advertisement to intercept the packets to the link_local_address of a registered mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1538, RQ_001_1538, RQ_001_1539, RQ_001_1540, RQ_001_1541, RQ_001_1542, RQ_001_1543, RQ_001_1544		

```
with {      QE1 at_home
          and QE1 using the same Interface_ID for its link_local_address
                                                and its home_address
          and QE3 having established communication
                    to the link_local_address of QE1
       }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE3 updates the Neighbor_Discovery_cache_entry
                     for the link_local_address of QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 uses the same Interface ID for both its link-local address and its home address QE3 has already sent packets to the link-local address of QE1 QE4 is configured with the same link-layer address as QE1 Configure R2 to advertise Prefix A instead of Prefix C		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds (for QE1 to register with its Home Agent EUT)		
3	Disconnect EUT		
4	Connect Home Network 1 and Home Network 2 together		
5	Cause QE3 to send an Echo Request to the link-local address of QE1		
6	Check: does EUT receive an Echo Reply?	No	Yes
Observations:	Explanation: QE1 sends Binding Updates with the L flag activated (same interface ID for link-local and home addresses). If EUT behaves correctly (i.e. sends a NA) then QE3 will send the Echo Request to the link-layer address of EUT and will not receive any Echo Reply. If EUT behaves incorrectly (i.e. sends no NA) then QE3 will send the Echo Request to the link-layer address of QE1, which is equal to QE4's one and will receive any Echo Reply. HN1 and HN2 are connected together to avoid QE4 performing DAD and advertising itself.		

Test Description			
Identifier:	TD_MOB_1547_01	**Test Purpose:**	TP_MOB_1547_01
Summary:	'HA act as proxy for a registered mobile node and reply to any received Neighbour Solicitations for it'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1547, RQ_001_1547, RQ_001_1548, RQ_001_1549		

```
with {      QE3 'is disconnected'
          and QE1 away_from_home
        and QE1 registered to EUT
       }
  ensure that
       { when {      QE3 is connected to its home_network
              and QE3 is requested to send a packet to QE1
                                indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE3 is disconnected QE1 is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Connect QE3 to Home Network 1		
2	Cause QE3 to send an Echo Request to the Home Address of QE1		
3	Check: does QE3 receive an Echo Reply?	Yes	No
Observations:	Very Similar to TD_MOBILITY_1536_01.		

10.4.2 Processing Intercepted Packets

Test Description			
Identifier:	TD_MOB_1536_01	**Test Purpose:**	TP_MOB_1536_01
Summary:	'HA intercepts packets for registered mobile node on the home network and uses reverse tunnel encapsulation'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1364, RQ_001_1536, RQ_001_1551, RQ_001_1568		

```
with {    QE1 away_from_home
       and QE1 registered to EUT
       and QE1 configured not to perform Route_Optimization
      }
  ensure that
      { when { QE3 is requested to send a packet to QE1
                              indicating that a response is required }
        then { QE3 indicates receipt of the response }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and has registered to its Home Agent EUT QE1 is configured NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the Home Address of QE1		
2	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:	Very Similar to TD_MOBILITY_1547_01.		

Test Description			
Identifier:	TD_MOB_1552_01	**Test Purpose:**	TP_MOB_1552_01
Summary:	'HA does not tunnel packets to the link_local_address of mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1552, RQ_001_1552, RQ_001_1553		

```
with {     QE1 away_from_home
        and QE1 registered to EUT
      }
  ensure that
      { when { QE3 is requested to send a packet
                              to the link_local_address of QE1
                              indicating that a response is required }
        then { QE3 indicates no receipt of the response }
      }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Reply to the link-layer address of QE1		
2	Check: does QE3 receive an Echo Reply?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1555_01	**Test Purpose:**	TP_MOB_1555_01
Summary:	'HA does not tunnel multicast packets with a scope smaller than global to the mobile node '		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1555, RQ_001_1556, RQ_001_1555, RQ_001_1556		

```
with {     QE1 subscribed to a link_local_multicast_group
         and QE1 subscribed to a organization_local_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {     QE3 is requested to send a packet
                                  to the link_local_multicast_group
                                  indicating that a response is required
              and QE3 is requested to send a packet
                                  to the organization_local_multicast_group
                                  indicating that a response is required }
         then { QE3 indicates no receipt of any response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 QE1 has registered to multicast group FF12::1 QE1 has registered to multicast group FF18::1

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the multicast address FF12::1		
2	Check: does QE3 receive an Echo Reply	No	Yes
3	Cause QE3 to send an Echo Request to the multicast address FF18::1		
4	Check: does QE3 receive an Echo Reply?	No	Yes
Observations:	FF12::1 is a link-local scope, non reserved, multicast address FF18::1 is an organization-local scope, non reserved, multicast address		

Test Description			
Identifier:	TD_MOB_1555_01	**Test Purpose:**	TP_MOB_1555_01
Summary:	'HA does not tunnel multicast packets with a scope smaller than global to the mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1555, RQ_001_1556, RQ_001_1555, RQ_001_1556		

```
with {     QE1 subscribed to a link_local_multicast_group
         and QE1 subscribed to a organization_local_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {     QE3 is requested to send a packet
                                  to the link_local_multicast_group
                                  indicating that a response is required
              and QE3 is requested to send a packet
                                  to the organization_local_multicast_group
                                  indicating that a response is required }
         then { QE3 indicates no receipt of any response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 QE1 has registered to multicast group FF12::1 QE1 has registered to multicast group FF18::1

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the multicast address FF12::1		
2	Check: does QE3 receive an Echo Reply?	No	Yes
3	Cause QE3 to send an Echo Request to the multicast address FF18::1		
4	Check: does QE3 receive an Echo Reply?	No	Yes
Observations:	FF12::1 is a link-local scope, non reserved, multicast address. FF18::1 is an organization-local scope, non reserved, multicast address.		

Test Description			
Identifier:	TD_MOB_1557_01	**Test Purpose:**	TP_MOB_1557_01
Summary:	'HA tunnels multicast packets with a global scope to the mobile node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1372, RQ_001_1557, RQ_001_1557		

```
with {      QE1 subscribed to a global_multicast_group
          and QE1 away_from_home
          and QE1 registered to EUT
       }
  ensure that
       { when { QE3 is requested to send a packet to the global_multicast_group
                                indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Home Network 1 QE1 has registered to multicast group FF1E::1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Cause QE3 to send an Echo Request to the multicast address FF1E::1		
3	Check: does QE3 receives an Echo Reply?	Yes	No
4	Disconnect EUT		
5	Cause QE3 to send an Echo Request to the multicast address FF1E::1		
6	Check: does QE3 receives an Echo Reply?	No	Yes
Observations:	FF1E::1 is a global scope, non reserved, multicast group.		

10.4.3 Multicast Membership Control

Test Description			
Identifier:	TD_MOB_1560_01	**Test Purpose:**	TP_MOB_1560_01
Summary:	'if multicast forwarding is not supported, HA ignores multicast group membership control messages'		
Roles:	Home_Agent	**Configuration**:	CF_MOB_02
References:	RQ_001_1560		

```
with {      QE1 subscribed to a global_multicast_group
          and EUT unable to support multicast_data_packet_forwarding
          and QE1 away_from_home
          and QE1 registered to EUT
       }
  ensure that
       { when { QE3 is requested to send a packet
                                to the global_multicast_group }
         then { QE3 indicates no receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 QE1 has registered with its Home Agent EUT EUT does not support multicast data packet forwarding		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE1 to register to multicast group FF1E::1		
2	Cause QE3 to send an Echo Reply to multicast address FF1E::1		
3	Check: does QE3 receive an Echo Reply?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1807_01	**Test Purpose:**	TP_MOB_1807_01
Summary:	'If HA support the multicast group membership control protocols, HA must be capable to determine which multicast data packets to forward via the tunnel to MN'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1561, RQ_001_1807		

```
-- Config-HA
  with {      QE1 connected to home_network
           and QE1 subscribed to a global_multicast_group
           and QE3 subscribed to another global_multicast_group
       }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 registers to EUT }
         then {     QE1 receives packets from its global_multicast_group }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEANS You do not know if the Home Agent do not forward the multicast packets or if the Mobile Node drops this packets		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

10.4.5 Handling ReverseTunnelled Packets

Test Description			
Identifier:	TD_MOB_1536_01	**Test Purpose:**	TP_MOB_1536_01
Summary:	'HA intercepts packets for registered mobile node on the home network and uses reverse tunnel encapsulation'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1364, RQ_001_1536, RQ_001_1551, RQ_001_1568		

```
with {    QE1 away_from_home
        and QE1 registered to EUT
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE3 is requested to send a packet to QE1
                                indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and has registered to its Home Agent EUT QE1 is configured NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the Home Address of QE1		
2	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:	Very Similar to TD_MOBILITY_1547_01.		

Test Description			
Identifier:	TD_MOB_1536_01	**Test Purpose:**	TP_MOB_1536_01
Summary:	'HA intercepts packets for registered mobile node on the home network and uses reverse tunnel encapsulation'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1364, RQ_001_1536, RQ_001_1551, RQ_001_1568		

```
with {    QE1 away_from_home
        and QE1 registered to EUT
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE3 is requested to send a packet to QE1
                                indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and has registered to its Home Agent EUT QE1 is configured NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the Home Address of QE1		
2	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:	Very Similar to TD_MOBILITY_1547_01.		

10.4.6 Protecting Return Routability packets

Test Description			
Identifier:	TD_MOB_1371_01	**Test Purpose:**	TP_MOB_1371_01
Summary:	'Home Agent supports ESP protection of Home Test and Home Test Init messages'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1371		

```
with {     QE4 configured to perform route_optimization
         and QE1 configured to perform route_optimization
         and QE1 configured to protect return_routability_packets
                              using ESP
       }
  ensure that
       { when {        QE1 moves to a Foreign_Network
                 before QE4 sends a packet to QE1
                             indicating that a response is required }
         then {        QE4 indicates receipt of the response directly from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is configured to perform route optimization QE1 is configured to perform route optimization QE1 is configured to protect return routability packets using ESP QE1 connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few seconds		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive Echo Reply?	Yes	No
5	Disconnect EUT		
6	Cause QE4 to send an Echo Request to QE1		
7	Check: does QE4 receive Echo Reply?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2014_01	**Test Purpose:**	TP_MOB_2014_01
Summary:	'Home Agent supports Home Test Init secured using tunnel_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1571, RQ_001_2034, RQ_001_2035		

```
with {     QE1 configured to protect any Home_Test_Init to EUT
                           using tunnel_mode_ESP
         and EUT configured to accept only Home_Test_Init secured
                           using tunnel_mode_ESP from QE1
         and QE1 configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE1 is away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Home Test Init sent to EUT using tunnel mode ESP EUT is configured to accept only Home Test Init protected using tunnel mode ESP from QE1 QE1 configured to perform Route Optimization QE4 configured to perform Route Optimization QE1 is away from Home and registered to its Home Agent EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of QE1		
2	Wait a few second (to let QE1 register to QE4)		
3	Disconnect EUT from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1574_01	**Test Purpose:**	TP_MOB_1574_01
Summary:	'Home Agent supports the protection of Home Test and Home Test Init messages'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1574		

```
with {     QE4 configured to perform route_optimization
         and QE1 configured to perform route_optimization
         and QE1 away_from_home
         and QE1 registered to EUT
         and QE1 configured to protect return_routability_packets
       }
  ensure that
       { when { QE1 receives a packet from QE4
                indicating that a response is required }
         then { QE1 sends the response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE4 is configured to perform route optimization QE1 is configured to perform route optimization QE1 is away from home and registered to EUT QE1 is configured to protect return routability packets with QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of QE1		
2	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
3	Wait a few seconds (to let QE4 and QE1 process the return routability procedure)		
4	Disconnect EUT from Home Network 1		o
5	Cause QE4 to send an Echo Request to the Home Address of QE1		
6	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:	RRP is performed during first echo exchange.		

11 Mobile Node Operation

Test Description			
Identifier:	TD_MOB_1003_01	**Test Purpose:**	TP_MOB_1003_01
Summary:	'MN continues to communicate after moving to a new link'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1002, RQ_001_1003, RQ_001_1004		

```
with {     EUT connected in home_network
         and EUT registered to QE1
         and EUT and QE4 established in communication
       }
  ensure that
       { when {      EUT moves to a foreign_network
                and QE4 is requested to send a packet to EUT
                    indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause EUT to send an Echo Request to QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Move EUT to Foreign Network 1		
4	wait a few seconds (to let the EUT register with its Home Agent).		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.3 Packet Processing

Test Description			
Identifier:	TD_MOB_1003_01	**Test Purpose:**	TP_MOB_1003_01
Summary:	'MN continues to communicate after moving to a new link'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1002, RQ_001_1003, RQ_001_1004		

```
with {     EUT connected in home_network
         and EUT registered to QE1
         and EUT and QE4 established in communication
       }
  ensure that
       { when {      EUT moves to a foreign_network
                and QE4 is requested to send a packet to EUT
                        indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause EUT to send an Echo Request to QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Move EUT to Foreign Network 1		
4	wait a few seconds (to let the EUT register with its Home Agent)		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.3.1 Sending Packets While Away From Home

Test Description			
Identifier:	TD_MOB_1820_01	**Test Purpose:**	TP_MOB_1820_01
Summary:	'When Home Agent receives a packet from Mobile Node using reverse tunnelling, it forwards the encapsulated packet to Correspondent Node'		
Roles:	Home_Agent, Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1820, RQ_001_1820		

```
with {    QE1 away_from_home
        and QE1 registered to EUT
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT is disconnected }
         then { QE1 and QE4 are unable to communicate with QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to EUT QE1 is configured NOT to perform Route Optimization

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address to QE1		
2	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
3	Disconnect EUT from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address to QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1003_01	**Test Purpose:**	TP_MOB_1003_01
Summary:	'MN continues to communicate after moving to a new link'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1002, RQ_001_1003, RQ_001_1004		

```
with {     EUT connected in home_network
         and EUT registered to QE1
         and EUT and QE4 established in communication
       }
  ensure that
       { when {      EUT moves to a foreign_network
                and QE4 is requested to send a packet to EUT
                     indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to send an Echo Request to QE4		
2	Check: does EUT receive an Echo Reply from QE4?	Yes	No
3	Move EUT to Foreign Network 1		
4	wait a few seconds (to let the EUT register with its Home Agent).		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1612_01	**Test Purpose:**	TP_MOB_1612_01
Summary:	'When MN starts a transport-level connections at Home, it conserves that connection away_from_home'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1612		

```
with {     EUT connected in home_network
         and EUT having established a transport_connection to QE3
       }
  ensure that
       { when { EUT moves to a foreign_network }
         then { EUT and QE3 are able to communicate
                using the transport_connection }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause EUT to establish a transport-level connection (telnet, ftp, ssh, etc.) with QE3		
2	Move EUT to foreign network 1		
3	Check: Is the transport-level connection established in step 1 still alive?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1613_01	**Test Purpose:**	TP_MOB_1613_01
Summary:	'When MN starts a transport-level connections in Foreign network, it conserves this connection into another Foreign network'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1613		

```
with {      EUT away_from_home
          and EUT registered to QE1
          and EUT having established a transport_connection to QE3
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { EUT and QE3 are able to communicate
                    using the transport_connection }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause EUT to establish a transport-level (telnet, ftp, ssh or similar) connection with QE1		
2	Move EUT to foreign network 2		
3	Check: Is the transport level connection established in step 1 still alive?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1614_01	**Test Purpose:**	TP_MOB_1614_01
Summary:	'MN sends packets directly to the correspondent node when a binding exists'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1614		

```
with {      EUT away_from_home
          and EUT registered to QE1
          and EUT configured to perform route_optimization
          and QE4 configured to perform route_optimization
          and EUT having established a binding to QE4
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends the response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected in Foreign Network 1 and registered to its Home Agent QE1 EUT configured to perform route optimization QE4 configured to perform route optimization EUT has established a binding to QE4		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnect QE1		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1615_01	**Test Purpose:**	TP_MOB_1615_01
Summary:	'MN uses reverse tunnelling to the correspondent node when no binding exists'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1615, RQ_001_1615, RQ_001_1819		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected in Foreign Network 1 and registered to its Home Agent QE1 EUT configured to perform route optimization QE4 configure NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
3	Disconnect QE1		
4	Cause QE4 to send an Echo Request to the Home Address of EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
Observations:			

11.4 Home Agent and Prefix Management

Test Description			
Identifier:	TD_MOB_1082_01	**Test Purpose:**	TP_MOB_1082_01
Summary:	'Home Agent uses a Security Association to protect integrity and authenticity of Mobile_Prefix_Solicitations and Advertisements'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1014, RQ_001_1082, RQ_001_1607		

```
with {     QE1 at_home
         and QE1 configured to protect
                        Mobile_Prefix_Solicitations
         and QE1 configured to accept only secured
                        Mobile_Prefix_Advertisements
         and QE1 configured to have a short prefix_lifetime --(suggest 30s)
       }
  ensure that
       { when {        QE1 moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to QE1
                            indicating that a response is required }
         then {        QE4 indicates receipt of the response from QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is connected to Foreign Network 1 and registered to its Home Agent EUT QE1 is configured to protect Mobile Prefix Solicitation messages QE1 is configured to only accept protected Mobile Prefix Advertisement messages EUT is configured to advertise a preferred prefix lifetime of 20 sec EUT is configured to advertise a valid prefix lifetime of 30 sec		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Wait 30 sec so that initial lifetime of the home address of QE1 has expired		
2	Cause QE4 to send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1082_02	**Test Purpose:**	TP_MOB_1082_02
Summary:	'Mobile Node uses a Security Association to protect integrity and authenticity of Mobile Prefix Solicitations and Advertisements'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1014, RQ_001_1082, RQ_001_1663, RQ_001_1664		

```
with {     EUT at_home
         and QE1 configured to accept only secured
                          Mobile_Prefix_Solicitations
         and QE1 configured to protect Mobile_Prefix_Advertisements
         and EUT configured to have a short prefix_lifetime (--suggest 30s)'
       }
  ensure that
       { when {        EUT moves to a foreign_network
                 before its prefix_lifetime expires
                 before QE4 sends a packet to the EUT
                           indicating that response is required }
         then {        QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE1 is configured to only accept protected Mobile Prefix Solicitation messages QE1 is configured to protect Mobile Prefix Advertisement messages QE1 is configured to advertise a preferred prefix lifetime of 20 sec QE1 is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that EUT has send a Mobile Prefix Solicitation message and received a Mobile Prefix Advertisement message		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.4.1 Dynamic Home Agent Address Discovery

Test Description			
Identifier:	TD_MOB_1655_01	**Test Purpose:**	TP_MOB_1655_01
Summary:	'Mobile Node already registered to a Home Agent uses that Home Agent for any new registrations'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_06
References:	RQ_001_1655, RQ_001_1655		

```
with
    {     EUT configured 'to use DHAAD'
      and EUT away_from_home
      and EUT registered to QE1
      and QE2 disconnected from Home_Network 1
    }
  ensure that
       { when {     QE1 is disconnected from Home_Network 1
              and QE2 is connected to Home_Network 1
              and EUT moves to another foreign_network }
         then {    QE4 and EUT are unable to communicate
                    within MAX_BINDACK_TIMEOUT
              -- EUT tries to register with QE1
              and QE4 is able to communicate after MAX_BINDACK_TIMEOUT }
              -- EUT registered to QE2
    }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to use DHAAD QE2 is configured to serve as home agent for home network 1 QE2 is disconnected EUT connected to foreign network 1 and registered to QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE1		
2	Connect QE2 to Home Network 1		
3	Move EUT to foreign network 2		
4	Cause QE4 to continuously send ICMP ECHO requests to EUT		
5	Check: does QE4 receive ICMP ECHO replies from EUT?	No	Yes
6	Wait for MAX_BINDACK_TIMEOUT seconds or more (EUT should still be trying to register to QE1 until this delay expires)		
7	Check: does QE4 receive ICMP ECHO replies from EUT? (EUT should now have successfully registered to QE2)	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1657_01	**Test Purpose:**	TP_MOB_1657_01
Summary:	'Mobile Node attempts Dynamic Home Agent Address Discovery if it cannot contact its current Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_06
References:	RQ_001_1657		

```
with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     QE1 is disconnected from the Home_Network
                and EUT moves to another foreign_network }
         then {     QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE2 is configured to be Home Agent on Home Network 1 EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE1 is configured to have a Binding Lifetime at one minute

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect QE1 from Home Network 1		
2	Move EUT to Foreign Network 2		
3	Wait a few seconds (for EUT to try to bind to Home Agent)		
4	Cause QE3 to send an Echo Request to the Home Address of EUT		
5	Check: does QE3 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1385_01	**Test Purpose:**	TP_MOB_1385_01
Summary:	'Mobile Node supports Dynamic Home Agent Address Discovery'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1385		

```
with {     EUT registered to QE1
         and EUT disconnected
         and EUT configured not to perform Route_Optimization
         and QE4 configured not to perform Route_Optimization
       }
  ensure that
       { when {            QE1 is requested to modify its IP_address
                before the EUT is connected to a foreign_network }
         then {            QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Disconnect EUT from Home Network 1		
2	Configure QE1 with a new IP address on prefix_B		
3	Connect EUT to Foreign Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of EUT		
5	Check: does QE4 receives an Echo Reply?	Yes	No
Observations:			

11.4.2 Sending Mobile Prefix Solicitations

Test Description			
Identifier:	TD_MOB_1661_01	**Test Purpose:**	TP_MOB_1661_01
Summary:	'MN sends Mobil Prefix Solicitation to HA when its home address becomes invalid'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1661, RQ_001_1661, RQ_001_1662, RQ_001_1665		

```
with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { EUT 'has its home address which becomes invalid' }
         then { EUT 'update the prefix lifetime information for its home address' }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign network 1		
2	Wait for original Home address lifetime to expire		
3	Cause QE4 to send an Echo Request to EUT		
4	Check: does QE4 receive an Echo Reply?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1082_02	**Test Purpose:**	TP_MOB_1082_02
Summary:	'Mobile Node uses a Security Association to protect integrity and authenticity of Mobile Prefix Solicitations and Advertisements'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1014, RQ_001_1082, RQ_001_1663, RQ_001_1664		

```
with {     EUT at_home
         and QE1 configured to accept only secured
                          Mobile_Prefix_Solicitations
         and QE1 configured to protect Mobile_Prefix_Advertisements
         and EUT configured to have a short prefix_lifetime (--suggest 30s)'
       }
  ensure that
       { when {        EUT moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to the EUT
                           indicating that response is required }
         then {        QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE1 is configured to only accept protected Mobile Prefix Solicitation messages QE1 is configured to protect Mobile Prefix Advertisement messages QE1 is configured to advertise a preferred prefix lifetime of 20 sec QE1 is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that EUT has send a Mobile Prefix Solicitation message and received a Mobile Prefix Advertisement message		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1082_02	**Test Purpose:**	TP_MOB_1082_02
Summary:	'Mobile Node uses a Security Association to protect integrity and authenticity of Mobile Prefix Solicitations and Advertisements'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1014, RQ_001_1082, RQ_001_1663, RQ_001_1664		

```
with {     EUT at_home
         and QE1 configured to accept only secured
                          Mobile_Prefix_Solicitations
         and QE1 configured to protect Mobile_Prefix_Advertisements
         and EUT configured to have a short prefix_lifetime (--suggest 30s)'
       }
  ensure that
       { when {        EUT moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to the EUT
                           indicating that response is required }
         then {        QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 QE1 is configured to only accept protected Mobile Prefix Solicitation messages QE1 is configured to protect Mobile Prefix Advertisement messages QE1 is configured to advertise a preferred prefix lifetime of 20 sec QE1 is configured to advertise a valid prefix lifetime of 30 sec

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait 30 sec so that EUT has send a Mobile Prefix Solicitation message and received a Mobile Prefix Advertisement message		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.4.3 Receiving Mobile Prefix Advertisements

Test Description			
Identifier:	TD_MOB_1384_01	**Test Purpose:**	TP_MOB_1384_01
Summary:	'MN reconfigures its home address according to the prefix information in a Mobile_Prefix_Advertisement'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1384		

```
with { EUT away_from_home
           and EUT registered to QE1
       }
  ensure that
       { when { QE1 is requested to modify its home_network prefix }
         then { QE4 and the EUT are able to communicate
                    using the new home_address }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT connected to foreign network 1 EUT registered to QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE1 to change the prefix on its home subnet interface		
2	Wait for some seconds to allow EUT to retrieve new home address		
3	Check: does EUT indicate a new home address?	Yes	No
Observations:			

11.5 Movement

Test Description			
Identifier:	TD_MOB_1809_01	**Test Purpose:**	TP_MOB_1809_01
Summary:	'Mobile Nodes support returning home'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1808, RQ_001_1809		

```
with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT returns home }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Home Network 1		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1809_01	**Test Purpose:**	TP_MOB_1809_01
Summary:	'Mobile Nodes support returning home'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1808, RQ_001_1809		

```
with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT returns home }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Home Network 1		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1380_01	**Test Purpose:**	TP_MOB_1380_01
Summary:	'Mobile Node supports care-of address formation'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1380		

```
with {     EUT connected to home_network
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT moves to a foreign_network }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds (to let the EUT register with its Home Agent)		
3	Cause QE4 to send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1683_01	**Test Purpose:**	TP_MOB_1683_01
Summary:	'Mobile Node generates a new primary Care of Address after having moved'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1380, RQ_001_1683, RQ_001_1690		

```
with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 2		
2	Wait a few seconds (to let the EUT register with its Home Agent)		
3	Cause QE4 to send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.5.1 Movement Detection

Test Description			
Identifier:	TD_MOB_1678_01	**Test Purpose:**	TP_MOB_1678_01
Summary:	' MN discover a new default router when it detects the default router is no longer bi-directionally reachable '		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1678		

```
with {     EUT connected in home_network
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and EUT receives a packet from QE3
                        indicating that a response is required
                    and containing a destination_address
                        set to the Care_of_Address of the EUT }
         then { QE3 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is disconnected EUT is connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Cause QE3 to send an Echo Request to the new Care-of Address of EUT		
3	Check: does QE3 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1824_01	**Test Purpose:**	TP_MOB_1824_01
Summary:	'When Mobile Node detects a L3 handover, it constructs a new Care-of Address, it select a new default router and register the new address to its Home Agent and the Correspondent Nodes which it is performing route optimization		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1682, RQ_001_1824		

```
with {    EUT away_from_home
       and EUT registered to QE1
       and EUT configured to perform route_optimization
       and QE4 configured to perform route_optimization
       and EUT having established a binding to QE4
      }
  ensure that
     { when { EUT moves to another foreign_network }
       then { QE4 and EUT are able to communicate directly }
     }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT is configured to perform Route Optimization QE4 is configured to perform Route Optimization EUT has established a binding with the Correspondent Node QE4		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Moves EUT to Foreign Network 2		
2	Wait a few seconds (to let EUT update its bindings to QE1 and QE4)		
3	Disconnect QE1 from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address to EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1824_01	**Test Purpose:**	TP_MOB_1824_01
Summary:	'When Mobile Node detects a L3 handover, it constructs a new Care-of Address, it select a new default router and register the new address to its Home Agent and the Correspondent Nodes which it is performing route optimization		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1682, RQ_001_1824		

```
with {    EUT away_from_home
       and EUT registered to QE1
       and EUT configured to perform route_optimization
       and QE4 configured to perform route_optimization
       and EUT having established a binding to QE4
      }
  ensure that
     { when { EUT moves to another foreign_network }
       then { QE4 and EUT are able to communicate directly }
     }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT is configured to perform Route Optimization QE4 is configured to perform Route Optimization EUT has established a binding with the Correspondent Node QE4		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Moves EUT to Foreign Network 2		
2	Wait a few seconds (to let EUT update its bindings to QE1 and QE4)		
3	Disconnect QE1 from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address to EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.5.2 Forming New Care-of Addresses

Test Description			
Identifier:	TD_MOB_1683_01	**Test Purpose:**	TP_MOB_1683_01
Summary:	'Mobile Node generates a new primary Care of Address after having moved'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1380, RQ_001_1683, RQ_001_1690		

```
with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 2		
2	Wait a few seconds (to let the EUT register with its Home Agent)		
3	Cause QE4 to send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1684_01	**Test Purpose:**	TP_MOB_1684_01
Summary:	'Mobile Node generates a new primary Care of Address when the current one is deprecated'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_07
References:	RQ_001_1684		

```
with {     EUT away_from_home
         and QE3 configured as default_router for EUT
         and QE2 able to communicate with EUT
       }
  ensure that
       { when {         QE3 stops advertizing the old prefix
                before (QE3 starts advertizing a new prefix
                before  QE2 is requested to send a packet to the EUT
                           indicating that a response is required) }
         then {         QE2 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN Impossible to know when the Care-of address registration will take place Depending of the implementation to reduce the handover		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1005_01	**Test Purpose:**	TP_MOB_1005_01
Summary:	'When away, Mobile Node is addressable at one or more Care of Addresses'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1005		

```
with
    {
          EUT at_home
      and QE6 'advertising a second prefix on Foreign Network 1'
    }
  ensure that
       { when { EUT moves to Foreign_Network 1 }
         then { QE4 and EUT are able to communicate
                    using first and second Care_of_address }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1 Both R1 and R3 have two global addresses, one on Prefix B and one on Prefix E R3 advertises prefix B and prefix E on Foreign Network 1 QE1, QE4 and R2 have a static route for prefix E and are able to reach a node with a prefix E on Foreign Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Cause QE4 to send an Echo Request to EUT on its care-of address generated using prefix E		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
4	Cause QE4 to send an Echo Request to EUT on its care-of address generated using prefix B		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.5.3 Using Multiple Care-of Addresses

Test Description			
Identifier:	TD_MOB_1683_01	**Test Purpose:**	TP_MOB_1683_01
Summary:	'Mobile Node generates a new primary Care of Address after having moved'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1380, RQ_001_1683, RQ_001_1690		

```
with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE4 and the EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 2		
2	Wait a few seconds (to let the EUT register with its Home Agent)		
3	Cause QE4 to send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1693_01	**Test Purpose:**	TP_MOB_1693_01
Summary:	' MN accepts packets at its previous CoA after registering its new primary CoA '		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1693		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and QE8 advertizing a new prefix   -- thus causing a new care-of
                                            -- address to be generated
       }
  ensure that
       { when {         QE8 stops advertizing the old prefix
                before (QE8 starts advertizing a new prefix
                before  EUT receives a packet from QE3
                            indicating that a response is required
                        and containing a destination_address
                            set to the old Care_Of_Address of the EUT) }
         then { QE3 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN Impossible to know when the Care-of address registration will take place Depending of the implementation to reduce the handover		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1694_01	**Test Purpose:**	TP_MOB_1694_01
Summary:	'Mobile Node invalidates all associated Care of Addresses when the corresponding router is unreachable'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1694		

```
with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE1 and the EUT are unable to communicate
                    using the old Care_of_Address }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 2		
2	Cause QE4 to send an Echo Request to EUT on its old care-of address of the Foreign Network 1		
3	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
Observations:			

11.5.4 Returning Home

Test Description			
Identifier:	TD_MOB_1702_01	**Test Purpose:**	TP_MOB_1702_01
Summary:	'Mobile Node replies to Neighbour Solicitations for its Home Address after returning home and having sent a Binding Update'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1695, RQ_001_1702, RQ_001_1702, RQ_001_1703		

```
with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     EUT returns home
                and QE1 is disconnected from the home_network }
         then { QE4 and the EUT are unable to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	QE1 is disconnected from Home Network 1		
2	Move EUT to Home Network 1		
3	Cause QE4 send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
Observations:			

Test Description			
Identifier:	TD_MOB_1700_01	**Test Purpose:**	TP_MOB_1700_01
Summary:	'Mobile Node does not perform Duplicate Address Detection when returning home before the expiry of its bindings'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1700		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and QE1 is disconnected
         and QE3 having the same Home_Address as EUT
         and QE3 having the same ethernet_address as QE1
       }
  ensure that
       { when {     EUT returns home before the binding_lifetime expires
                and QE3 is disconnected
                and QE1 is connected to the Home_Network }
         then {     EUT and QE4 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 QE3 is disconnected from Home Network 1 QE3 is configured with the same address as EUT and the same Ethernet address as QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Disconnected QE1 from Home Network 1		
2	Connected QE3 to Home Network 1		
3	Move EUT to Home Network 1		
4	Wait a very few seconds (two seconds, to let Mobile Node send Binding Update to its Home Agent)		
5	Disconnect QE3 to Home Network 1		
6	Connect QE1 to Home network 1		
7	Wait a few seconds (to let EUT de-register with its Home Agent QE1)		
8	Cause QE4 to send an Echo Request to EUT		
9	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1701_01	**Test Purpose:**	TP_MOB_1701_01
Summary:	'Mobile Node performs Duplicate Address Detection when returning home after its bindings have expired'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1701		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT binding_lifetime expired at QE1
         and QE4 connected to home_network
                 using the same Home_Address as EUT
       }
  ensure that
       { when { EUT returns home }
         then { EUT is unable to communicate } -- DAD fails
    }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to have a Binding Lifetime at one minute EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 QE4 is manually configured with the same address as EUT QE4 is disconnected

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Disconnect QE1		
2	Wait one minute (to let the Binding of EUT expire)		
3	Connect QE4 to Home Network 1		
4	Move EUT to Home Network 1		
5	QE4 is disconnected from Home Network 1		
6	Cause QE1 to send an Echo Request to the Home Address of EUT		
7	Check: does QE1 receive an Echo Reply from EUT?	No	Yes
Observations:			

11.6.3 Protecting Return Routability Packets

Test Description			
Identifier:	TD_MOB_1725_01	**Test Purpose:**	TP_MOB_1725_01
Summary:	'Mobile Node support the protection of Home Test and Home Test Init messages'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1725		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE1 configured to protect return_routability_packets
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends a response directly to QE4 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT is configured to perform route optimization QE4 is configured to perform route optimization QE1 is configured to protect return routability packets by IPSec with EUT

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Check: does QE4 receive an Echo Reply from EUT	Yes	No
Observations:			

11.6 Return Routability Procedure

Test Description			
Identifier:	TD_MOB_1063_01	**Test Purpose:**	TP_MOB_1063_01
Summary:	'Mobile Node is able to participate in Return Routability Procedure'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1048, RQ_001_1063, RQ_001_1709, RQ_001_1063, RQ_001_1064, RQ_001_1744, RQ_001_1745, RQ_001_1750, RQ_001_1751, RQ_001_1754, RQ_001_1759		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 EUT is configured to perform route optimization QE4 is configured to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request message to the Home Address of EUT		
2	Check: does QE4 receive Echo Replies from EUT?	Yes	No
3	Wait a few seconds to ensure that EUT and QE4 have had time to complete the Return Routability Procedure		
4	Disconnect QE1 from the network		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:	If the Mobile Node still manages to communicate with the Corresponding Node once the Home Agent is disconnected from the network, this shows that is using Route Optimization and hence that it managed to perform the Return Routability Procedure.		

11.6.1 Sending Test Init Messages

Test Description			
Identifier:	TD_MOB_1063_01	**Test Purpose:**	TP_MOB_1063_01
Summary:	'Mobile Node is able to participate in Return Routability Procedure'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1048, RQ_001_1063, RQ_001_1709, RQ_001_1063, RQ_001_1064, RQ_001_1744, RQ_001_1745, RQ_001_1750, RQ_001_1751, RQ_001_1754, RQ_001_1759		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1 EUT is configured to perform route optimization QE4 is configured to perform route optimization

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request message to the Home Address of EUT		
2	Check: does QE4 receive Echo Replies from EUT?	Yes	No
3	Wait a few seconds to ensure that EUT and QE4 have had time to complete the Return Routability Procedure		
4	Disconnect QE1 from the network		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:	If the Mobile Node still manages to communicate with the Corresponding Node once the Home Agent is disconnected from the network, this shows that is using Route Optimization and hence that it managed to perform the Return Routability Procedure.		

11.6.2 Receiving Test Messages

Test Description			
Identifier:	TD_MOB_1724_01	**Test Purpose:**	TP_MOB_1724_01
Summary:	'Mobile Node falls back to tunnelling if correspondent refuses Route Optimization'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1724		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and has registered to its Home Agent QE1 EUT is configured to perform route optimization QE4 is configured NOT to perform route optimization		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:	As QE4 is configured to refuse Route Optimization, EUT can only fall back to tunnelling.		

11.7.1 Sending Binding Updates To The Home Agent

Test Description			
Identifier:	TD_MOB_1727_01	**Test Purpose:**	TP_MOB_1727_01
Summary:	'Mobile Node registers its new primary Care-Of Address after changing it'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1727		

```
with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     EUT moves to another foreign_network
                and EUT receives a packet from QE4
                        indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and is registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign Network 2		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receives an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1768_01	**Test Purpose:**	TP_MOB_1768_01
Summary:	'Mobile Node sends Binding Update before expiry of the binding lifetime'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1728, RQ_001_1768		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT able to communicate with QE4
       }
  ensure that
       { when { QE4 (is requested to send a packet to the EUT
                     indicating that a response is required)
                     after the binding_lifetime }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait a period equal to the Binding Lifetime		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1733_01	**Test Purpose:**	TP_MOB_1733_01
Summary:	'Mobile Node sends Binding Update messages to its Home Agent until it receives a matching Binding_Acknowledgement message, before MAX_BINDACK_TIMEOUT seconds'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1733		

```
with {     EUT at_home
         and EUT configured to have its Home_Agent_address available
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE1 is connected after 15 seconds
                     -- ~ MAX_BINDACK_TIMEOUT / 2
           }
         then {     QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1 EUT is aware of the address of its Home Agent QE1 is disconnected from Home Network 1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign Network 1		
2	Wait 20 seconds (EUT will try to bind to QE1)		
3	Connect QE1 to Home Network 1		
4	Wait a few seconds (to let EUTde-register to QE1)		
5	Cause QE4 to send an Echo Request to the Home Address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1734_01	**Test Purpose:**	TP_MOB_1734_01
Summary:	'When Mobile Node knows several Home Agent (returned during DHAAD), Mobile Node tries to register with other Home Agents when registration fails with the first one'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_06
References:	RQ_001_1734		

```
with {     EUT not configured to have its Home_Agent_address available
         and QE1 configured to have a Home_Agent_Preference greater than QE2
       }
  ensure that
       { when {     QE1 is disconnected
                and EUT moves to a foreign_network }
         then {     QE4 and EUT are able to communicate
                  after  MAX_BINDACK_TIMEOUT
           }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is unaware the address of its Home Agent QE2 is configured to serve as Home Agent in the Home Network 1 QE1 is configured with a Home Agent Preference higher than QE2		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds		
3	Cause QE4 to send an Echo Request to the Home Address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	No	Yes
5	Wait two minutes		
6	Cause QE4 to send an Echo Request to the Home Address of EUT		
7	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1740_01	**Test Purpose:**	TP_MOB_1740_01
Summary:	'Mobile Node attempts to register all of its home addresses'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1740		

```
with {     EUT disconnected
         and EUT configured to have 2 home_addresses
       }
  ensure that
       { when {     EUT is connected to a foreign_network
            and QE4 is requested to send a packet
                              indicating that a response is required
                           to the first home_address of EUT
            and QE4 is requested to send a packet
                              indicating that a response is required
                           to the second home_address of EUT }
         then {     QE4 indicates receipt of the first response
                and QE4 indicates receipt of the second response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1 EUT is configured to have two different home addresses		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds (to let EUT register to QE1)		
3	Cause QE4 to send an Echo Request to the first home address of EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
5	Cause QE4 to send an Echo Request to EUT to the second home address of EUT		
6	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1742_01	**Test Purpose:**	TP_MOB_1742_01
Summary:	'Mobile Node does not attempt to use its home address if registration with Home Agent fails due to DAD'		
Roles:	Mobile_Node, Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1742, RQ_001_1742, RQ_001_1770		

```
with { QE3 configured to have the home_address of EUT }
  ensure that
       { when { EUT moves to a foreign_network }
              -- the registering of EUT has failed
         then { EUT does not use its Home_Address to communicate }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN It is not possible to control that Mobile Node doesn't use its Home Address		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1743_01	**Test Purpose:**	TP_MOB_1743_01
Summary:	'Mobile Node goes on registering home addresses when the registration of one of them fails'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1743		

```
with {     EUT disconnected
         and EUT configured to have 2 home_addresses
         and QE3 configured to use the first home_address of EUT
       }
  ensure that
       { when {     EUT is connected to a foreign_network
                and EUT receives a packet from QE4
                        containing a destination_address
                        set to EUT home_address 2
                    and indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN You do not know how the Home Addresses are listed, so you can't guarantee that the address which failed is not the first or the last		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1814_01	**Test Purpose:**	TP_MOB_1814_01
Summary:	'Mobile Node sends Binding Update messages to its Home Agent until it receives a matching Binding_Acknowledgement message, without time limit if it knows only one Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1814		

```
with {     EUT at_home
         and EUT configured to have its Home_Agent_address available
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE1 is connected after  MAX_BINDACK_TIMEOUT
                and QE4 is requested to send a packet to QE1
                                    indicating that a response is required }
         then {     QE4 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Home Network 1 EUT is aware of the address of its Home Agent QE1 QE1 is disconnected

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign Network 1		
2	Wait 2 minutes (more than MAX_BINDACK_TIMEOUT seconds)		
3	Connect QE1 to Home Network 1		
4	Wait a few seconds (to let EUT establish a binding with QE1)		
5	Cause QE4 to send an Echo Request to the home address of EUT		
6	Check: does QE4 receives an Echo reply from EUT?	Yes	No
Observations:			

11.7.2 Correspondent registration

Test Description			
Identifier:	TD_MOB_1744_01	**Test Purpose:**	TP_MOB_1744_01
Summary:	'Mobile Node updates its bindings with correspondent nodes after changing its primary care-of address'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1744		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding to QE4
       }
  ensure that
       { when {     EUT moves to another foreign_network
                and QE4 is requested to send a packet to EUT
                    indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT having established a binding with QE4

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Foreign Network 2		
2	Cause QE4 to send directly an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive directly an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_1753_01	**Test Purpose:**	TP_MOB_1753_01
Summary:	'Mobile Node is able to delete a binding with a correspondent node'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1753		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     EUT returns home
                and QE4 is requested to send a packet to EUT
                    indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT}
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1 EUT having established a binding with QE4

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Home Network 1		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.7.3 Receiving Binding Acknowledgements

Test Description			
Identifier:	TD_MOB_1768_01	**Test Purpose:**	TP_MOB_1768_01
Summary:	'Mobile Node sends Binding Update before expiry of the binding lifetime'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_1728, RQ_001_1768		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT able to communicate with QE4
       }
  ensure that
       { when { QE4 (is requested to send a packet to the EUT
                     indicating that a response is required)
                     after the binding_lifetime }
         then { QE4 indicates receipt of the response from the EUT }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is connected to Foreign Network 1 and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Wait a period equal to the Binding Lifetime		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

11.7.4 Receiving Binding Refresh Requests

Test Description			
Identifier:	TD_MOB_1776_01	**Test Purpose:**	TP_MOB_1776_01
Summary:	'Mobile Node updates binding with correspondent node upon receipt of a Binding Refresh Request'		
Roles:	Correspondent_Node	**Configuration**:	CF_MOB_04
References:	RQ_001_1383, RQ_001_1776		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
      { when { QE4 sends a Binding_Refresh_Request }
        then { QE4 and the EUT are able to communicate directly
                    after the original binding_lifetime }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN depending of the implementation to reduce the handover

Step	Test Sequence	Verdict	
		Pass	**Fail**
Observations:			

Test Description			
Identifier:	TD_MOB_1776_01	**Test Purpose:**	TP_MOB_1776_01
Summary:	'Mobile Node updates binding with correspondent node upon receipt of a Binding Refresh Request'		
Roles:	Correspondent_Node	**Configuration:**	CF_MOB_04
References:	RQ_001_1383, RQ_001_1776		

```
with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
      { when { QE4 sends a Binding_Refresh_Request }
        then { QE4 and the EUT are able to communicate directly
                    after the original binding_lifetime }
       }
```

Pre-test conditions:	NOT TESTABLE BY INTEROPERABILITY MEAN depending of the implementation to reduce the handover		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
Observations:			

4.1.3 TDs extracted from RFC 3776 "Using IPsec to Protect Mobile IPv6 Signalling Between Mobile Nodes and Home Agents"

4.1 Mandatory Support

Test Description			
Identifier:	TD_MOB_2013_01	**Test Purpose:**	TP_MOB_2013_01
Summary:	'Mobile Node sends Binding Update secured using transport_mode_ESP to its Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2013, RQ_001_2031		

```
with {     EUT configured to protect any Binding_Update to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only Binding_Update secured
                          using transport_mode_ESP from EUT
       }
  ensure that
      { when {   a Security_Association is established between EUT and QE1
               and EUT moves to a foreign_network }
        then { QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Binding Update sent to QE1 using transport mode ESP QE1 is configured to accept only Binding Update protected using transport mode ESP from EUT EUT is connected to Home Network 1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds (to let EUT register to QE1)		
3	Cause QE4 to send an Echo Request to EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2013_02	**Test Purpose:**	TP_MOB_2013_02
Summary:	'Home Agent sends Binding_Acknowledgement secured using transport_mode_ESP to a Mobile Node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2013, RQ_001_2031		

```
with {     EUT configured to protect any Binding_Acknowledgement to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only a Binding_Acknowledgement secured
                          using transport_mode_ESP from EUT
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
              and QE1 moves to a foreign_network
            }
         then { QE4 is able to communicate with QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Binding Acknowledgement sent to QE1 using transport mode ESP QE1 is configured to accept only Binding Acknowledgement protected using transport mode ESP from EUT QE1 is connected to Home Network 1

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few second (to let QE1 register to EUT)		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2016_01	**Test Purpose:**	TP_MOB_2016_01
Summary:	'Home Agent supports Mobile_Prefix_Solicitation secured using transport_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2016, RQ_001_2031		

```
with {     QE1 configured to protect any Mobile_Prefix_Solicitation to EUT
                          using transport_mode_ESP
         and EUT configured to accept only a Mobile_Prefix_Solicitation secured
                          using transport_mode_ESP from QE1
         and QE6  configured to have a prefix_lifetime of 1 minute
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 is able to communicate with QE1 after the prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Mobile Prefix Solicitation sent to EUT using transport mode ESP EUT is configured to accept only Mobile Prefix Solicitation protected using transport mode ESP from QE1 R1 is configured to have a valid prefix lifetime at 1 minute QE1 is away from Home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Wait a few minutes (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2016_02	**Test Purpose:**	TP_MOB_2016_02
Summary:	'Mobile Node supports Mobile_Prefix_Advertisement secured using transport_mode_ESP'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2016, RQ_001_2031		

```
with {     QE1 configured to protect any Mobile_Prefix_Advertisement to EUT
                          using transport_mode_ESP
         and EUT configured to accept only Mobile_Prefix_Advertisement
                           secured using transport_mode_ESP from QE1
         and EUT away_from_home
         and EUT registered to QE1
         and QE6 configured to have a prefix_lifetime of 1 minute
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 and EUT are able to communicate after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Mobile Prefix Advertisement sent to EUT using transport mode ESP EUT is configured to accept only Mobile Prefix Advertisement protected using transport mode ESP from QE1 R1 is configured to have a valid prefix lifetime at 1 minute EUT is away from Home and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Wait a few minutes (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2017_01	**Test Purpose:**	TP_MOB_2017_01
Summary:	'Mobile Node sends Mobile_Prefix_Solicitation secured using transport_mode_ESP to its Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2017, RQ_001_2030		

```
with {     EUT configured to protect any Mobile_Prefix_Solicitation to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only a Mobile_Prefix_Solicitation
                          secured using transport_mode_ESP from EUT
         and QE6 configured to have a prefix_lifetime of 1 minute
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 and EUT are able to communicate after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Mobile Prefix Solicitation sent to QE1 using transport mode ESP QE1 is configured to accept only Mobile Prefix Solicitation protected using transport mode ESP from EUT R1 is configured to have a valid prefix lifetime at 1 minute EUT is away from Home and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Wait a few more one minute (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2017_02	**Test Purpose:**	TP_MOB_2017_02
Summary:	'Home Agent sends Mobile_Prefix_Advertisement secured using transport_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2017, RQ_001_2030		

```
with {     EUT configured to protect any Mobile_Prefix_Advertisement to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only a Mobile_Prefix_Advertisement
                           secured using transport_mode_ESP from EUT
         and QE1 away_from_home
         and QE1 registered to EUT
         and QE6 configured to have a prefix_lifetime of 1 minute
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 is able to communicate with QE1 after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Mobile Prefix Advertisement sent to QE1 using transport mode ESP QE1 is configured to accept only Mobile Prefix Advertisement protected using transport mode ESP from EUT R1 is configured to have a valid prefix lifetime at 1 minute QE1 is away from Home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait a few more one minute (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 to send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2018_01	**Test Purpose:**	TP_MOB_2018_01
Summary:	'Mobile Node sends payload_packet tunnelled to its Home Agent secured using tunnel_mode_ESP'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2018		

```
with {     EUT configured to protect any payload_packet to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only a payload_packet secured
                          using tunnel_mode_ESP from EUT
         and EUT configured not to perform Route_Optimization
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
                and EUT moves to a foreign_network
           }
         then { QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect payload packets sent to QE1 using tunnel mode ESP QE1 is configured to accept only payload packets protected using tunnel mode ESP from EUT EUT configured not to perform Route Optimization EUT is away from home and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2018_02	**Test Purpose:**	TP_MOB_2018_02
Summary:	'Home Agent sends payload_packet tunnelled to Mobile Node secured using tunnel_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2018		

```
with {     EUT configured to protect any payload_packet to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only a payload_packet secured
                          using tunnel_mode_ESP from EUT
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect payload packets sent to QE1 using tunnel mode ESP QE1 is configured to accept only payload packets protected using tunnel mode ESP from EUT QE1 configured not to perform Route Optimization QE1 is away from home and registered to its Home Agent EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of the QE1		
2	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2018_03	**Test Purpose:**	TP_MOB_2018_03
Summary:	'Mobile Node supports payload_packet secured using tunnel_mode_ESP'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2018		

```
with {     QE1 configured to protect any payload_packet to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only a payload_packet secured
                          using tunnel_mode_ESP from QE1
         and EUT configured not to perform Route_Optimization
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and EUT moves to a foreign_network }
         then { QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect payload packets sent to EUT using tunnel mode ESP EUT is configured to accept only payload packets protected using tunnel mode ESP from QE1 EUT configured not to perform Route Optimization EUT is away from home and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2018_04	**Test Purpose:**	TP_MOB_2018_04
Summary:	'Home Agent supports payload_packet tunnelled from Mobile Node and secured using tunnel_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2018		

```
with {     QE1 configured to protect any payload_packet to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only a payload_packet secured
                          using tunnel_mode_ESP from QE1
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 is able to communicate with QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect payload packets sent to EUT using tunnel mode ESP EUT is configured to accept only payload packets protected using tunnel mode ESP from QE1 QE1 configured not to perform Route Optimization QE1 is away from home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address of QE1		
2	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2019_01	**Test Purpose:**	TP_MOB_2019_01
Summary:	'Mobile Node supports multicast group membership control packets tunnelled from its Home Agent protected by ESP with the tunnel mode'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2019		

```
with {     QE1 configured to protect any
                          multicast_group_membership_control_packet to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only a
                          multicast_group_membership_control_packet from QE1
                          secured using tunnel_mode_ESP
         and EUT configured to subscribed to a global_multicast_group
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { QE3 sends a packet to the global_multicast_group
                                  indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to secured multicast group membership control packets tunnelled to EUT by ESP with tunnel mode EUT is configured to accept only protected multicast group membership control packets tunnelled from QE1 by ESP with tunnel mode EUT is configured to subscribed to a multicast group MG1 with a global scope EUT away from home and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE3 to send an Echo Request to the Address of multicast group MG1		
2	Check: does QE3 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2019_02	**Test Purpose:**	TP_MOB_2019_02
Summary:	'Home Agent supports multicast group membership control packets tunnelled from Mobile Node protected by ESP with the tunnel mode'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2019		

```
with {     QE1 configured to protect any
                          multicast_group_membership_control_packet to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only
                         a multicast_group_membership_control_packet from QE1
                         secured using tunnel_mode_ESP
         and QE1 configured to subscribed to a global_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { EUT sends a packet to the global_multicast_group
                                  indicating that a response is requested }
         then { QE3 indicates receipt of the response }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to secured multicast group membership control packets tunnelled to EUT by ESP with tunnel mode EUT is configured to accept only protected multicast group membership control packets tunnelled from QE1 by ESP with tunnel mode QE1 is configured to subscribed to a multicast group MG1 with a global scope QE1 away from home and registered to its Home Agent EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Cause QE3 to send an Echo Request to the Address of multicast group MG1		
2	Check: does QE3 receive an Echo Reply from QE1?	Yes	No
Observations:			

4.2 Policy Requirements

Test Description			
Identifier:	TD_MOB_2026_01	**Test Purpose:**	TP_MOB_2026_01
Summary:	'When Mobile Node returns home and is configured with manually established Security Associations, it makes inactive security policies concerning tunnelled traffic with Home Agent without deleting Security Associations'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2026		

```
with {     EUT configured 'with manually established Security Associations
                           and Policies'
         and EUT configured to protect packets to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only a packets secured
                          using tunnel_mode_ESP
         and EUT configured not to perform Route_Optimization
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {       EUT moves to another foreign_network
                after EUT returns home }
         then {       QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments). EUT is configured with manually established Security Associations and Policies. EUT is configured to secure packets tunnelled to QE1 using tunnel mode ESP. QE1 is configured to accept only packets protected using tunnel mode ESP. EUT is configured not to perform Route Optimization. EUT is away from home and has registered to its Home Agent QE1.

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move EUT to Home Network 1		
2	Wait a few seconds		
3	Move EUT to Foreign Network 1		
4	Cause QE4 to send an Echo Request to EUT		
5	Check: does QE4 receive an Echo Reply?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2026_02	**Test Purpose:**	TP_MOB_2026_02
Summary:	'When Mobile Node returns home and is configured with manually established Security Associations, Home Agent makes inactive security policies concerning tunnelled traffic with Mobile Node without deleting Security Associations		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2026		

```
with {     EUT configured 'with manually established Security Associations
                           and Policies'
         and EUT configured to protect packets to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only a packets secured
                          using tunnel_mode_ESP
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {       QE1 moves to another foreign_network
                after QE1 returns home }
         then {       QE4 is able to communicate with QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments EUT is configured with manually established Security Associations and Policies EUT is configured to secure packets tunnelled to QE1 using tunnel mode ESP QE1 is configured to accept only packets protected using tunnel mode ESP QE1 is configured not to perform Route Optimization QE1 is away from home and has registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Move QE1 to Home Network 1		
2	Wait a few seconds		
3	Move QE1 to Foreign Network 1		
4	Cause QE4 to send an Echo Request to QE1		
5	Check: does QE4 receive an Echo Reply?	Yes	No
Observations:			

4.3 IPsec Protocol Processing

Test Description			
Identifier:	TD_MOB_2017_01	**Test Purpose:**	TP_MOB_2017_01
Summary:	'Mobile Node sends Mobile_Prefix_Solicitation secured using transport_mode_ESP to its Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2017, RQ_001_2030		

```
with {     EUT configured to protect any Mobile_Prefix_Solicitation to QE1
                         using transport_mode_ESP
         and QE1 configured to accept only a Mobile_Prefix_Solicitation
                         secured using transport_mode_ESP from EUT
         and QE6 configured to have a prefix_lifetime of 1 minute
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 and EUT are able to communicate after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Mobile Prefix Solicitation sent to QE1 using transport mode ESP QE1 is configured to accept only Mobile Prefix Solicitation protected using transport mode ESP from EUT R1 is configured to have a valid prefix lifetime at 1 minute EUT is away from Home and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait a few more one minute (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 to send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2017_02	**Test Purpose:**	TP_MOB_2017_02
Summary:	'Home Agent sends Mobile_Prefix_Advertisement secured using transport_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2017, RQ_001_2030		

```
with {     EUT configured to protect any Mobile_Prefix_Advertisement to QE1
                         using transport_mode_ESP
         and QE1 configured to accept only a Mobile_Prefix_Advertisement
                          secured using transport_mode_ESP from EUT
         and QE1 away_from_home
         and QE1 registered to EUT
         and QE6 configured to have a prefix_lifetime of 1 minute
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 is able to communicate with QE1 after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Mobile Prefix Advertisement sent to QE1 using transport mode ESP QE1 is configured to accept only Mobile Prefix Advertisement protected using transport mode ESP from EUT R1 is configured to have a valid prefix lifetime at 1 minute QE1 is away from Home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Wait a few more one minute (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 to send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2013_01	**Test Purpose:**	TP_MOB_2013_01
Summary:	'Mobile Node sends Binding Update secured using transport_mode_ESP to its Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2013, RQ_001_2031		

```
with {     EUT configured to protect any Binding_Update to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only Binding_Update secured
                          using transport_mode_ESP from EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and EUT moves to a foreign_network }
         then { QE4 and EUT are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Binding Update sent to QE1 using transport mode ESP QE1 is configured to accept only Binding Update protected using transport mode ESP from EUT EUT is connected to Home Network 1

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move EUT to Foreign Network 1		
2	Wait a few seconds (to let EUT register to QE1)		
3	Cause QE4 to send an Echo Request to EUT		
4	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2013_02	**Test Purpose:**	TP_MOB_2013_02
Summary:	'Home Agent sends Binding_Acknowledgement secured using transport_mode_ESP to a Mobile Node'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2013, RQ_001_2031		

```
with {     EUT configured to protect any Binding_Acknowledgement to QE1
                          using transport_mode_ESP
         and QE1 configured to accept only a Binding_Acknowledgement secured
                          using transport_mode_ESP from EUT
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
              and QE1 moves to a foreign_network
           }
         then { QE4 is able to communicate with QE1 }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Binding Acknowledgement sent to QE1 using transport mode ESP QE1 is configured to accept only Binding Acknowledgement protected using transport mode ESP from EUT QE1 is connected to Home Network 1

Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Wait a few second (to let QE1 register to EUT)		
3	Cause QE4 to send an Echo Request to QE1		
4	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2016_01	**Test Purpose:**	TP_MOB_2016_01
Summary:	'Home Agent supports Mobile_Prefix_Solicitation secured using transport_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2016, RQ_001_2031		

```
with {     QE1 configured to protect any Mobile_Prefix_Solicitation to EUT
                          using transport_mode_ESP
         and EUT configured to accept only a Mobile_Prefix_Solicitation secured
                          using transport_mode_ESP from QE1
         and QE6  configured to have a prefix_lifetime of 1 minute
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 is able to communicate with QE1 after the prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Mobile Prefix Solicitation sent to EUT using transport mode ESP EUT is configured to accept only Mobile Prefix Solicitation protected using transport mode ESP from QE1 R1 is configured to have a valid prefix lifetime at 1 minute QE1 is away from Home and registered to its Home Agent EUT		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Wait a few minutes (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 send an Echo Request to the Home Address of QE1		
3	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2016_02	**Test Purpose:**	TP_MOB_2016_02
Summary:	'Mobile Node supports Mobile_Prefix_Advertisement secured using transport_mode_ESP'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2016, RQ_001_2031		

```
with {     QE1 configured to protect any Mobile_Prefix_Advertisement to EUT
                          using transport_mode_ESP
         and EUT configured to accept only Mobile_Prefix_Advertisement
                           secured using transport_mode_ESP from QE1
         and EUT away_from_home
         and EUT registered to QE1
         and QE6 configured to have a prefix_lifetime of 1 minute
       }
  ensure that
       { when { a Security_Association is established between EUT and QE1 }
         then { QE4 and EUT are able to communicate after prefix_lifetime }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Mobile Prefix Advertisement sent to EUT using transport mode ESP EUT is configured to accept only Mobile Prefix Advertisement protected using transport mode ESP from QE1 R1 is configured to have a valid prefix lifetime at 1 minute EUT is away from Home and registered to its Home Agent QE1		
Step	**Test Sequence**	**Verdict**	
		Pass	**Fail**
1	Wait a few minutes (to let the exchange of MPA / MPS be carried out)		
2	Cause QE4 send an Echo Request to the Home Address of EUT		
3	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2014_01	**Test Purpose:**	TP_MOB_2014_01
Summary:	'Home Agent supports Home Test Init secured using tunnel_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_1571, RQ_001_2034, RQ_001_2035		

```
with {     QE1 configured to protect any Home_Test_Init to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only Home_Test_Init secured
                          using tunnel_mode_ESP from QE1
         and QE1 configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE1 is away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Home Test Init sent to EUT using tunnel mode ESP EUT is configured to accept only Home Test Init protected using tunnel mode ESP from QE1 QE1 configured to perform Route Optimization QE4 configured to perform Route Optimization QE1 is away from Home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address of QE1		
2	Wait a few second (to let QE1 register to QE4)		
3	Disconnect EUT from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE1?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2014_02	**Test Purpose:**	TP_MOB_2014_02
Summary:	'Mobile Node supports Home Test secured using tunnel_mode_ESP'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2034, RQ_001_2035		

```
with {     QE1 configured to protect any Home_Test to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only a Home_Test secured
                          using tunnel_mode_ESP from QE1
         and EUT configured to perform Route_Optimization
         and QE4 configured to perform route_optimization
         and EUT is away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and EUT moves to a foreign_network }
         then {     QE4 and EUT are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) QE1 is configured to protect Home Test sent to EUT using tunnel mode ESP EUT is configured to accept only Home Test protected using tunnel mode ESP from QE1 EUT configured to perform Route Optimization QE4 configured to perform Route Optimization EUT is away from Home and has registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address of EUT		
2	Wait a few second (to let EUT register to QE4)		
3	Disconnect QE1 from Home Network		
4	Cause QE4 to send an Echo Request to the Home Address of EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2015_01	**Test Purpose:**	TP_MOB_2015_01
Summary:	'Mobile Node sends Home Test Init secured using tunnel_mode_ESP to its Home Agent'		
Roles:	Mobile_Node	**Configuration:**	CF_MOB_03
References:	RQ_001_2034		

```
with {     EUT configured to protect any Home_Test_Init to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only Home_Test_Init secured
                          using tunnel_mode_ESP from EUT
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
              and EUT moves to a foreign_network }
         then { QE4 and EUT are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Home Test Init sent to QE1 using tunnel mode ESP QE1 is configured to accept only Home Test Init protected using tunnel mode ESP from EUT EUT configured to perform Route Optimization QE4 configured to perform Route Optimization EUT is away from Home and registered to its Home Agent QE1

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address to EUT		
2	Wait a few second (to let QE4 register to EUT)		
3	Disconnect QE1 from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address to EUT		
5	Check: does QE4 receive an Echo Reply from EUT?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2015_02	**Test Purpose:**	TP_MOB_2015_02
Summary:	'Home Agent sends Home Test secured using tunnel_mode_ESP'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2034		

```
with {     EUT configured to protect any Home_Test to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only Home_Test secured
                          using tunnel_mode_ESP from EUT
         and QE1 configured to perform Route_Optimization
         and QE4 configured to perform Route_Optimization
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate directly }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to protect Home Test sent to QE1 using tunnel mode ESP QE1 is configured to accept only Home Test protected using tunnel mode ESP from EUT QE1 configured to perform Route Optimization QE4 configured to perform Route Optimization QE1 is away from Home and registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	Fail
1	Cause QE4 to send an Echo Request to the Home Address of QE1		
2	Wait a few second (to let QE4 register to QE1)		
3	Disconnect EUT from Home Network 1		
4	Cause QE4 to send an Echo Request to the Home Address of QE1		
5	Check: does QE4 receive an Echo Reply from QE4?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2039_01	**Test Purpose:**	TP_MOB_2039_01
Summary:	'Home Agent update its security association entries when the Mobile Node change of foreign network'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2039		

```
with {     EUT configured to protect any Binding_Acknowledgement to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only a Binding_Acknowledgement
                          secured using tunnel_mode_ESP from EUT
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { QE1 moves to another foreign_network }
         then { QE4 and QE1 are able to communicate }
       }
```

Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to secure Binding Acknowledgement using ESP transport mode QE1 is configured to accept only Binding Acknowledgement protected by ESP transport mode QE1 is connected to Foreign Network 1 and has registered to its Home Agent EUT

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Move QE1 to Foreign Network 2		
2	Cause QE4 to send an Echo Request to QE1		
3	Check: does QE4 receive an Echo Reply?	Yes	No
Observations:			

Test Description			
Identifier:	TD_MOB_2040_01	**Test Purpose:**	TP_MOB_2040_01
Summary:	'Home Agent does not take into account a Binding Update not secured'		
Roles:	Home_Agent	**Configuration:**	CF_MOB_02
References:	RQ_001_2040		

```
with { and EUT configured to accept only a Binding_Update
                          secured using tunnel_mode_ESP
         and QE1 configured to send any Binding_Update not secured
       }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and QE1 are unable to communicate }
       }
```

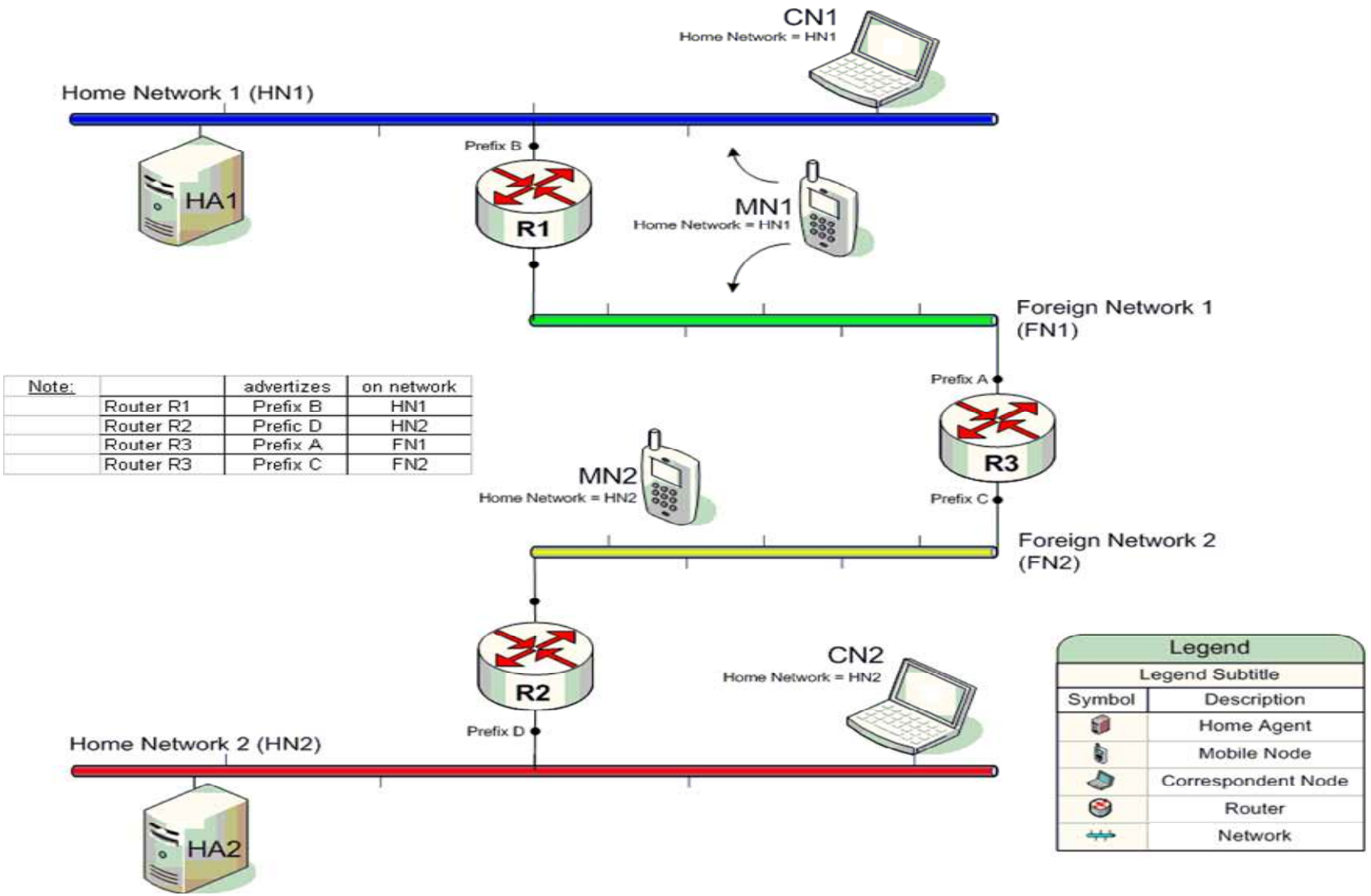
Pre-test conditions:	The cleaning procedure has been run (see configuration comments) EUT is configured to accept only Binding Update protected by ESP with transport mode QE1 is configured to send unprotected Binding Update

Step	Test Sequence	Verdict	
		Pass	**Fail**
1	Move QE1 to Foreign Network 1		
2	Cause QE4 to send an Echo Request to QE1		
3	Check: does QE4 receive an Echo Reply?	No	Yes
Observations:			

Annex A (informative): Interoperability Testing Configurations

The following architectural configurations are referenced in the IPv6 *Mobility* Interoperability Test Descriptions specified in the present document. They are intended to give a general rather than specific view of the possible roles of the EUT and its associated QE(s) and the relationships between them.

The CF_Mob-Basic_01 configuration is not referenced in any of the Test Description, however all other configurations are subset of the architecture described in CF_Mob-Basic_01. Hence, replicating CF_Mob-Basic_01 in a test bed will save a lot of time to a tester willing to test all the Test Descriptions of the present document.



Note:		advertizes	on network
	Router R1	Prefix B	HN1
	Router R2	Prefic D	HN2
	Router R3	Prefix A	FN1
	Router R3	Prefix C	FN2

Figure A.1: CF_Mob-Basic_01

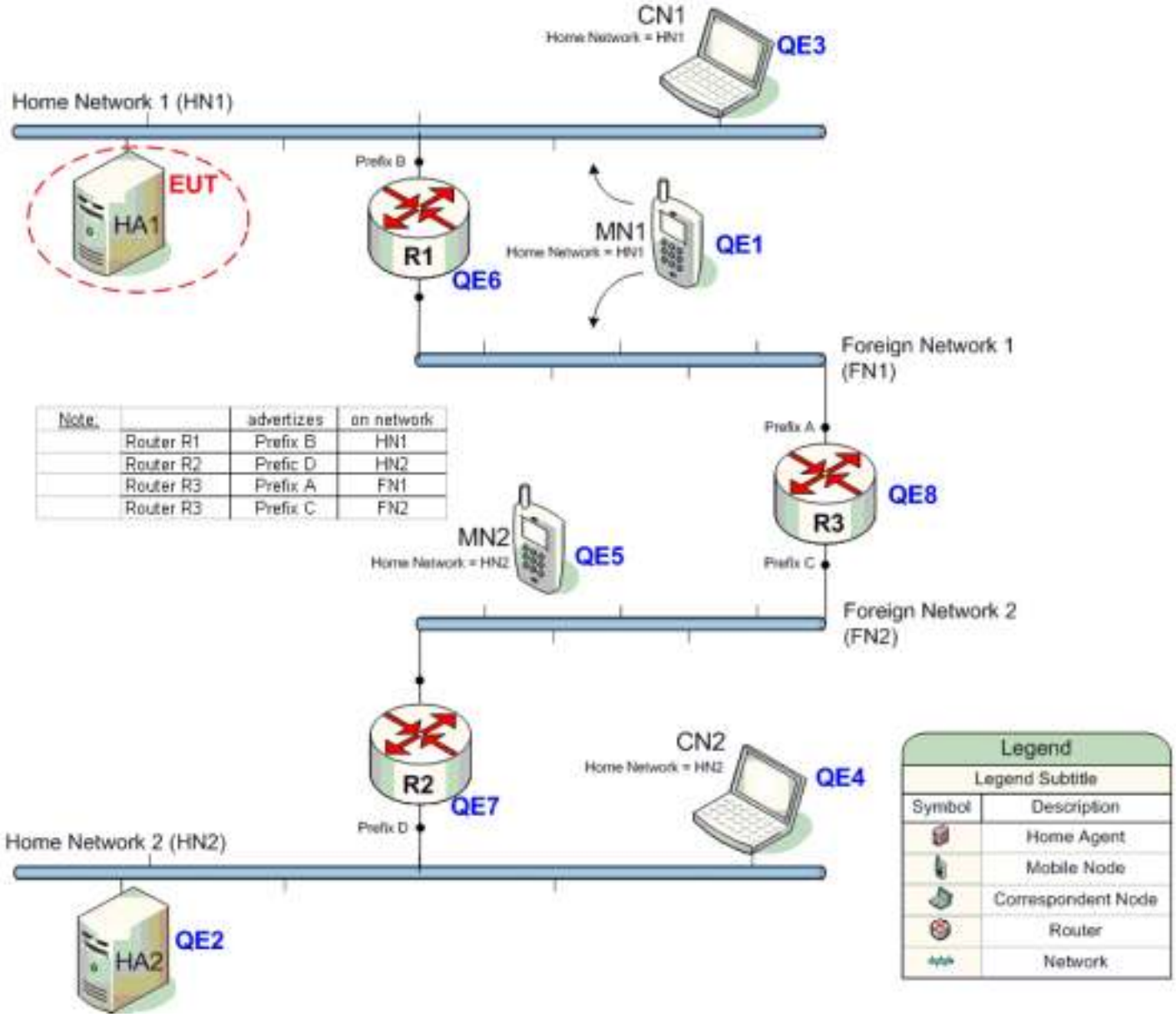
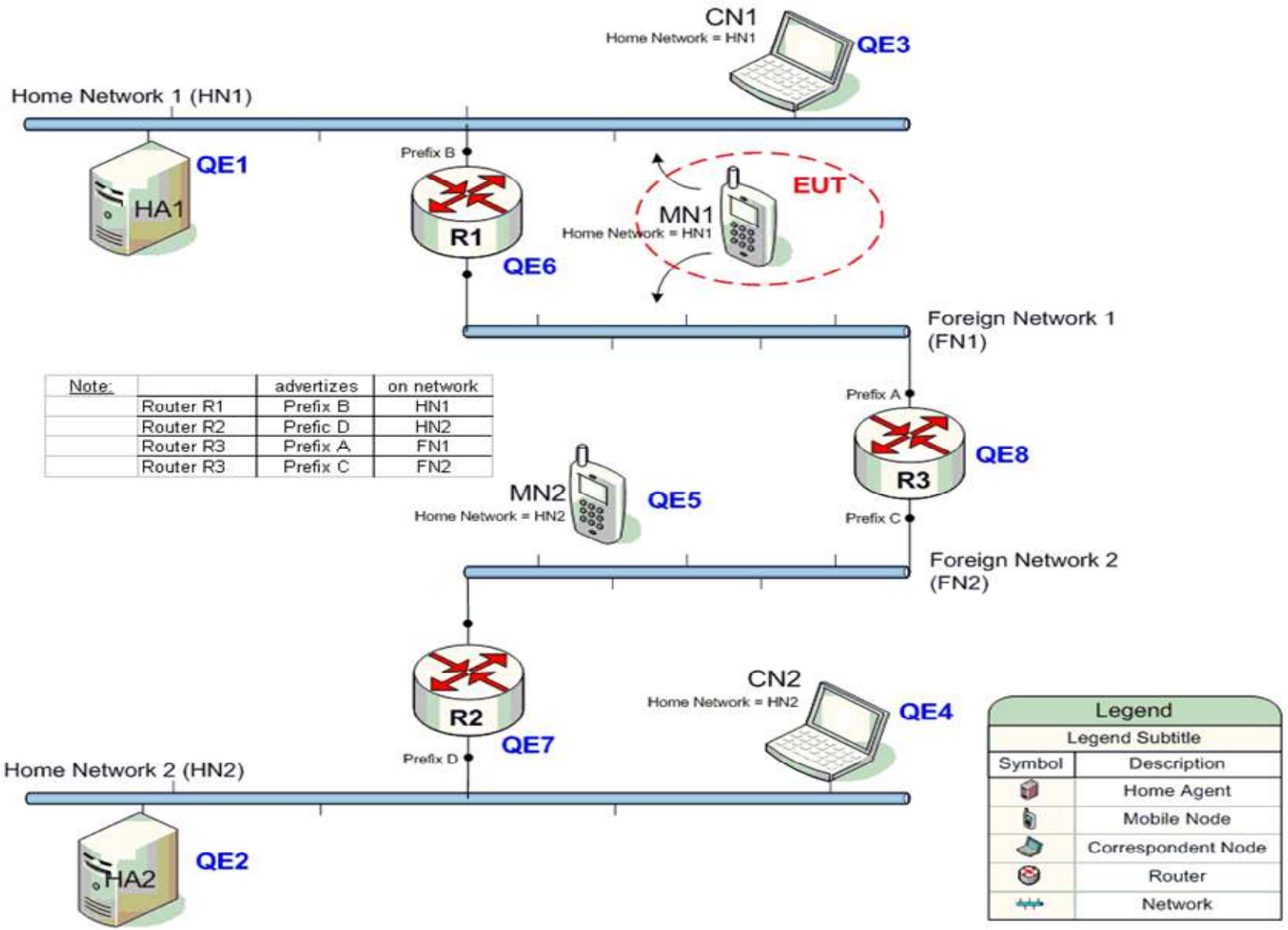
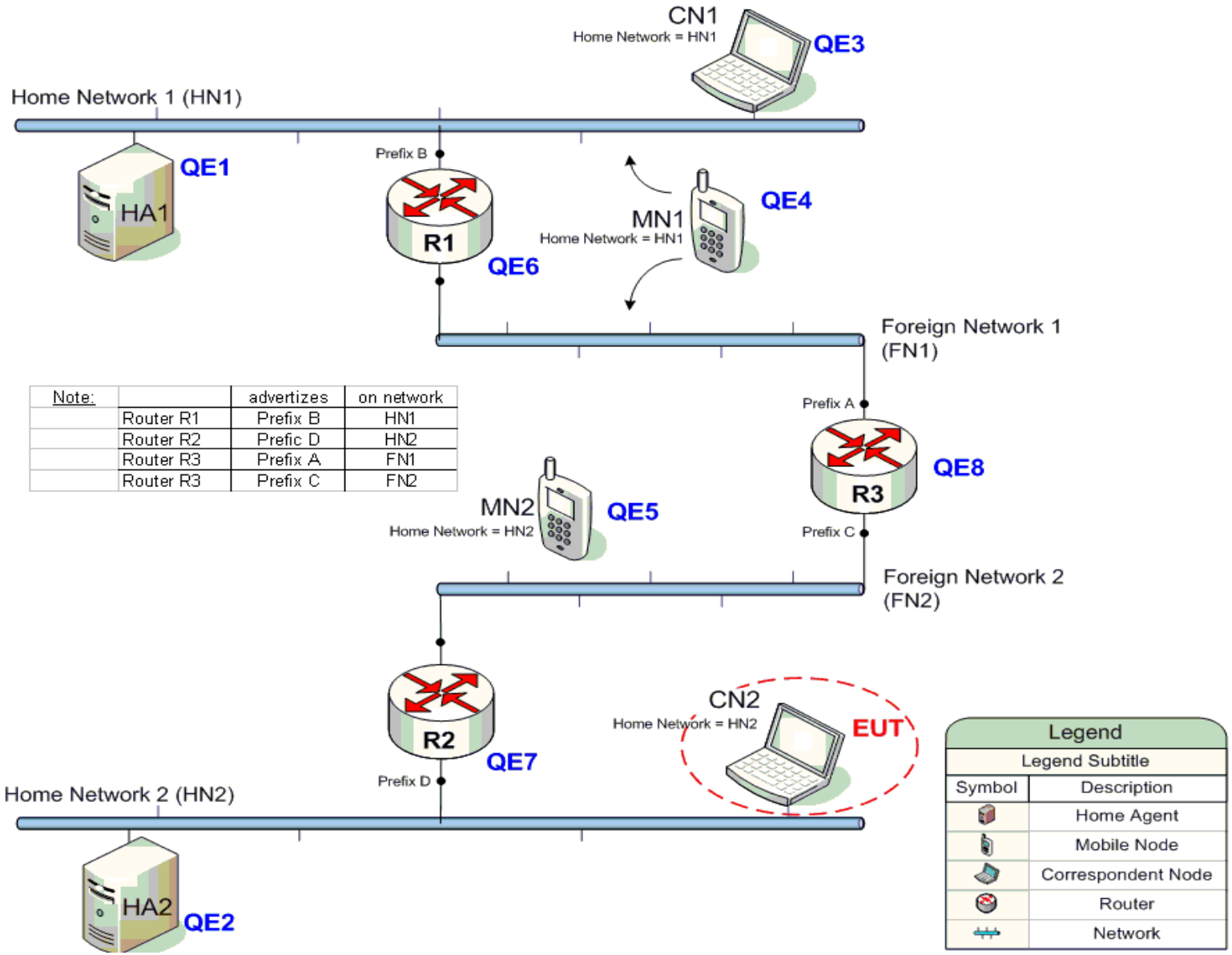


Figure A.2: CF_MOB_02



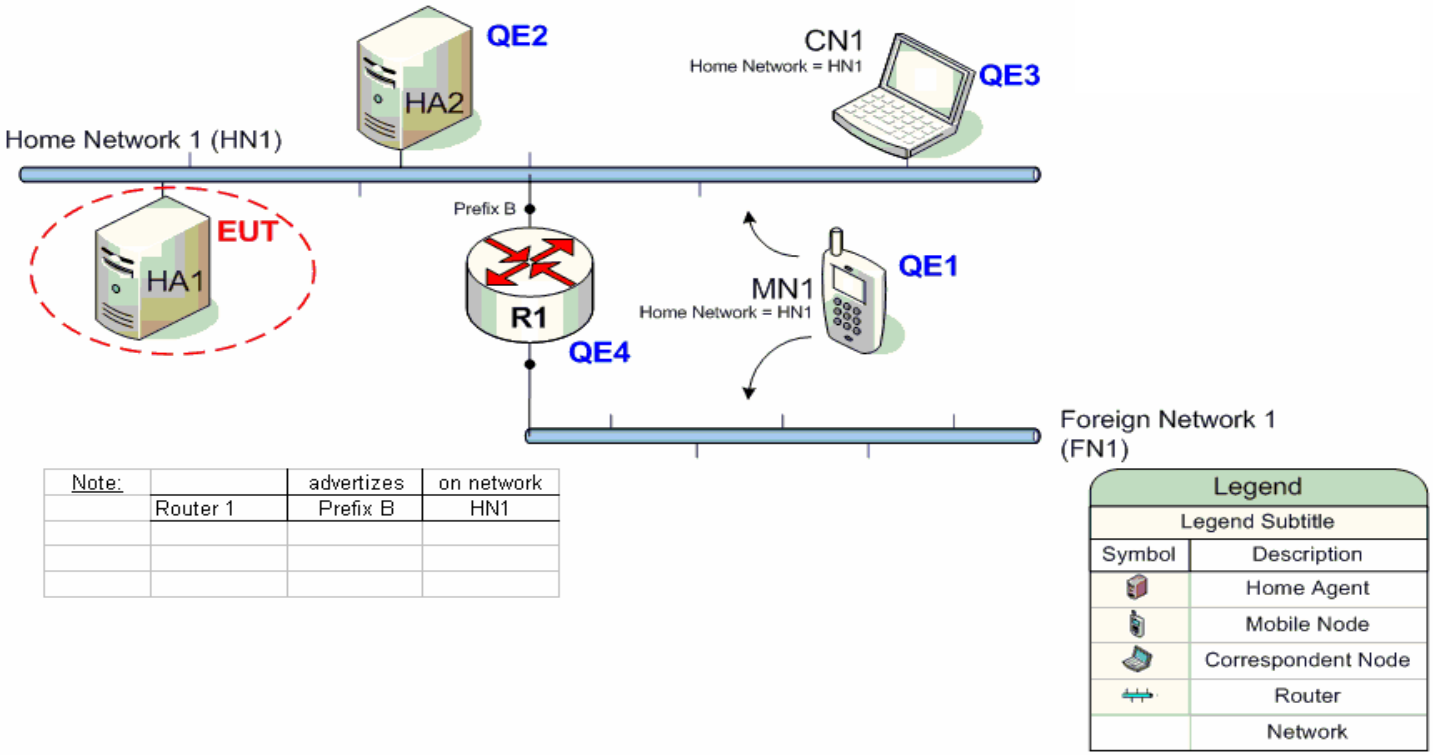
Note:		advertizes	on network
	Router R1	Prefix B	HN1
	Router R2	Prefic D	HN2
	Router R3	Prefix A	FN1
	Router R3	Prefix C	FN2

Figure A.3: CF_MOB_03



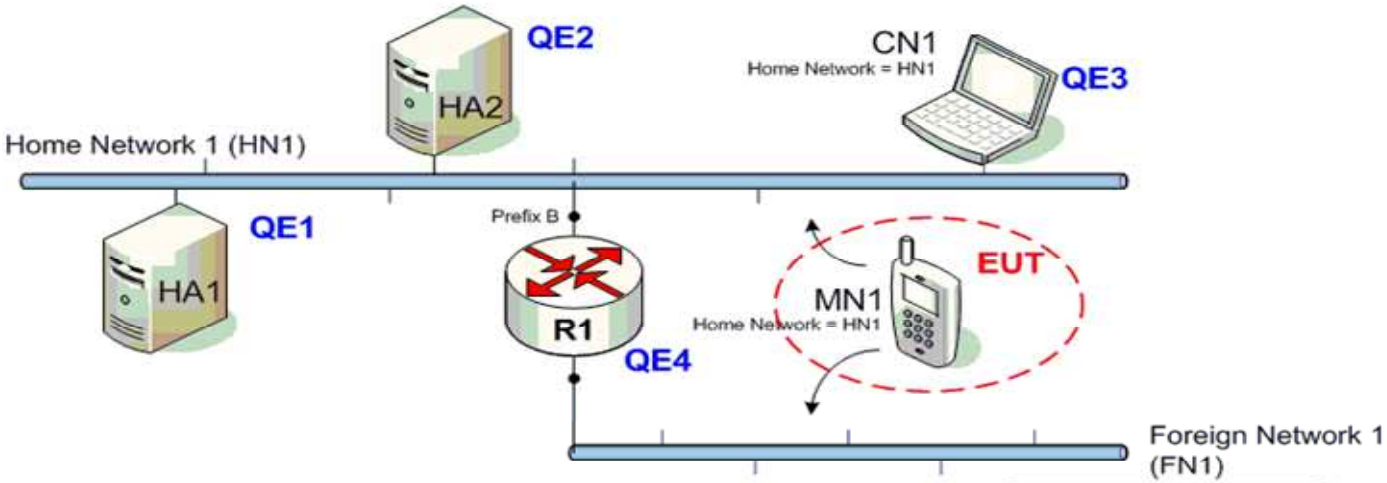
Note:		advertizes	on network
	Router R1	Prefix B	HN1
	Router R2	Prefic D	HN2
	Router R3	Prefix A	FN1
	Router R3	Prefix C	FN2

Figure A.4: CF_MOB_04



Note:		advertizes	on network
	Router 1	Prefix B	HN1

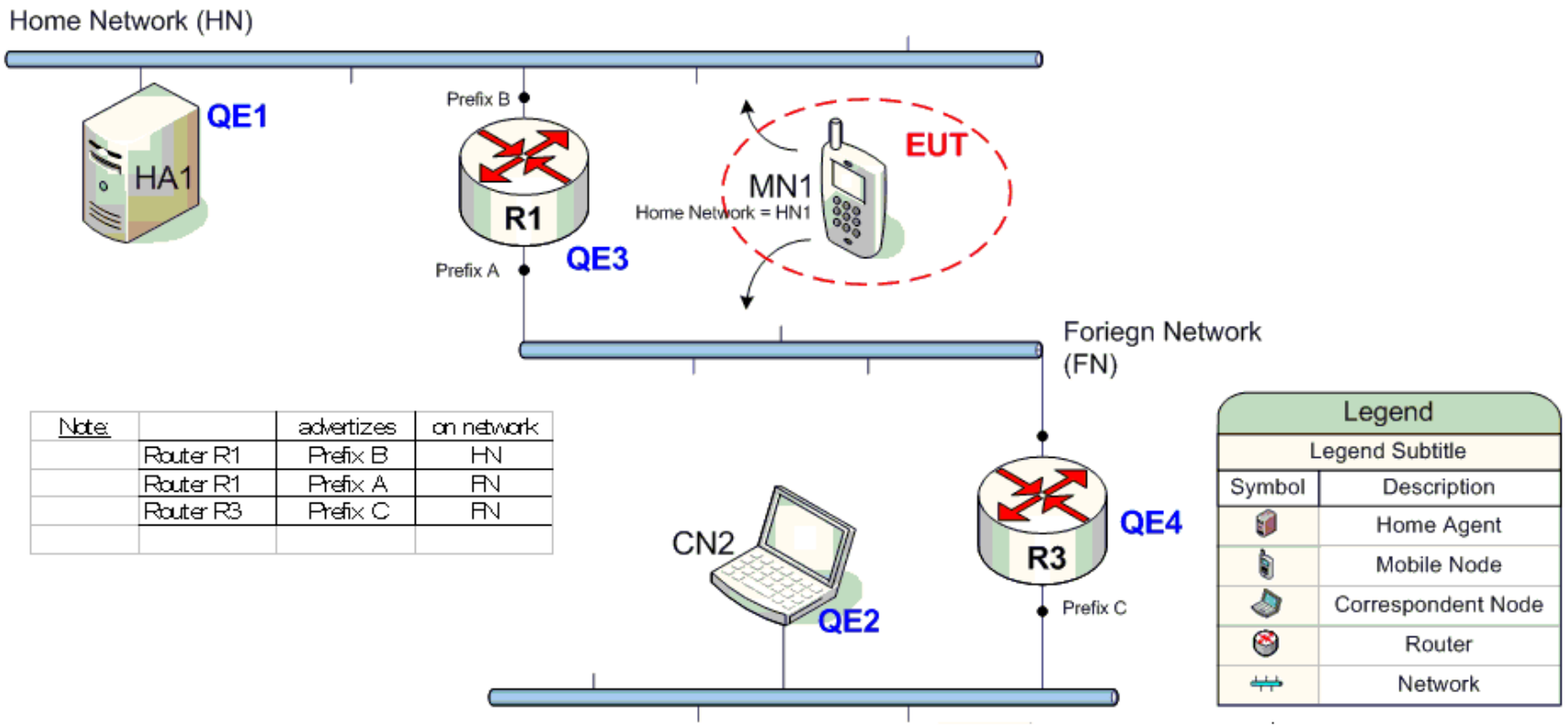
Figure A.5: CF_MOB_05



Note:		advertizes	on network
	Router 1	Prefix B	HN1

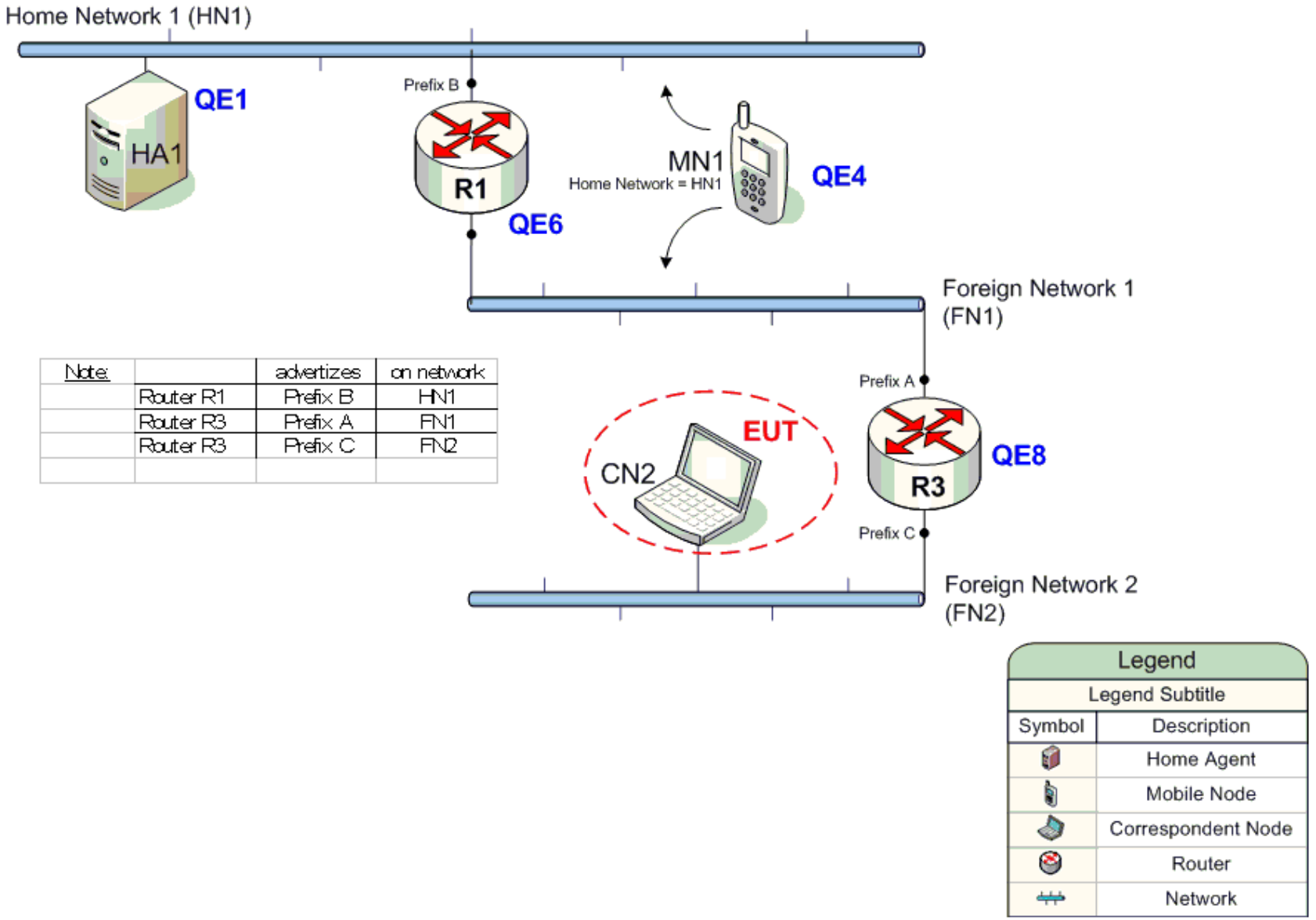
Legend	
Legend Subtitle	
Symbol	Description
	Home Agent
	Mobile Node
	Correspondent Node
	Router
	Network

Figure A.6: CF_MOB_06



Note:		advertizes	on network
	Router R1	Prefix B	HN
	Router R1	Prefix A	FN
	Router R3	Prefix C	FN

Figure A.7: CF_MOB_07



Note:		advertizes	on network
	Router R1	Prefix B	HN1
	Router R3	Prefix A	FN1
	Router R3	Prefix C	FN2

Figure A.8: CF_MOB_08

Annex B (informative): IPv6 Interoperability Test Purposes

The Test Suite Structure is based on the IPv6 Mobility RFCs and the IPv6 Requirements Catalogue nodes. It is defined by the groups within the following TPLan specification of test purposes. The numbering is not contiguous so that new TPs can be added at a later date without the need to completely renumber the TSS groups.

```
TSS     : MOB
Title   : 'IOP - Mobility'
Version : 1.0.0
Date    : 29.11.2006
Author  : 'STF276 - Task 4'

-- Last $Rev: 430 $
-- Last $Author: vreck $
-- $Date: 2007-03-15 16:25:18 +0100 (Thu, 15 Mar 2007) $


--***Cross references***
xref RQ_001 {RFC3775,
             RFC3776}

xref CF_MOB_02 {Configs_IOP_SEC.pdf}
xref CF_MOB_03 {Configs_IOP_SEC.pdf}
xref CF_MOB_04 {Configs_IOP_SEC.pdf}
xref CF_MOB_05 {Configs_IOP_SEC.pdf}
xref CF_MOB_06 {Configs_IOP_SEC.pdf}
xref CF_MOB_07 {Configs_IOP_SEC.pdf}

--***Definitions***

-- Entities
---- Primary Configuration entities
def entity EUT
def entity QE1
def entity QE2
def entity QE3
def entity QE4
def entity QE5
def entity QE6
def entity QE7
def entity QE8

---- Supplementary entities
def entity binding
def entity Binding_Cache_entry
def entity default_router
def entity foreign_network
def entity global_multicast_group
def entity home
def entity home_network
def entity link_local_multicast_group
def entity Neighbor_Discovery_cache_entry
def entity network
def entity organization_local_multicast_group
def entity Security_Association
def entity Security_Policy
def entity transport_connection

-- Messages
def event Binding_Acknowledgement
def event Binding_Refresh_Request
def event Binding_Update
          { Home_Address,
            Home_Address_Destination_Option }
def event Home_Test
def event Home_Test_Init
def event Mobile_Prefix_Advertisements {Mobile_Prefix_Advertisement}
def event Mobile_Prefix_Solicitations {Mobile_Prefix_Solicitation}
def event multicast_group_membership_control_packet
def event packets {packet}
```

```
def event payload_packet
def event return_routability_packets

-- Procedures
def event IKEv2
def event multicast_data_packet_forwarding
def event route_optimization

-- Values
def value accept
def value binding_lifetime
def value care_of_address
def value ESP
def value ethernet_address
def value global_address
def value global_addresses
def value home_address
def value home_addresses
def value Home_Agent_address
def value Home_Agent_Preference
def value Interface_ID
def value IP_address
def value IPsec
def value lifetime
def value link_local_address
def value MAX_BINDACK_TIMEOUT
def value MAX_RR_BINDING_LIFETIME
def value on_link
def value packet {Authentication_Header,
                  destination_address,
                  ESP_Header,
                  Home_Address_Option,
                  sequence_number,
                  source_address}
def value prefix
def value prefix_lifetime
def value transport_mode_ESP
def value tunnel_mode_ESP

-- Units
def unit minute
def unit seconds

-- Conditions
def condition at_home
def condition available
def condition away_from_home
def condition connected
def context {~connected [to]}
def condition configured
def condition disconnected
def context {~disconnected [from]}
def condition registered
def context {~registered to}
def condition required
def condition subscribed
def context {~subscribed to}
def condition unreachable
def context {~unreachable from}

-- Keywords - Preconditions
def word communication
def word established
def word implement
def word perform
def word protect
def word use

-- Keywords - Stimuli
def word advertizing
def word communicate
def word configure
def word detects
def word expired
def word generates
def word modify
def word moves
def word reboot
```

```
def word registers
def word requested
def context {is ~requested to}
def context {is ~requested to ~configure}
def context {is ~requested to ~modify}
def word returns
def context {~returns ~home}
def word send
def context {is requested to ~send}
def word set
def context {~set to}
def word starts
def context {~starts advertizing}
def word stops
def context {~stops advertizing}
def word support

-- Keywords - Responses
def word accepts
def word deletes
def word discards
def word establishes
def word expires
def word implemented
def word indicate
def word indicates
def word only
def word receipt
def context {~indicates ~receipt of}
def word response
def context {sends no ~response}
def word secured
def word updates
def word using

-- Keywords - Glue
def word able
def word another
def word any
def word are
def word at
def word between
def word by
def word directly
def word does
def word even
def word first
def word for
def word greater
def word have
def word having
def word its
def word least
def word more
def word new
def word offered
def word old
def word on
def word original
def word own
def context {~its ~own}
def word previously
def word same
def word second
def word short
def word than
def word unable
def word used



--**********************************--
--* RFC3775 Mobility Support in IPv6 *--
--**********************************--
Group 1 'RFC 3775'
Group 1.1 'Group A'

TP id   : TP_MOB_1724_01
```

```
summary : 'Mobile Node falls back to tunnelling if correspondent refuses
           Route Optimization'
RQ ref  : RQ_001_1724
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1724_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1744_01
summary : 'Mobile Node updates its bindings with correspondent nodes
           after changing its primary care-of address'
RQ ref  : RQ_001_1744
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1744_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding to QE4
       }
  ensure that
       { when {     EUT moves to another foreign_network
                and QE4 is requested to send a packet to EUT
                        indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1457_01
summary : 'Correspondent Node creates a new entry in its binding cache
           when receiving a valid Binding Update'
RQ ref  : RQ_001_1457
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1457_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1753_01
summary : 'Mobile Node is able to delete a binding with a correspondent node'
RQ ref  : RQ_001_1753
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1753_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     EUT returns home
                and QE4 is requested to send a packet to EUT
                        indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT}
```

```
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1486_01
summary : 'Correspondent Node deletes any Binding Cache entry
           after the expiration of the binding lifetime'
RQ ref  : RQ_001_1486 ,RQ_001_1461
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1486_01

  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
       }
  ensure that
       { when {    EUT establishes a binding to QE4}
         then {    EUT and QE4 are able to communicate directly
                             within the binding_lifetime
              and EUT and QE4 are unable to communicate directly
                             after the binding_lifetime }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1068_01
summary : 'Correspondent Node does not establish any binding with
           a lifetime greater than MAX_RR_BINDING_LIFETIME'
RQ ref  : RQ_001_1068
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1068_01

  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
    }
  ensure that
       { when {     EUT establishes a binding to QE4}
         then {     EUT and QE4 are able to communicate directly
                             within the binding_lifetime
                and EUT and QE4 are unable to communicate directly
                             after MAX_RR_BINDING_LIFETIME }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1068_02
summary : 'Mobile Node does not establish any binding with
           a lifetime greater than MAX_RR_BINDING_LIFETIME'
RQ ref  : RQ_001_1068
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1068_02
--NOT interop TESTABLE
-- ==> there is no Interoperability method to guarantee that a
-- Mobile Node (on a foreign network) will use its Home Address
-- to send a new packet (instead of using its Care-of-Address).

  with {     EUT configured to perform route_optimization
         and EUT away_from_home
         and EUT registered to QE1
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when {     EUT establishes a binding to QE4 }
         then {     EUT and QE4 are able to communicate directly
                             within the binding_lifetime
                and EUT and QE4 are unable to communicate directly
                             after MAX_RR_BINDING_LIFETIME }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1483_01
```

```
summary : 'Correspondent Node tries to refresh a binding cache entry
           before it expires'
RQ ref  : RQ_001_1483
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1483_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
         and EUT established 'in continuous unbounded communication with QE4'
       }
  ensure that
       { when { QE4 sends a packet to the EUT
                    after the binding_lifetime
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1390_01
summary : 'Correspondent Node maintains a separate binding cache for each
           of its unicast routable addresses'
RQ ref  : RQ_001_1390
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1390_01

  with {     EUT configured to have 2 global_addresses
         and QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding to QE4
                  using its first global_address
         and QE1 disconnected from the network
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                    containing a source_address
                    set to its second global_address }
         then { QE4 does not indicate receipt of the packet  }
       }


TP id   : TP_MOB_1458_01
summary : 'Correspondent Node updates a binding cache entry on receipt
           of a valid Binding Update'
RQ ref  : RQ_001_1458
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1458_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     QE4 moves to another foreign_network
                and EUT is requested to send a packet to QE4
                        indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1776_01
summary : 'Mobile Node updates binding with correspondent node upon receipt
           of a Binding Refresh Request'
RQ ref  : RQ_001_1776 , RQ_001_1383
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1776_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
```

```
      { when { QE4 sends a Binding_Refresh_Request }
        then { QE4 and the EUT are able to communicate directly
               after the original binding_lifetime }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1348_01
summary : 'Correspondent Node is able to participate in Return
           Routability Procedure'
RQ ref  : RQ_001_1348 , RQ_001_1050, RQ_001_1051
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1348_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1063_01
summary : 'Mobile Node is able to participate in Return Routability
           Procedure'
RQ ref  : RQ_001_1063 , RQ_001_1048 , RQ_001_1709,
          RQ_001_1007, RQ_001_1723, RQ_001_1750
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1063_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                indicating that a response is required }
         then { QE4 indicates receipt of the response directly from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1047_01
summary : 'Correspondent Node completes Return Routability Procedure
           before accepting a Binding Update from a mobile node'
RQ ref  : RQ_001_1047
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1047_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
         and QE1 unreachable from QE4 -- QE4 cannot complete RRP
       }
  ensure that
       { when { QE4 is requested to send a packet to the EUT
                indicating that a response is required }
         then { EUT sends no response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1485_01
summary : 'Correspondent Node stops retransmitting Binding Refresh Requests
           messages when it receives a Binding Update'
RQ ref  : RQ_001_1485
Role    : Correspondent_Node
config  : CF_MOB_04
```

```
TD ref  : TD_MOB_1485_01

--*********************************************************************--
--*  However we wrap this up, it is a conformance test, not interop      *--
--*********************************************************************--

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT having established a binding with QE4
       }
  ensure that
       { when {     EUT 'tries to refresh its binding with QE4'
                and EUT receives 'a message indicating that
                                  the refresh is successful' }
         then { EUT 'stops sending refresh requests to QE4' }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1725_01
summary : 'Mobile Node support the protection of Home Test and
           Home Test Init messages'
RQ ref  : RQ_001_1725
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1725_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE1 configured to protect return_routability_packets
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends a response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1065_01
summary : 'Correspondent Node sends a Binding_Acknowledgement message upon
           receipt of a Binding Update message containing A-Flag set to 1'
RQ ref  : RQ_001_1065, RQ_001_1350
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1065_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and QE4 configured 'to request Binding_Acknowledgements from
                             correspondent nodes'
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends the response directly to QE4 }
       }

End Group -- Group A

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

Group 1.2 'Group B'

TP id   : TP_MOB_1382_01
summary : 'Mobile Node is able to receive and process Binding Acknowledgements'
RQ ref  : RQ_001_1382
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1382_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and QE1 disconnected
```

```
        and QE3 having the same Home_Address as EUT
        and QE3 having the same ethernet_address as QE1
      }
  ensure that
      { when {     EUT returns home before the binding_lifetime expires
               and QE3 is disconnected
               and QE1 is connected to the Home_Network }
        then {     EUT and QE4 are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1700_01
summary : 'Mobile Node does not perform Duplicate Address Detection
           when returning home before the expiry of its bindings'
RQ ref  : RQ_001_1700
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1700_01

  with {     EUT away_from_home
        and EUT registered to QE1
        and QE1 is disconnected
        and QE3 having the same Home_Address as EUT
        and QE3 having the same ethernet_address as QE1
      }
  ensure that
      { when {     EUT returns home before the binding_lifetime expires
               and QE3 is disconnected
               and QE1 is connected to the Home_Network }
        then {     EUT and QE4 are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1693_01
summary : ' MN accepts packets at its previous CoA after registering
            its new primary CoA '
RQ ref  : RQ_001_1693
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1693_01

  with {     EUT away_from_home
        and EUT registered to QE1
        and QE8 advertizing a new prefix   -- thus causing a new care-of
                                           -- address to be generated
      }
  ensure that
      { when {         QE8 stops advertizing the old prefix
               before (QE8 starts advertizing a new prefix
               before  EUT receives a packet from QE3
                           indicating that a response is required
                       and containing a destination_address
                           set to the old Care_Of_Address of the EUT) }
        then { QE3 indicates receipt of the response from the EUT }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1678_01
summary : ' MN discover a new default router when it detects the default
            router is no longer bi-directionally reachable '
RQ ref  : RQ_001_1678
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1678_01

  with {     EUT connected in home_network
        and QE1 disconnected
      }
  ensure that
      { when {     EUT moves to a foreign_network
               and EUT receives a packet from QE3
                       indicating that a response is required
                   and containing a destination_address
                       set to the Care_of_Address of the EUT }
        then { QE3 indicates receipt of the response from the EUT }
```

```
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1507_01
summary : 'Home Agent removes binding when the associated prefix valid
           lifetime expires'
RQ ref  : RQ_001_1507
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1507_01

  with {     QE1 away_from_home
         and QE1 registered to the EUT
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { the prefix_lifetime expires at the EUT }
         then { QE4 and QE1 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1501_01
summary : 'Home Agent performs Duplicate Address Detection on home link
           before sending a Binding Acknowledgment'
RQ ref  : RQ_001_1501 , RQ_001_1502 , RQ_001_1503
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1501_01

  with {     QE1 at_home
         and QE3 disconnected from home_network
         and QE3 having the same home_address as QE1
       }
  ensure that
       { when {     QE1 is disconnected from the home_network
                and QE3 is connected to the home_network
                and QE1 is connected to a foreign_network }
         then { QE4 and QE1 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1500_01
summary : 'Home Agent maintains a Binding Cache entry from receipt of a
           Binding Update until the Binding Lifetime expires'
RQ ref  : RQ_001_1497, RQ_001_1500 , RQ_001_1522
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1500_01

  with {     QE1 at_home
         and QE1 configured not to perform Route_Optimization
    }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate
                        within the Binding_Lifetime }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1497_01
summary : 'Home Agent creates a new entry in its Binding Cache
           upon receipt of a Binding Update'
RQ ref  : RQ_001_1497
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1497_01

  with
    {
          QE1 at_home
      and QE1 configured not to perform Route_Optimization
    }
  ensure that
       { when { QE1 moves to a foreign_network }
```

```
        then { QE4 is able to communicate with QE1 }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1826_01
summary : 'Home Agent updates the entry in its Binding Cache
           upon receipt of a Binding Update'
RQ ref  : RQ_001_1826
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1826_01

  with {     QE1 away_from_home
         and QE1 registered to EUT
         and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { QE1 moves to another foreign_network }
         then { QE3 and QE1 are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1494_01
summary : 'Home Agent returns a type "132" Binding Acknowledgment
           upon receipt of a Binding Update which Home Address
           is not on link'
RQ ref  : RQ_001_1494
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1494_01

  with {    QE1 disconnected from the home_network
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when {     QE1 is connected to a foreign_network
                and EUT receives a Binding_Update
                        containing a Home_Address not on_link }
         then {     QE4 and QE1 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1503_01
summary : 'Home Agent returns a type "134" Binding Acknowledgement when
           Duplicate Address Detection for the Home Address / Link
           Local Address fails'
RQ ref  : RQ_001_1503
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1503_01

  with {     QE1 disconnected from home_network
         and QE3 connected to home_network
         and QE3 configured to have the same link_local_address as QE1
       }
  ensure that
       { when { QE1 is connected to a foreign_network }
         then { QE4 and QE1 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1684_01
summary : 'Mobile Node generates a new primary Care of Address
           when the current one is deprecated'
RQ ref  : RQ_001_1684
Role    : Mobile_Node
config  : CF_MOB_07
TD ref  : TD_MOB_1684_01

  with {     EUT away_from_home
         and QE3 configured as default_router for EUT
         and QE2 able to communicate with EUT
       }
  ensure that
```

```
       { when {         QE3 stops advertizing the old prefix
                before (QE3 starts advertizing a new prefix
                before  QE2 is requested to send a packet to the EUT
                           indicating that a response is required) }
         then {         QE2 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1683_01
summary : 'Mobile Node generates a new primary Care of Address after
           having moved'
RQ ref  : RQ_001_1683 , RQ_001_1380 , RQ_001_1690
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1683_01

  with {     EUT away_from_home
         and QE4 able to communicate with EUT
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE4 and the EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1465_01
summary : 'Correspondent Node deletes any existing related bindings
           on receipt of a Binding Update requesting
           to delete a binding'
RQ ref  : RQ_001_1465
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1465_01

  with {     EUT away_from_home
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT returns home }
         then { QE4 and the EUT are able to communicate with EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1496_01
summary : 'Correspondent Node sends a type "131" Binding Acknowledgment
           when receiving a Home Registration Binding Update'
RQ ref  : RQ_001_1496
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1496_01

  with {     EUT at_home
         and QE5 at_home
         and QE5 configured 'to use EUT as Home Agent'
       }
  ensure that
       { when { QE5 moves to a foreign_network }
         then { QE3 and QE5 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1528_01
summary : 'When accepting Care of Address de-registration Binding
           Update, Home Agent deletes any existing entry in its
           Binding Cache for the Mobile Node'
RQ ref  : RQ_001_1528 , RQ_001_1529
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1528_01

  with {     QE1 away_from_home
         and QE1 registered to EUT
```

```
        and QE4 able to communicate with QE1
        and QE1 configured not to perform Route_Optimization
      }
  ensure that
      { when { QE1 returns home }
        then { QE4 and QE1 are able to communicate directly }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1526_01
summary : 'Home Agent rejects De-Registration Binding Update if no
           entry exists in its Binding Cache for the Mobile Node'
RQ ref  : RQ_001_1526
Role    : Home_Agent
config  : CF_MOB_05
TD ref  : TD_MOB_1526_01

  with {     EUT disconnected from Home_Network
        and QE2 configured to have the same IP_address as EUT
        and QE1 connected to a foreign_network
        and QE1 registered to QE2
      }
  ensure that
      { when {     QE2 is disconnected from Home_Network
               and EUT is connected to Home_Network
               and QE1 returns home }
        then {     QE4 and QE1 are unable to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1380_01
summary : 'Mobile Node supports care-of address formation'
RQ ref  : RQ_001_1380
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1380_01

  with {     EUT connected to home_network
        and QE4 able to communicate with EUT
      }
  ensure that
      { when { EUT moves to a foreign_network }
        then { QE4 and the EUT are able to communicate }
      }

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TP id   : TP_MOB_1809_01
summary : 'Mobile Nodes support returning home'
RQ ref  : RQ_001_1809 , RQ_001_1808
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1809_01

  with {     EUT away_from_home
        and QE4 able to communicate with EUT
      }
  ensure that
      { when { EUT returns home }
        then { QE4 and the EUT are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1005_01
summary : 'When away, Mobile Node is addressable at one or
           more Care of Addresses'
RQ ref  : RQ_001_1005
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1005_01

  with
    {
         EUT at_home
      and QE6 'advertising a second prefix on Foreign Network 1'
```

```
    }
  ensure that
       { when { EUT moves to Foreign_Network 1 }
         then { QE4 and EUT are able to communicate
                    using first and second Care_of_address }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1702_01
summary : 'Mobile Node replies to Neighbour Solicitations for
           its Home Address after returning home
           and having sent a Binding Update'
RQ ref  : RQ_001_1702 , RQ_001_1695
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1702_01

  with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     EUT returns home
                and QE1 is disconnected from the home_network }
         then { QE4 and the EUT are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1701_01
summary : 'Mobile Node performs Duplicate Address Detection
           when returning home after its bindings have expired'
RQ ref  : RQ_001_1701
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1701_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT binding_lifetime expired at QE1
         and QE4 connected to home_network
                 using the same Home_Address as EUT
       }
  ensure that
       { when { EUT returns home }
         then { EUT is unable to communicate } -- DAD fails
    }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1768_01
summary : 'Mobile Node sends Binding Update before expiry of the
           binding lifetime'
RQ ref  : RQ_001_1768 , RQ_001_1728
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1768_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT able to communicate with QE4
       }
  ensure that
       { when { QE4 (is requested to send a packet to the EUT
                     indicating that a response is required)
                     after the binding_lifetime }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1694_01
summary : 'Mobile Node invalidates all associated Care of Addresses
           when the corresponding router is unreachable'
RQ ref  : RQ_001_1694
Role    : Mobile_Node
```

```
config  : CF_MOB_03
TD ref  : TD_MOB_1694_01

  with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE1 and the EUT are unable to communicate
                    using the old Care_of_Address }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1510_01
summary : 'Home Agent returns a Binding Acknowledgment in response to
           a Binding Update'
RQ ref  : RQ_001_1510
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1510_01

  with { QE1 at_home }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE4 and the EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1534_01
summary : 'Home Agent stops intercepting packets for the mobile
           node on acceptance of the Primary Care of Address
           De-Registration Binding Update'
RQ ref  : RQ_001_1534
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1534_01

  with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {     QE1 returns home
                and EUT is disconnected from the home_network }
         then { QE4 and QE1 are able to communicate }
       }

End Group -- Group B

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

Group 1.3 'Group C'

TP id   : TP_MOB_1344_01
summary : 'Correspondent Node discard received packets containing
           Type 2 Routing Headers'
RQ ref  : RQ_001_1344
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1344_01

  with {     EUT disconnected
         and QE4 configured to have the same link_local_address as EUT
         and QE4 connected to Home_Network 2
         and QE4 registered to QE1
         and QE4 having established a binding with QE3
       }
  ensure that
       { when {     QE4 is disconnected
                and EUT is connected to Home_Network 1
                and QE3 is requested to send a packet to the EUT
                        indicating that a response is required
                        -- QE3 sends packets containing
                        -- Routing Headers Type 2
                and QE5 is requested to send a packet to the EUT
                        indicating that a response is required }
                        -- QE5 sends packets not containing
```

```
                    -- Routing Headers Type 2
        then {     QE3 indicates no receipt of the response from the EUT
               and QE5 indicates receipt of the response from the EUT }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1003_01
summary : 'MN continues to communicate after moving to a new link'
RQ ref  : RQ_001_1003 , RQ_001_1004 , RQ_001_1002
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1003_01

  with {     EUT connected in home_network
         and EUT registered to QE1
         and EUT and QE4 established in communication
       }
  ensure that
       { when {      EUT moves to a foreign_network
                and QE4 is requested to send a packet to EUT
                        indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1614_01
summary : 'MN sends packets directly to the correspondent node when
           a binding exists'
RQ ref  : RQ_001_1614
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1614_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and EUT having established a binding to QE4
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { EUT sends the response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1412_01
summary : 'Correspondent Node receives packets directly from the Mobile Node
           when a binding exists or Mobile Node believes in its existence'
RQ ref  : RQ_001_1412
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1412_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT is requested to send a packet to QE4
                    indicating that a response is required }
         then { EUT indicates receipt of the response directly from QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1423_01
summary : 'Correspondent Node sends packets directly to Mobile Node
           when a binding already exists'
RQ ref  : RQ_001_1423
Role    : Correspondent_Node
```

```
config  : CF_MOB_04
TD ref  : TD_MOB_1423_01

  with {     QE4 away_from_home
         and QE4 registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 having established a binding with EUT
         and EUT having established a binding with QE4
       }
  ensure that
       { when { QE4 sends a packet to the EUT
                    indicating that a response is required }
         then { QE4 indicates receipt of the response directly from EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1615_01
summary : 'MN uses reverse tunnelling to the correspondent node when
           no binding exists'
RQ ref  : RQ_001_1615
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1615_01

  with {     EUT away_from_home
         and EUT registered to QE1
         and EUT configured to perform route_optimization
         and QE4 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT receives a packet from QE4
                    indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1592_01
summary : 'HA schedules the delivery of the prefix information when a new
           prefix is added to its home subnet interface'
RQ ref  : RQ_001_1592 , RQ_001_1370 , RQ_001_1805
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1592_01

  with {     QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {         EUT generates a new prefix to the home_address of QE1
                before (QE4 sends a packet to QE1
                            using the new home_address
                        and indicating that a response is required) }
         then {         QE4 indicates receipt of a response from QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1600_01
summary : 'HA retransmits heroically unsolicited Advertisement until
             the receipt of a Mobile_Prefix_Solicitation from MN'
RQ ref  : RQ_001_1600 , RQ_001_1813
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1600_01

  with {     QE1 away_from_home
         and QE1 registered to EUT }
  ensure that
       { when {        QE1 is disconnected from the foreign_network
                before EUT generates a new prefix to the home_address of QE1
                before QE1 is connected to the foreign_network after 1 minute
                before QE4 sends a packet to QE1
                            using the new home_address
                       and indicating that a response is required }
         then {        QE4 indicates receipt of a response from QE1 }
```

```
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1812_01
summary : 'HA deliveries new prefix information when lifetime information
           (valid or preferred lifetime) of the prefix changes'
RQ ref  : RQ_001_1812
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1812_01

  with {     QE1 away_from_home
         and QE1 registered to EUT
         and EUT configured to have a short prefix_lifetime on the
                         Home_Network of QE1 --(suggest 30s)
      }
  ensure that
      { when { EUT is requested to configure a new prefix_lifetime
                  for QE1 }
        then { QE1 and QE4 are able to communicate
                  using the new prefix
                  after the original prefix_lifetime expires }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1603_01
summary : 'HA transmits again unsolicited Mobile_Prefix_Advertisement after the
           prolongation of the binding of the Mobile Node'
RQ ref  : RQ_001_1603
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1603_01

  with {      QE1 away_from_home
         and QE1 is registered to the EUT
         and QE1 and the EUT configured to have a short binding_lifetime
                                                -- (Suggest 30s)
      }
  ensure that
      { when {        QE1 is disconnected from the foreign_network
               before the binding_lifetime expires between QE1 and the EUT
               before QE1 is connected to another foreign_network }
        then { QE1 and QE4 are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1612_01
summary : 'When MN starts a transport-level connections at Home, it conserves
           that connection away_from_home'
RQ ref  : RQ_001_1612
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1612_01

  with {     EUT connected in home_network
         and EUT having established a transport_connection to QE3
      }
  ensure that
      { when { EUT moves to a foreign_network }
        then { EUT and QE3 are able to communicate
                  using the transport_connection }
      }

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TP id   : TP_MOB_1613_01
summary : 'When MN starts a transport-level connections in Foreign network,
           it conserves this connection into another Foreign network'
RQ ref  : RQ_001_1613
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1613_01

  with {      EUT away_from_home
         and EUT registered to QE1
```

```
          and EUT having established a transport_connection to QE3
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { EUT and QE3 are able to communicate
                     using the transport_connection }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1413_01
summary : 'Correspondent Node drops packets containing Home Address
           option (but no Binding Update) when there is no corresponding
           Binding Cache entry'
RQ ref  : RQ_001_1413
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1413_01

  with {      EUT configured to perform route_optimization
          and QE4 configured to perform route_optimization
          and QE4 away_from_home
          and QE4 registered to QE1
          and EUT having no Binding_Cache_entry for QE4
          and QE4 having a Binding_Cache_entry for EUT
       }
  ensure that
       { when { EUT receives a packet directly from QE4
                     indicating that a response is required }
         then { EUT sends no response to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1413_02
summary : 'Correspondent Node drops packets containing Home Address
           option (but no Binding Update) and send a Binding Error
           message to the Mobile Node, when there is no corresponding
           Binding Cache entry'
RQ ref  : RQ_001_1413
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1413_02

  with {      EUT configured to perform route_optimization
          and QE4 configured to perform route_optimization
          and QE4 away_from_home
          and QE4 registered to QE1
          and EUT having no Binding_Cache_entry for QE4
          and QE4 having a Binding_Cache_entry for EUT
              }
  ensure that
       { when { EUT receives a packet directly from QE4
                     indicating that a response is required }
      then {        EUT sends no response to QE4
             before EUT and QE4 are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1414_01
summary : 'When using Route Optimization, a Correspondent Node drops any
           packet in which the Home Address option (but not Binding Update)
           does not match the Home Address recorded in its Binding Cache'
RQ ref  : RQ_001_1414
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1414_01

  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
       }
  ensure that
       { when { EUT receives a packet from QE4
                     containing a Home_Address_Option
```

```
                    not set to the Home_Address of QE4 }
          then { EUT discards the packet }
        }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1414_02
summary : 'When using Route Optimization, a Correspondent Node drops any
           packet in which the Home Address option (but not Binding Update)
           does not match the Home Address recorded in its Binding Cache and
           send a Binding Error message to the Mobile Node'
RQ ref  : RQ_001_1414
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1414_02

  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
        }
  ensure that
        { when {        EUT receives a packet from QE4
                            containing a Home_Address_Option
                            not set to the Home_Address of QE4 }
          then {        EUT discards the packet
                  before EUT and QE4 are able to communicate }
        }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1415_01
summary : 'A Correspondent Node drops packets in which the Source Address does
           not match the care-of address recorded in the corresponding
           Binding Cache Entry'
RQ ref  : RQ_001_1415
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1415_01


  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
        }
  ensure that
        { when { EUT receives a packet from QE4
                    containing a Home_Address_Option
                    set to the Home_Address of QE4
                 and containing a Source_Address
                    not set to the Care_Of_Address of QE4 }
          then { EUT discards the packet }
        }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1415_02
summary : 'A Correspondent Node drops packets in which the Source Address does
           not match the care-of address recorded in the corresponding
           Binding Cache Entry and sends a Binding Error message to the
           Mobile Node'
RQ ref  : RQ_001_1415
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1415_02


  with {     EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE4 away_from_home
         and QE4 registered to QE1
         and QE4 having established a binding with EUT
        }
  ensure that
        { when {        EUT receives a packet from QE4
```

```
                        containing a Home_Address_Option
                        set to the Home_Address of QE4
                    and containing a Source_Address
                        not set to the Care_Of_Address of QE4 }
        then {      EUT discards the packet
               before QE4 and the EUT are able communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1384_01
summary : 'MN reconfigures its home address according to the prefix information
           in a Mobile_Prefix_Advertisement'
RQ ref  : RQ_001_1384
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1384_01

  with { EUT away_from_home
           and EUT registered to QE1
       }
  ensure that
       { when { QE1 is requested to modify its home_network prefix }
         then { QE4 and the EUT are able to communicate
                    using the new home_address }
       }

End Group -- Group C

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--
--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

Group 1.4 'Group D'


Group 1.4.2

TP id   : TP_MOB_1574_01
summary : 'Home Agent supports the protection of Home Test and
           Home Test Init messages'
RQ ref  : RQ_001_1574
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1574_01

  with {     QE4 configured to perform route_optimization
         and QE1 configured to perform route_optimization
         and QE1 away_from_home
         and QE1 registered to EUT
         and QE1 configured to protect return_routability_packets
       }
  ensure that
       { when { QE1 receives a packet from QE4
                    indicating that a response is required }
         then { QE1 sends the response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1371_01
summary : 'Home Agent supports ESP protection of Home Test and
           Home Test Init messages'
RQ ref  : RQ_001_1371
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1371_01

  with {     QE4 configured to perform route_optimization
         and QE1 configured to perform route_optimization
         and QE1 configured to protect return_routability_packets
                          using ESP
       }
  ensure that
       { when {      QE1 moves to a Foreign_Network
               before QE4 sends a packet to QE1
                          indicating that a response is required }
```

```
        then {        QE4 indicates receipt of the response directly from QE1 }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1082_01
summary : 'Home Agent uses a Security Association to protect integrity and
           authenticity of Mobile_Prefix_Solicitations and Advertisements'
RQ ref  : RQ_001_1082 , RQ_001_1014 , RQ_001_1607
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1082_01

  with {     QE1 at_home
        and QE1 configured to protect
                          Mobile_Prefix_Solicitations
        and QE1 configured to accept only secured
                          Mobile_Prefix_Advertisements
        and QE1 configured to have a short prefix_lifetime --(suggest 30s)
      }
  ensure that
      { when {        QE1 moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to QE1
                          indicating that a response is required }
        then {        QE4 indicates receipt of the response from QE1 }
      }
--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1082_02
summary : 'Mobile Node uses a Security Association to protect integrity and
           authenticity of Mobile Prefix Solicitations and Advertisements'
RQ ref  : RQ_001_1082 , RQ_001_1014 , RQ_001_1663 , RQ_001_1664
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1082_02

  with {     EUT at_home
        and QE1 configured to accept only secured
                           Mobile_Prefix_Solicitations
        and QE1 configured to protect Mobile_Prefix_Advertisements
        and EUT configured to have a short prefix_lifetime (--suggest 30s)'
      }
  ensure that
      { when {        EUT moves to a foreign_network
                before its prefix_lifetime expires
                before QE4 sends a packet to the EUT
                          indicating that response is required }
        then {        QE4 indicates receipt of the response from the EUT }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1385_01
summary : 'Mobile Node supports Dynamic Home Agent Address Discovery'
RQ ref  : RQ_001_1385
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1385_01

  with {     EUT registered to QE1
        and EUT disconnected
        and EUT configured not to perform Route_Optimization
        and QE4 configured not to perform Route_Optimization
      }
  ensure that
      { when {            QE1 is requested to modify its IP_address
                before the EUT is connected to a foreign_network }
        then {            QE4 and the EUT are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1657_01
summary : 'Mobile Node attempts Dynamic Home Agent Address Discovery
           if it cannot contact its current Home Agent'
RQ ref  : RQ_001_1657
Role    : Mobile_Node
```

```
config  : CF_MOB_06
TD ref  : TD_MOB_1657_01

  with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     QE1 is disconnected from the Home_Network
                and EUT moves to another foreign_network }
         then {     QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1743_01
summary : 'Mobile Node goes on registering home addresses when
           the registration of one of them fails'
RQ ref  : RQ_001_1743
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1743_01

  with {     EUT disconnected
         and EUT configured to have 2 home_addresses
         and QE3 configured to use the first home_address of EUT
       }
  ensure that
       { when {     EUT is connected to a foreign_network
                and EUT receives a packet from QE4
                        containing a destination_address
                        set to EUT home_address 2
                    and indicating that a response is required }
         then {     QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1727_01
summary : 'Mobile Node registers its new primary Care-Of Address
           after changing it'
RQ ref  : RQ_001_1727
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1727_01

  with {     EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {     EUT moves to another foreign_network
                and EUT receives a packet from QE4
                        indicating that a response is required }
         then { QE4 indicates receipt of the response from the EUT }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1742_01
summary : 'Mobile Node does not attempt to use its home address
           if registration with Home Agent fails due to DAD'
RQ ref  : RQ_001_1742
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1742_01

  with { QE3 configured to have the home_address of EUT }
  ensure that
       { when { EUT moves to a foreign_network }
              -- the registering of EUT has failed
        then { EUT does not use its Home_Address to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1740_01
summary : 'Mobile Node attempts to register all of its home addresses'
RQ ref  : RQ_001_1740
Role    : Mobile_Node
```

```
config  : CF_MOB_03
TD ref  : TD_MOB_1740_01

  with {     EUT disconnected
         and EUT configured to have 2 home_addresses
       }
  ensure that
       { when {     EUT is connected to a foreign_network
            and QE4 is requested to send a packet
                            indicating that a response is required
                          to the first home_address of EUT
            and QE4 is requested to send a packet
                            indicating that a response is required
                          to the second home_address of EUT }
         then {     QE4 indicates receipt of the first response
                and QE4 indicates receipt of the second response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1016_01
summary : 'Home Agent supports transport_mode_ESP protection of
           Binding Update messages'
RQ ref  : RQ_001_1016
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1016_01

  with {     QE1 disconnected
         and QE1 configured to protect any Binding_Update packet
                                    using transport_mode_ESP
       }
  ensure that
       { when {     QE1 is connected to a foreign_network
            and QE1 receives a packet from QE4
                            indicating that a response is required }
         then { QE1 sends response directly to QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1017_01
summary : 'Mobile Node uses transport_mode_ESP to protect Binding Update
           messages'
RQ ref  : RQ_001_1017
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1017_01

  with { QE1 configured to accept only a Binding_Update
            secured using transport_mode_ESP
       }
  ensure that
       { when {     EUT moves to a foreign_network
            and QE4 is requested to send a packet to EUT
                            indicating that a response is required }
         then { QE4 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1655_01
summary : 'Mobile Node already registered to a Home Agent uses that
           Home Agent for any new registrations'
RQ ref  : RQ_001_1655
Role    : Mobile_Node
config  : CF_MOB_06
TD ref  : TD_MOB_1655_01

  with
    {     EUT configured 'to use DHAAD'
      and EUT away_from_home
      and EUT registered to QE1
      and QE2 disconnected from Home_Network 1
    }
  ensure that
       { when {     QE1 is disconnected from Home_Network 1
            and QE2 is connected to Home_Network 1
```

```
            and EUT moves to another foreign_network }
        then {    QE4 and EUT are unable to communicate
                   within MAX_BINDACK_TIMEOUT
            -- EUT tries to register with QE1
            and QE4 is able to communicate after MAX_BINDACK_TIMEOUT }
            -- EUT registered to QE2
    }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1366_01
summary : 'Home Agent returns a Binding_Acknowledgement message in response
           to a Binding Update message'
RQ ref  : RQ_001_1366
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1366_01

  with {     QE1 at_home
       }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 receives a packet from QE4
                                indicating that a response is required }
         then {     QE4 indicates receipt of the response}
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1804_01
summary : 'Home Agent is able to participate in
           Dynamic Home Agent Address Discovery'
RQ ref  : RQ_001_1804 , RQ_001_1368
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1804_01

  with {     QE1 at_home
         and QE1 not configured to have its Home_Agent_address available
       }
  ensure that
       { when {     QE1 moves to a foreign_network
              and QE4 is requested to send a packet to QE1
                                  indicating that a response is required }
         then {    QE4 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1733_01
summary : 'Mobile Node sends Binding Update messages to its Home Agent
           until it receives a matching Binding_Acknowledgement message,
           before MAX_BINDACK_TIMEOUT seconds'
RQ ref  : RQ_001_1733
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1733_01

  with {     EUT at_home
         and EUT configured to have its Home_Agent_address available
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE1 is connected after 15 seconds
                     -- ~ MAX_BINDACK_TIMEOUT / 2
           }
         then {     QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1733_02
summary : 'If the Mobile Node knows just one Home Agent (no DHAAD),
           it continues to send Binding Update messages to its
           Home Agent after MAX_BINDACK_TIMEOUT seconds'
RQ ref  : RQ_001_1733
Role    : Mobile_Node
```

```
config  : CF_MOB_03
TD ref  : TD_MOB_1733_02

  with {     EUT at_home
         and EUT configured to have its Home_Agent_address available
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE1 is connected after MAX_BINDACK_TIMEOUT
           }
         then {     QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1734_01
summary : 'When Mobile Node knows several Home Agent (returned during DHAAD),
           Mobile Node tries to register with other Home Agents when
           registration fails with the first one'
RQ ref  : RQ_001_1734
Role    : Mobile_Node
config  : CF_MOB_06
TD ref  : TD_MOB_1734_01

  with {     EUT not configured to have its Home_Agent_address available
         and QE1 configured to have a Home_Agent_Preference greater than QE2
       }
  ensure that
       { when {     QE1 is disconnected
                and EUT moves to a foreign_network }
         then {     QE4 and EUT are able to communicate
                  after  MAX_BINDACK_TIMEOUT
           }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1814_01
summary : 'Mobile Node sends Binding Update messages to its Home Agent
           until it receives a matching Binding_Acknowledgement message,
           without time limit if it knows only one Home Agent'
RQ ref  : RQ_001_1814
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1814_01

  with {     EUT at_home
         and EUT configured to have its Home_Agent_address available
         and QE1 disconnected
       }
  ensure that
       { when {     EUT moves to a foreign_network
                and QE1 is connected after  MAX_BINDACK_TIMEOUT
                and QE4 is requested to send a packet to QE1
                                  indicating that a response is required }
         then {     QE4 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1013_01
summary : 'Home Agent supports the protection of Binding Update message'
RQ ref  : RQ_001_1013
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1013_01

  with
    {        QE1 at_home
         and QE1 configured to protect any Binding_Update
    }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE4 is requested to send a packet to QE1
                                  indicating that a response is required }
```

```
        then { QE4 indicates receipt of the response }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1013_02
summary : 'Mobile Node supports the protection of Binding Update message'
RQ ref  : RQ_001_1013
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1013_02

  with {     EUT at_home
        and QE1 configured to accept only a secured Binding_Update
      }
  ensure that
      { when {     EUT moves to a foreign_network
             and QE4 is requested to send a packet to EUT
                                indicating that a response is required }
        then { QE4 indicates receipt of the response }
      }

End Group

End Group

Group 1.5 'Group E'

TP id   : TP_MOB_1376_01
summary : 'Mobile Node is able to perform IPv6 encapsulation' -- miss decapsulation
RQ ref  : RQ_001_1376
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1376_01

  with {    EUT away_from_home
        and EUT registered to QE1
        and EUT configured not to perform Route_Optimization
      }
  ensure that
      { when { QE4 is requested to send a packet to EUT
                              indicating that a response is required }
        then { QE4 indicates receipt of the response }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1536_01
summary : 'HA intercepts packets for registered mobile node on the home
           network and uses reverse tunnel encapsulation'
RQ ref  : RQ_001_1536 , RQ_001_1551 , RQ_001_1568 , RQ_001_1364
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1536_01

  with {    QE1 away_from_home
        and QE1 registered to EUT
        and QE1 configured not to perform Route_Optimization
      }
  ensure that
      { when { QE3 is requested to send a packet to QE1
                              indicating that a response is required }
        then { QE3 indicates receipt of the response }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1537_01
summary : 'HA multicasts Neighbour Advertisement in Home network in order
            to intercept packets for mobile node'
RQ ref  : RQ_001_1537
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1537_01

  with {      QE1 at home
         and QE1 having established communication to QE3
      }
```

```
  ensure that
       { when {      QE1 moves to a foreign_network
          }
         then { QE3 updates its Neighbor_Discovery_cache_entry
                                for the link_local_address of QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1547_01
summary : 'HA act as proxy for a registered mobile node and reply to any
           received Neighbour Solicitations for it'
RQ ref  : RQ_001_1547
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1547_01

  with {      QE3 'is disconnected'
         and QE1 away_from_home
        and QE1 registered to EUT
       }
  ensure that
       { when {      QE3 is connected to its home_network
              and QE3 is requested to send a packet to QE1
                                 indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1557_01
summary : 'HA tunnels multicast packets with a global scope to the mobile node'
RQ ref  : RQ_001_1557 , RQ_001_1372
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1557_01

  with {      QE1 subscribed to a global_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when { QE3 is requested to send a packet to the global_multicast_group
                                 indicating that a response is required }
         then { QE3 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1552_01
summary : 'HA does not tunnel packets to the link_local_address of
           mobile node'
RQ ref  : RQ_001_1552
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1552_01

  with {     QE1 away_from_home
        and QE1 registered to EUT
       }
  ensure that
       { when { QE3 is requested to send a packet
                                 to the link_local_address of QE1
                                 indicating that a response is required }
         then { QE3 indicates no receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1803_01
summary : 'Home Agent discards packets received from
           a non registered mobile node'
RQ ref  : RQ_001_1803
Role    : Home_Agent
config  : CF_MOB_05
TD ref  : TD_MOB_1803_01

  with
```

```
    {
          EUT disconnected from home_network 1
      and QE2 configured to use the same IP_address as EUT
      and QE1 away_from_home
      and QE1 registered to QE2
      and QE1 configured to perform route_optimization
      and QE4 configured to perform route_optimization
    }
  ensure that
       { when {     QE2 is disconnected from home_network
                and EUT is connected to home_network
                and QE1 is requested to send a packet to QE4
                                indicating that a response is required }
         then {     QE3 indicates no receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1538_01
summary : 'HA multicast Neighbour Advertisement to intercept the packets
           to the link_local_address of a registered mobile node'
RQ ref  : RQ_001_1538
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1538_01

  with {      QE1 at_home
         and QE1 using the same Interface_ID for its link_local_address
                                              and its home_address
         and QE3 having established communication
                   to the link_local_address of QE1
       }
  ensure that
       { when { QE1 moves to a foreign_network }
         then { QE3 updates the Neighbor_Discovery_cache_entry
                    for the link_local_address of QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1555_01
summary : 'HA does not tunnel multicast packets with a scope smaller than
           global to the mobile node '
RQ ref  : RQ_001_1555 , RQ_001_1556
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1555_01

  with {     QE1 subscribed to a link_local_multicast_group
         and QE1 subscribed to a organization_local_multicast_group
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {     QE3 is requested to send a packet
                                    to the link_local_multicast_group
                                    indicating that a response is required
                and QE3 is requested to send a packet
                                    to the organization_local_multicast_group
                                    indicating that a response is required }
         then { QE3 indicates no receipt of any response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1560_01
summary : 'if multicast forwarding is not supported, HA ignores multicast
             group membership control messages'
RQ ref  : RQ_001_1560
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1560_01

  with {      QE1 subscribed to a global_multicast_group
         and EUT unable to support multicast_data_packet_forwarding
         and QE1 away_from_home
         and QE1 registered to EUT
       }
```

```
  ensure that
       { when { QE3 is requested to send a packet
                                to the global_multicast_group }
         then { QE3 indicates no receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1810_01
summary : 'MN is able to receive tunnelled multicast packets from HA'
RQ ref  : RQ_001_1810
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1810_01

  with {      EUT subscribed to a global_multicast_group
          and EUT away_from_home
          and EUT registered to QE1
       }
  ensure that
       { when { QE3 is requested to send a packet
                                 to the global_multicast_group }
         then { QE3 indicates no receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1817_01
summary : 'Correspondent Node registers the binding when it receives a BU
           from Mobile Node'
RQ ref  : RQ_001_1817
Role    : Correspondent_Node
config  : CF_MOB_04
TD ref  : TD_MOB_1817_01

  with { QE4 away_from_home
     and QE4 registered to QE1
         and QE4 configured to perform route_optimization
         and EUT configured to perform route_optimization
       }
  ensure that
       { when { QE4 is requested to send a packet directly to EUT
                                indicating that a response is required
           }
         then { QE4 indicates receipt of the response }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1818_01
summary : 'A Binding Update contains a Home Address destination option.'
RQ ref  : RQ_001_1818
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1818_01

  with
    { QE1 at_home
    }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 sends a Binding_Update
                     containing no Home_Address_Destination_Option }
         then { QE4 and QE1 are unable to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1820_01
summary : 'When Home Agent receives a packet from Mobile Node using reverse
           tunnelling, it forwards the encapsulated packet to Correspondent Node'
RQ ref  : RQ_001_1820
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1820_01

  with {    QE1 away_from_home
        and QE1 registered to EUT
```

```
        and QE1 configured not to perform Route_Optimization
       }
  ensure that
       { when { EUT is disconnected }
         then { QE1 and QE4 are unable to communicate with QE4 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1824_01
summary : 'When Mobile Node detects a L3 handover, it constructs a new
           Care-of Address, it select a new default router and register
           the new address to its Home Agent and the Correspondent Nodes
           which it is performing route optimization'
RQ ref  : RQ_001_1824, RQ_001_1682
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_1824_01

  with {    EUT away_from_home
        and EUT registered to QE1
        and EUT configured to perform route_optimization
        and QE4 configured to perform route_optimization
        and EUT having established a binding to QE4
       }
  ensure that
       { when { EUT moves to another foreign_network }
         then { QE4 and EUT are able to communicate directly }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1825_01
summary : 'When Home Agent processes a Binding Update secured by IPsec,
           the contents of Home Address Option is not checked'
RQ ref  : RQ_001_1825
Role    : Home_Agent
config  : CF_MOB_05
TD ref  : TD_MOB_1825_01

  with {     QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
         and EUT configured to accept any secured Binding_Update from QE1
         and QE1 configured to send only a secured Binding_Update to EUT
       }
  ensure that
       { when { QE1 sends a Binding_Update secured by IPsec
                           containing a Home_Address_Option
                                  not indicating its Home_Address
           }
         then { EUT accepts the packet }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_1807_01
summary : 'If HA support the multicast group membership control protocols,
           HA must be capable to determine which multicast data packets to
           forward via the tunnel to MN'
RQ ref  : RQ_001_1807 , RQ_001_1561
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_1807_01

-- Config-HA
  with {      QE1 connected to home_network
         and QE1 subscribed to a global_multicast_group
         and QE3 subscribed to another global_multicast_group
       }
  ensure that
       { when {     QE1 moves to a foreign_network
                and QE1 registers to EUT }
         then {     QE1 receives packets from its global_multicast_group }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--
```

```
End Group -- Group E

End Group -- Group 1


--*************************************************--
--* RFC3776  -  Security for Mobility
--*************************************************--

Group 5

TP id   : TP_MOB_2013_01
summary : 'Mobile Node sends Binding Update secured using transport_mode_ESP
           to its Home Agent'
RQ ref  : RQ_001_2013 , RQ_001_2031
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2013_01

  with {     EUT configured to protect any Binding_Update to QE1
                            using transport_mode_ESP
         and QE1 configured to accept only Binding_Update secured
                            using transport_mode_ESP from EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                  and EUT moves to a foreign_network }
         then { QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2013_02
summary : 'Home Agent sends Binding_Acknowledgement secured
           using transport_mode_ESP to a Mobile Node'
RQ ref  : RQ_001_2013 , RQ_001_2031
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2013_02

  with {     EUT configured to protect any Binding_Acknowledgement to QE1
                            using transport_mode_ESP
         and QE1 configured to accept only a Binding_Acknowledgement secured
                            using transport_mode_ESP from EUT
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
                   and QE1 moves to a foreign_network
           }
         then { QE4 is able to communicate with QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2014_01
summary : 'Home Agent supports Home Test Init secured using tunnel_mode_ESP'
RQ ref  :  RQ_001_2034, RQ_001_2035, RQ_001_1571
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2014_01

  with {     QE1 configured to protect any Home_Test_Init to EUT
                            using tunnel_mode_ESP
         and EUT configured to accept only Home_Test_Init secured
                            using tunnel_mode_ESP from QE1
         and QE1 configured to perform route_optimization
         and QE4 configured to perform route_optimization
         and QE1 is away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                  and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate directly }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--
```

```
TP id   : TP_MOB_2014_02
summary : 'Mobile Node supports Home Test secured using tunnel_mode_ESP'
RQ ref  : RQ_001_2034 , RQ_001_2035
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2014_02

  with {     QE1 configured to protect any Home_Test to EUT
                          using tunnel_mode_ESP
         and EUT configured to accept only a Home_Test secured
                          using tunnel_mode_ESP from QE1
         and EUT configured to perform Route_Optimization
         and QE4 configured to perform route_optimization
         and EUT is away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and EUT moves to a foreign_network }
         then {     QE4 and EUT are able to communicate directly }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2015_01
summary : 'Mobile Node sends Home Test Init secured
           using tunnel_mode_ESP to its Home Agent'
RQ ref  : RQ_001_2034
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2015_01

  with {     EUT configured to protect any Home_Test_Init to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only Home_Test_Init secured
                          using tunnel_mode_ESP from EUT
         and EUT configured to perform route_optimization
         and QE4 configured to perform route_optimization
       }
  ensure that
       { when {      a Security_Association is established between EUT and QE1
               and EUT moves to a foreign_network }
         then { QE4 and EUT are able to communicate directly }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2015_02
summary : 'Home Agent sends Home Test secured using tunnel_mode_ESP'
RQ ref  : RQ_001_2034
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2015_02

  with {     EUT configured to protect any Home_Test to QE1
                          using tunnel_mode_ESP
         and QE1 configured to accept only Home_Test secured
                          using tunnel_mode_ESP from EUT
         and QE1 configured to perform Route_Optimization
         and QE4 configured to perform Route_Optimization
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                and QE1 moves to a foreign_network }
         then { QE4 and QE1 are able to communicate directly }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2016_01
summary : 'Home Agent supports Mobile_Prefix_Solicitation secured
           using transport_mode_ESP'
RQ ref  : RQ_001_2016 , RQ_001_2031
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2016_01

  with {     QE1 configured to protect any Mobile_Prefix_Solicitation to EUT
```

```
                          using transport_mode_ESP
        and EUT configured to accept only a Mobile_Prefix_Solicitation secured
                          using transport_mode_ESP from QE1
        and QE6  configured to have a prefix_lifetime of 1 minute
        and QE1 away_from_home
        and QE1 registered to EUT
      }
  ensure that
      { when { a Security_Association is established between EUT and QE1 }
        then { QE4 is able to communicate with QE1 after the prefix_lifetime }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2016_02
summary : 'Mobile Node supports Mobile_Prefix_Advertisement secured
           using transport_mode_ESP'
RQ ref  : RQ_001_2016 , RQ_001_2031
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2016_02

  with {     QE1 configured to protect any Mobile_Prefix_Advertisement to EUT
                          using transport_mode_ESP
        and EUT configured to accept only Mobile_Prefix_Advertisement
                           secured using transport_mode_ESP from QE1
        and EUT away_from_home
        and EUT registered to QE1
        and QE6 configured to have a prefix_lifetime of 1 minute
      }
  ensure that
      { when { a Security_Association is established between EUT and QE1 }
        then { QE4 and EUT are able to communicate after prefix_lifetime }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2017_01
summary : 'Mobile Node sends Mobile_Prefix_Solicitation secured
           using transport_mode_ESP to its Home Agent'
RQ ref  : RQ_001_2017 , RQ_001_2030
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2017_01

  with {     EUT configured to protect any Mobile_Prefix_Solicitation to QE1
                          using transport_mode_ESP
        and QE1 configured to accept only a Mobile_Prefix_Solicitation
                          secured using transport_mode_ESP from EUT
        and QE6 configured to have a prefix_lifetime of 1 minute
        and EUT away_from_home
        and EUT registered to QE1
      }
  ensure that
      { when { a Security_Association is established between EUT and QE1 }
        then { QE4 and EUT are able to communicate after prefix_lifetime }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2017_02
summary : 'Home Agent sends Mobile_Prefix_Advertisement secured
           using transport_mode_ESP'
RQ ref  : RQ_001_2017 , RQ_001_2030
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2017_02

  with {     EUT configured to protect any Mobile_Prefix_Advertisement to QE1
                          using transport_mode_ESP
        and QE1 configured to accept only a Mobile_Prefix_Advertisement
                           secured using transport_mode_ESP from EUT
        and QE1 away_from_home
        and QE1 registered to EUT
        and QE6 configured to have a prefix_lifetime of 1 minute
      }
  ensure that
      { when { a Security_Association is established between EUT and QE1 }
```

```
        then { QE4 is able to communicate with QE1 after prefix_lifetime }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2018_01
summary : 'Mobile Node sends payload_packet tunnelled to its Home Agent
          secured using tunnel_mode_ESP'
RQ ref  : RQ_001_2018
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2018_01

  with {     EUT configured to protect any payload_packet to QE1
                           using tunnel_mode_ESP
        and QE1 configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from EUT
        and EUT configured not to perform Route_Optimization
        and EUT away_from_home
        and EUT registered to QE1
      }
  ensure that
      { when {      a Security_Association is established between EUT and QE1
               and EUT moves to a foreign_network
          }
        then { QE4 and EUT are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2018_02
summary : 'Home Agent sends payload_packet tunnelled to Mobile Node secured
          using tunnel_mode_ESP'
RQ ref  : RQ_001_2018
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2018_02

  with {     EUT configured to protect any payload_packet to QE1
                           using tunnel_mode_ESP
        and QE1 configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from EUT
        and QE1 configured not to perform Route_Optimization
        and QE1 away_from_home
        and QE1 registered to EUT
      }
  ensure that
      { when {   a Security_Association is established between EUT and QE1
               and QE1 moves to a foreign_network }
        then { QE4 and QE1 are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2018_03
summary : 'Mobile Node supports payload_packet secured
           using tunnel_mode_ESP'
RQ ref  : RQ_001_2018
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2018_03

  with {     QE1 configured to protect any payload_packet to EUT
                           using tunnel_mode_ESP
        and EUT configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from QE1
        and EUT configured not to perform Route_Optimization
        and EUT away_from_home
        and EUT registered to QE1
      }
  ensure that
      { when {   a Security_Association is established between EUT and QE1
               and EUT moves to a foreign_network }
        then { QE4 and EUT are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--
```

```
TP id   : TP_MOB_2018_04
summary : 'Home Agent supports payload_packet tunnelled from Mobile Node
           and secured using tunnel_mode_ESP'
RQ ref  : RQ_001_2018
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2018_04

  with {     QE1 configured to protect any payload_packet to EUT
                           using tunnel_mode_ESP
         and EUT configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from QE1
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {   a Security_Association is established between EUT and QE1
                  and QE1 moves to a foreign_network }
         then { QE4 is able to communicate with QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--


TP id   : TP_MOB_2025_01
summary : 'When Mobile Node returns home and uses Internet Key Exchange,
           it makes inactive its Security_Policy protecting
           tunnelled traffic with its Home Agent'
RQ ref  : RQ_001_2025
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2025_01

  with {     EUT configured to use IKEv2 to implement its Security_Policy
         and EUT configured to protect any payload_packet to QE1
                           using tunnel_mode_ESP
         and QE1 configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from EUT
         and EUT configured not to perform Route_Optimization
         and EUT away_from_home
         and EUT registered to QE1
         and EUT having established communication to QE4
       }
  ensure that
       { when { EUT returns home }
         then { QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2025_02
summary : 'When Mobile Node returns home and uses Internet Key Exchange,
           it makes inactive its Security_Policy protecting
           tunnelled traffic with its Home Agent'
RQ ref  : RQ_001_2025
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2025_02

  with {     QE1 configured to use IKEv2 to implement its Security_Policy
         and EUT configured to protect any payload_packet to QE1
                           using tunnel_mode_ESP
         and QE1 configured to accept only a payload_packet secured
                           using tunnel_mode_ESP from EUT by ESP
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
         and QE1 having established communication to QE4
       }
  ensure that
       { when { QE1 returns home }
         then { QE4 is able to communicate with QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2026_01
```

```
summary : 'When Mobile Node returns home and is configured with manually
           established Security Associations, it makes inactive security
           policies concerning tunnelled traffic with Home Agent without
           deleting Security Associations'
RQ ref  : RQ_001_2026
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2026_01

  with {     EUT configured 'with manually established Security Associations
                             and Policies'
         and EUT configured to protect packets to QE1
                            using tunnel_mode_ESP
         and QE1 configured to accept only a packets secured
                            using tunnel_mode_ESP
         and EUT configured not to perform Route_Optimization
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when {       EUT moves to another foreign_network
                after EUT returns home }
         then {       QE4 and EUT are able to communicate }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2026_02
summary : 'When Mobile Node returns home and is configured with manually
           established Security Associations, Home Agent makes inactive
           security policies concerning tunnelled traffic with Mobile Node
           without deleting Security Associations'
RQ ref  : RQ_001_2026
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2026_02

  with {     EUT configured 'with manually established Security Associations
                             and Policies'
         and EUT configured to protect packets to QE1
                            using tunnel_mode_ESP
         and QE1 configured to accept only a packets secured
                            using tunnel_mode_ESP
         and QE1 configured not to perform Route_Optimization
         and QE1 away_from_home
         and QE1 registered to EUT
       }
  ensure that
       { when {       QE1 moves to another foreign_network
                after QE1 returns home }
         then {       QE4 is able to communicate with QE1 }
       }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2019_01
summary : 'Mobile Node supports multicast group membership control packets
           tunnelled from its Home Agent protected by ESP with the tunnel mode'
RQ ref  : RQ_001_2019
Role    : Mobile_Node
config  : CF_MOB_03
TD ref  : TD_MOB_2019_01

  with {     QE1 configured to protect any
                            multicast_group_membership_control_packet to EUT
                            using tunnel_mode_ESP
         and EUT configured to accept only a
                            multicast_group_membership_control_packet from QE1
                            secured using tunnel_mode_ESP
         and EUT configured to subscribed to a global_multicast_group
         and EUT away_from_home
         and EUT registered to QE1
       }
  ensure that
       { when { QE3 sends a packet to the global_multicast_group
                                   indicating that a response is required }
```

```
        then { QE3 indicates receipt of the response }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2019_02
summary : 'Home Agent supports multicast group membership control packets
          tunnelled from Mobile Node protected by ESP with the tunnel mode'
RQ ref  : RQ_001_2019
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2019_02

  with {     QE1 configured to protect any
                           multicast_group_membership_control_packet to EUT
                           using tunnel_mode_ESP
        and EUT configured to accept only
                         a multicast_group_membership_control_packet from QE1
                         secured using tunnel_mode_ESP
        and QE1 configured to subscribed to a global_multicast_group
        and QE1 away_from_home
        and QE1 registered to EUT
      }
  ensure that
      { when { EUT sends a packet to the global_multicast_group
                                  indicating that a response is requested }
        then { QE3 indicates receipt of the response }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2039_01
summary : 'Home Agent update its security association entries when the
          Mobile Node change of foreign network'
RQ ref  : RQ_001_2039
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2039_01

  with {     EUT configured to protect any Binding_Acknowledgement to QE1
                           using tunnel_mode_ESP
        and QE1 configured to accept only a Binding_Acknowledgement
                           secured using tunnel_mode_ESP from EUT
        and QE1 away_from_home
        and QE1 registered to EUT
      }
  ensure that
      { when { QE1 moves to another foreign_network }
        then { QE4 and QE1 are able to communicate }
      }

--xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx--

TP id   : TP_MOB_2040_01
summary : 'Home Agent does not take into account a Binding Update not
          secured'
RQ ref  : RQ_001_2040
Role    : Home_Agent
config  : CF_MOB_02
TD ref  : TD_MOB_2040_01

  with { and EUT configured to accept only a Binding_Update
                           secured using tunnel_mode_ESP
        and QE1 configured to send any Binding_Update not secured
      }
  ensure that
      { when { QE1 moves to a foreign_network }
        then { QE4 and QE1 are unable to communicate }
      }

End Group 5
```

Annex C (informative): Bibliography

- ETSI TS 102 424: "Telecommunications and Internet converged Services and Protocols for Advanced Networking (TISPAN); Requirements of the NGN network to support Emergency Communication from Citizen to Authority".
- ETSI ES 202 553: "Methods for testing and Specification (MTS); TPLan: A notation for expressing test Purposes".

History

Document history		
V1.1.1	June 2007	Publication